KongohPlus ver. 1.0.0

User Manual

October 22nd, 2026

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1. Overview of KongohPlus

KongohPlus is a probabilistic genotyping software [1] designed to perform likelihood ratio analysis of crime stain profiles. Based on a statistical model that incorporates peak height information, the software calculates the probability of observing a DNA profile under competing hypotheses and evaluates the likelihood ratio (LR) as the ratio of these probabilities.

KongohPlus has been developed for forensic DNA analysis to support the interpretation of crime stain profiles. The statistical model accounts for factors that affect peak heights, including DNA quantity, contributor mixture proportions, DNA degradation, stutter, dropout, and drop-in. The software also allows users to configure the analysis conditions and laboratory-specific parameters, enabling analyses tailored to the characteristics of individual laboratory workflows.

This manual describes the installation of KongohPlus, the required settings, the analysis procedures, and the interpretation of the analysis results. It also explains the statistical models and computational principles underlying the software.

2. Initial Setup

1. Ensure that R (>= 4.5.2) is installed. It is available from the R Development Core Team website.
2. Begin an R session.
3. Execute the following command in R to install required packages.

```
install.packages('https://github.com/manabe0322/KongohPlus/releases/download/
    v1.0.0/KongohPlus_v1.0.0.zip', repos = NULL, type = 'win.binary')
install.packages(c("shiny", "cmaes", "combinat", "data.table", "dplyr", "DT",
    "ggplot2", "gridExtra", "gtools", "htmltools", "knitr", "magrittr",
    "parallel", "Rcpp", "rhandsontable", "rmarkdown", "shinyFeedback",
    "shinyFiles", "shinyjs", "shinythemes", "stringr", "waiter"))
```

2.1. Settings Required to Enable Report Output

Run this application from RStudio or install Pandoc manually.

How to install Pandoc (for Windows):

1. Visit https://pandoc.org/installing.html
2. Click 'Download the latest installer'
3. Download the Windows installer (*.msi)
4. Double-click the downloaded file and follow the installation wizard.

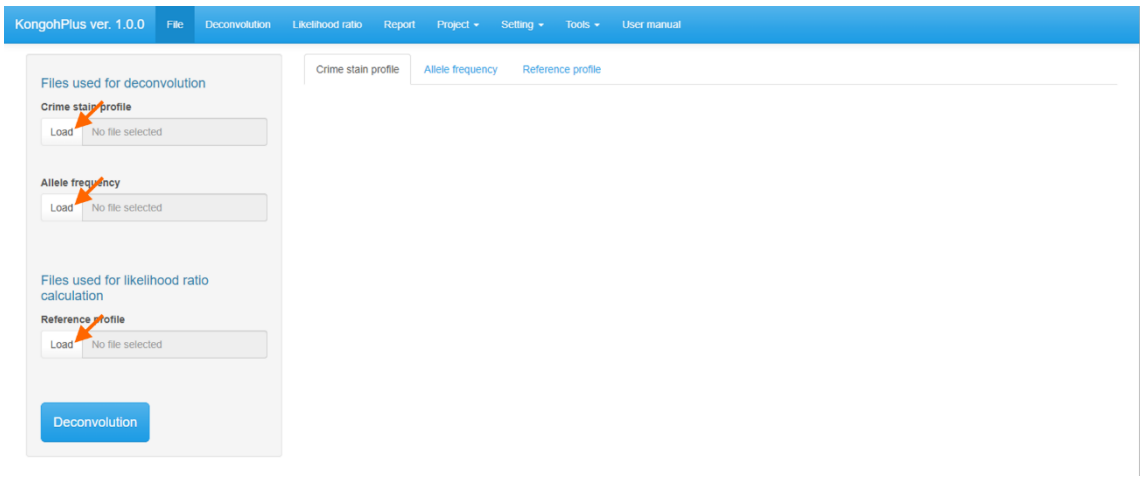
3. Quick guide

1. Enter the following command in the R console to launch the software.

```
library(KongohPlus)

KongohPlus()
```

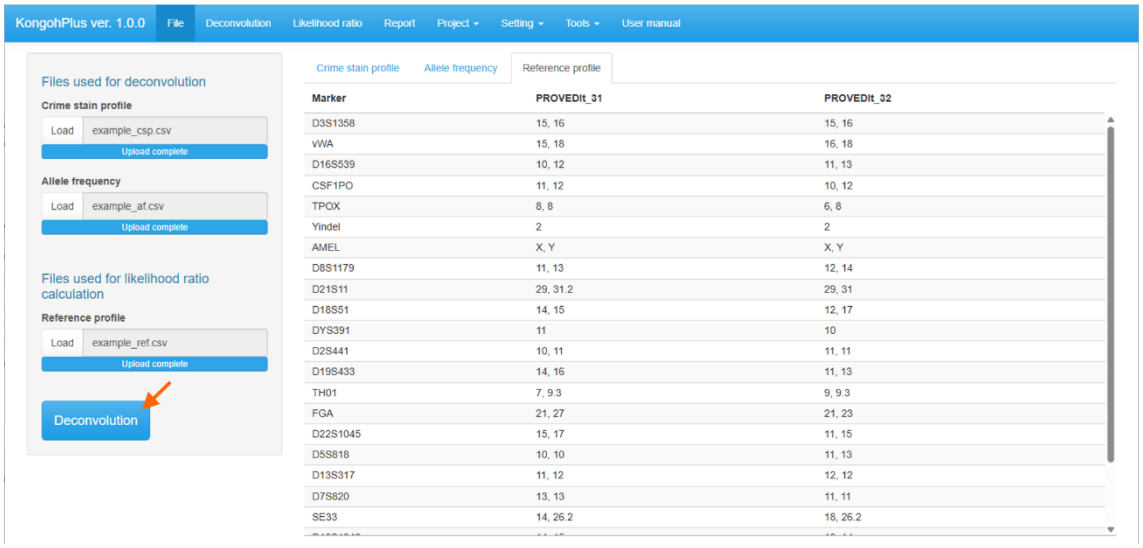
2. Enter the files required for analysis using the **“Load”** button.



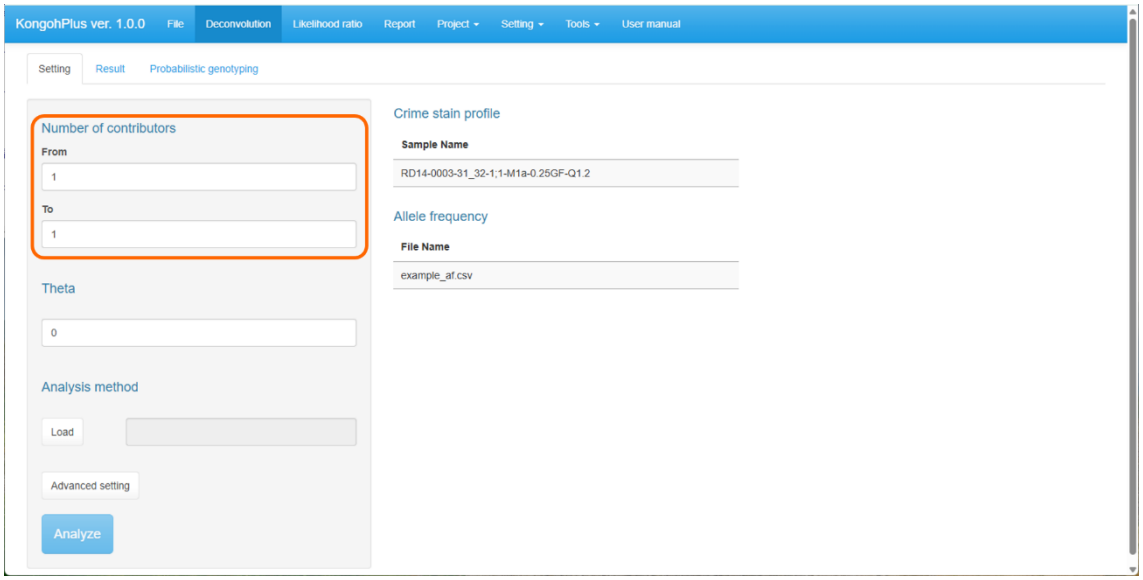
Note:

- Deconvolution requires the crime stain profile and allele frequency files.
- Likelihood ratio calculation requires the crime stain profile, allele frequency, and reference profile files.
- For the required file formats, see Section 4.

3. Click **“Deconvolution”**.



4. Assign the number of contributors.

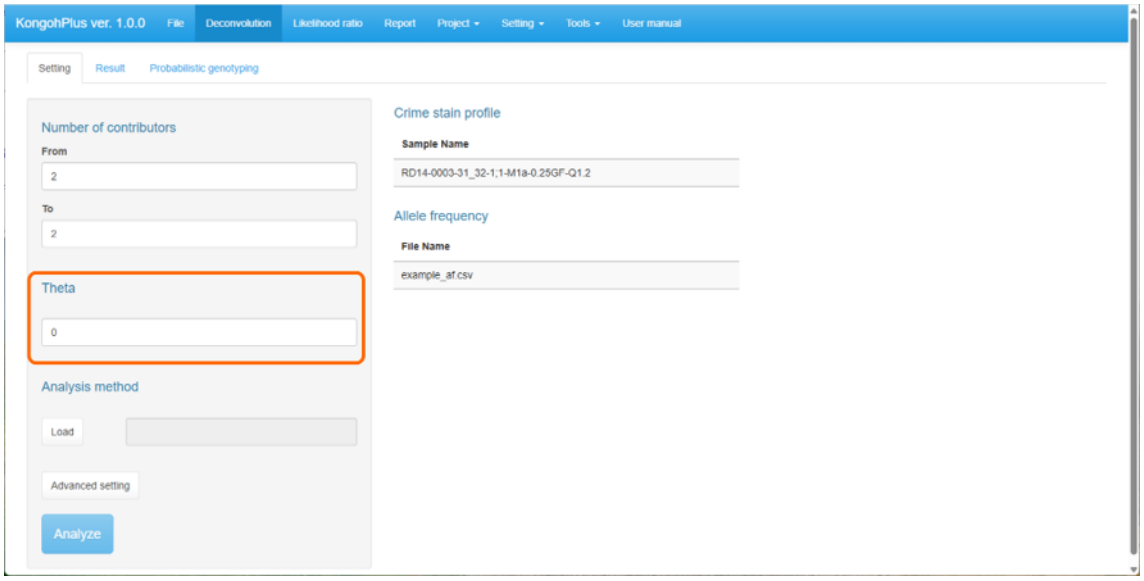


Note:

- You can specify the minimum (From) and maximum (To) numbers of contributors. For example, if From is set to 2 and To is set to 3, deconvolution will be performed twice: once assuming two contributors and once assuming three contributors.

- However, unlike the variable number of contributors method proposed by Taylor et al. [2], this function does not weight or combine the deconvolution results obtained under different contributor-number assumptions into a single likelihood ratio.

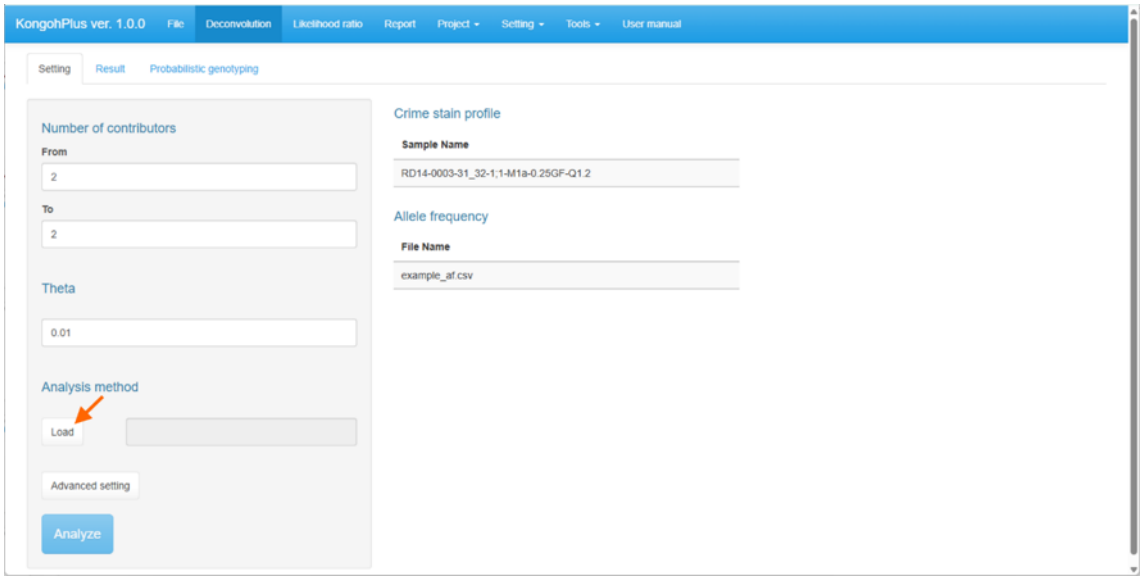
5. Set the theta value.



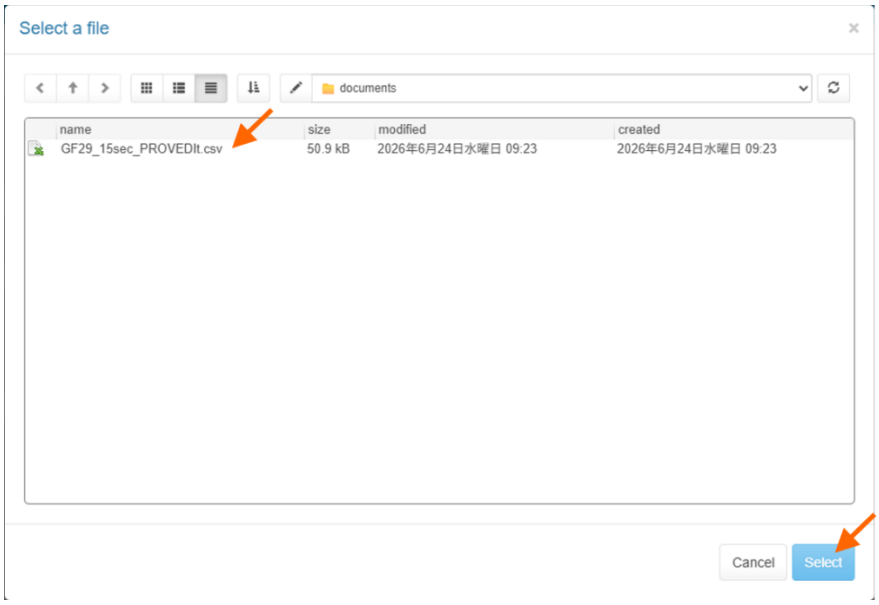
Note:

- Theta is a parameter used to calculate genotype frequencies while accounting for sub-population effects.

6. Select the analysis method using the **“Load”** button.



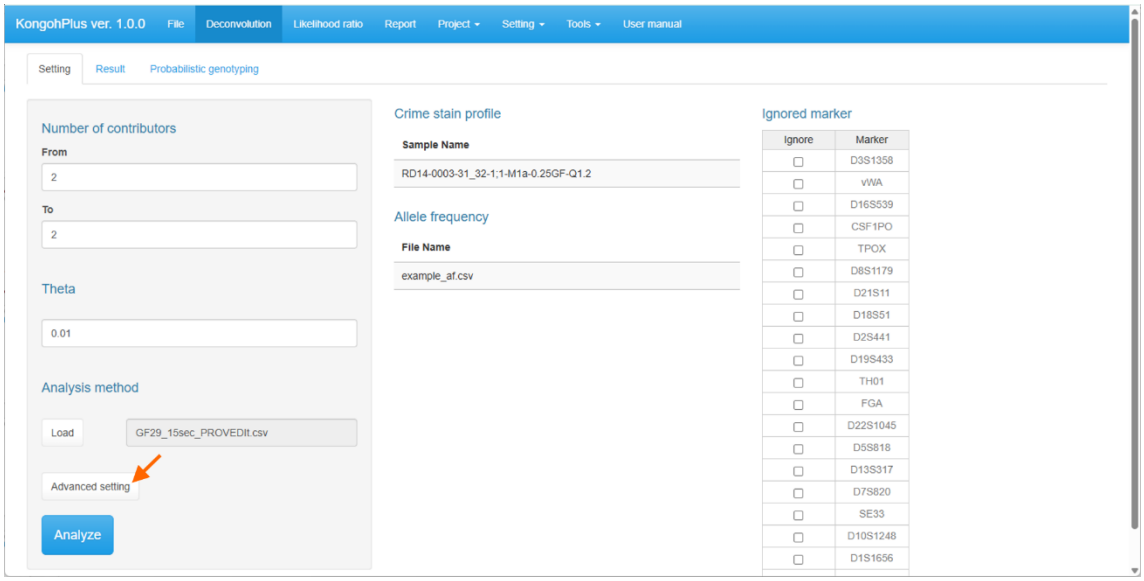
7. Select an analysis method, and then click **”Select”**.



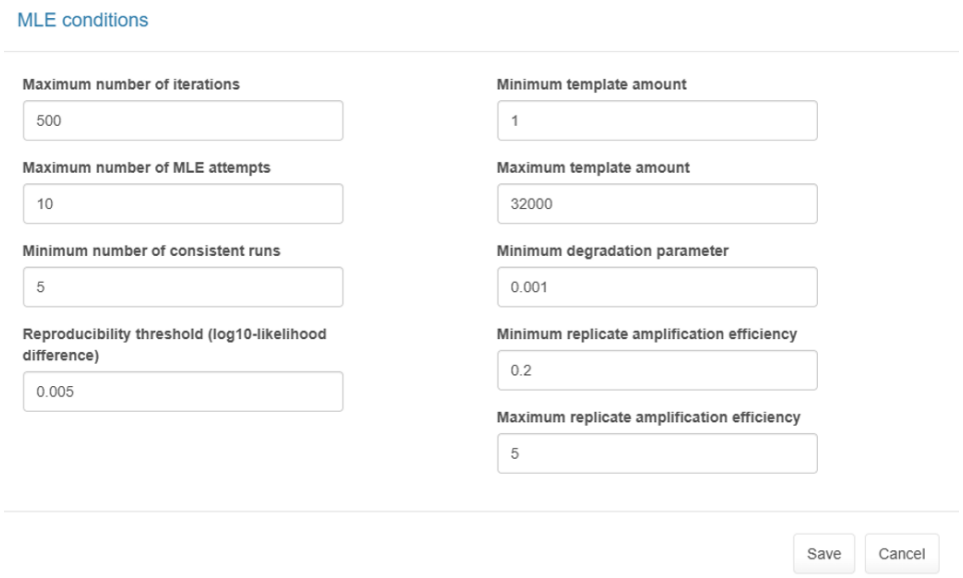
Note:

- The analysis method must be created in advance. For details, see Section 8.7.

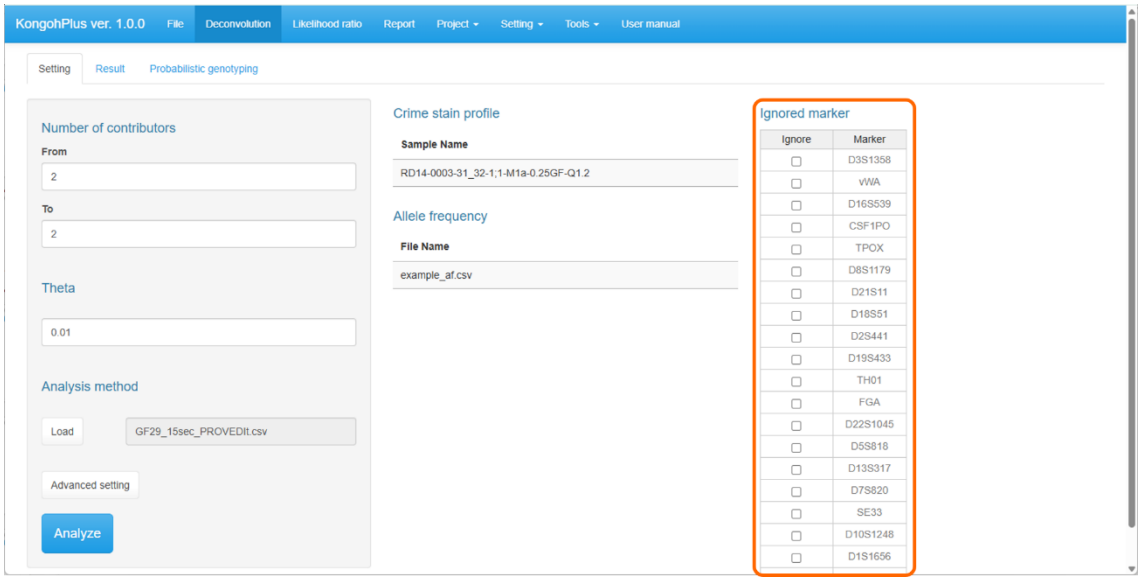
8. If necessary, click **”Advanced setting”** to configure the advanced settings.



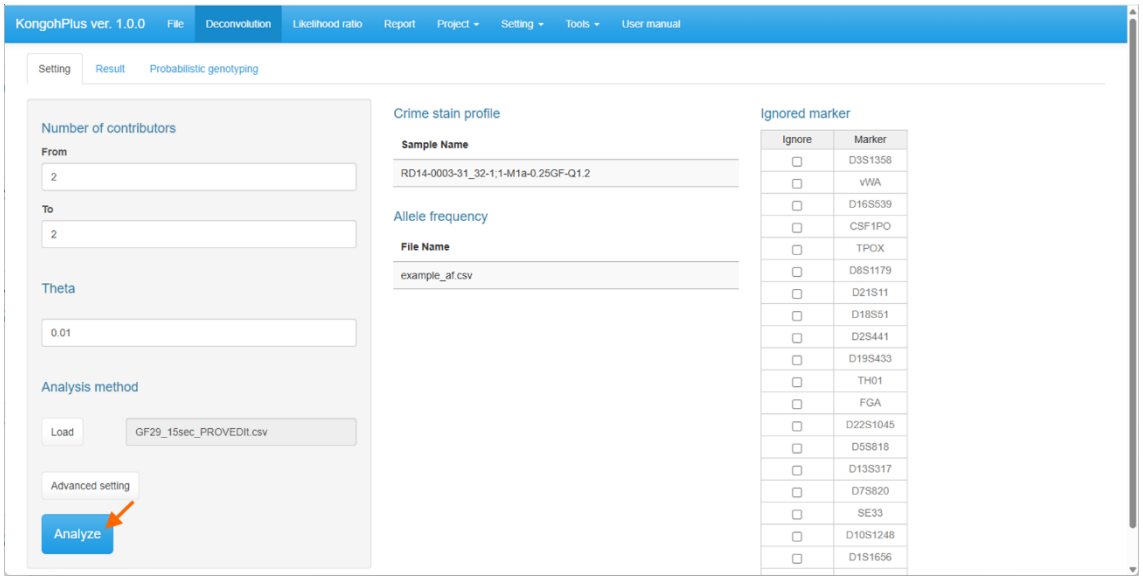
9. Here, configure the settings for the maximum likelihood estimation performed during the deconvolution process (see Section 10.4 for details). Confirm the values for each setting, and then click **“Save”**.



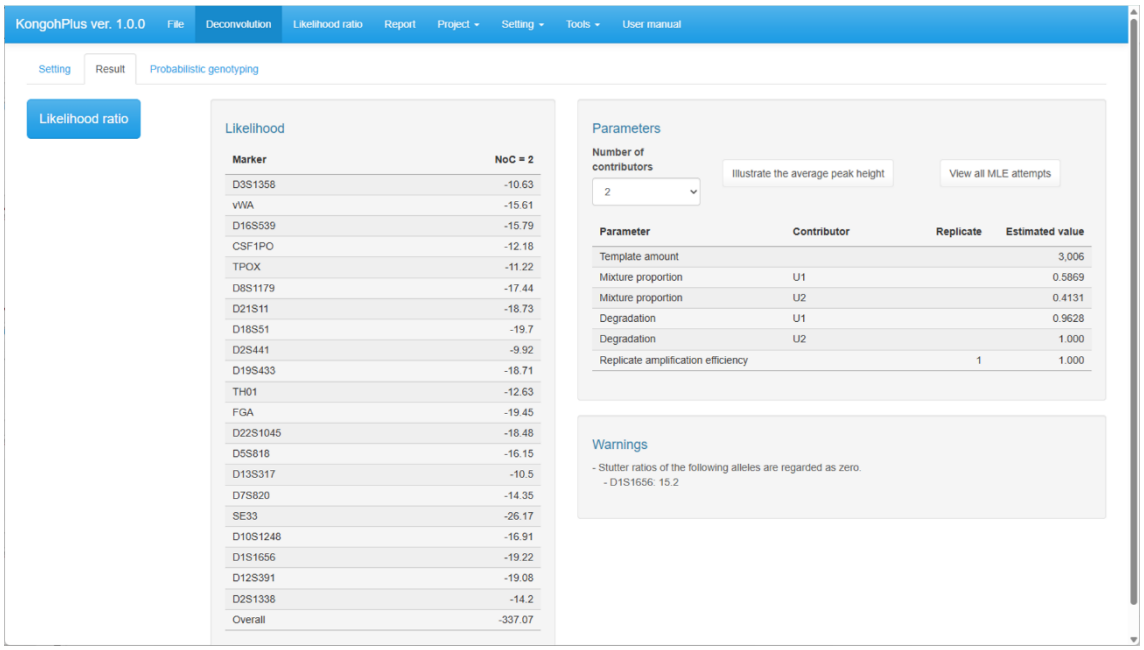
10. Check the markers to be excluded from the analysis, if any.



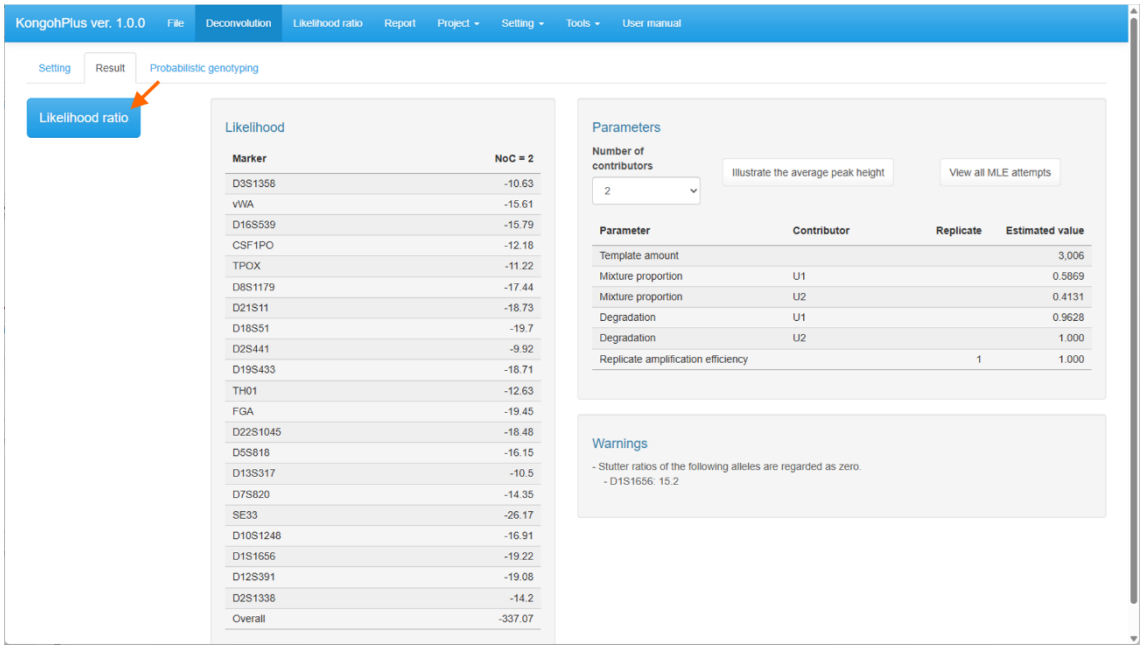
11. Click **“Analyze”** to start the deconvolution.



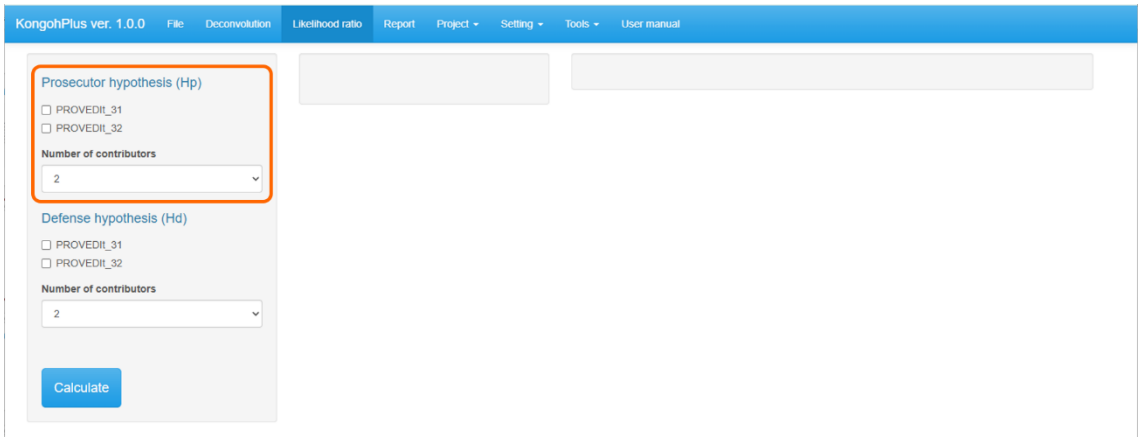
12. After the deconvolution is complete, the following analysis results are displayed. For details, see Section 5.



13. Next, proceed to the likelihood ratio calculation. Click **“Likelihood ratio”**.



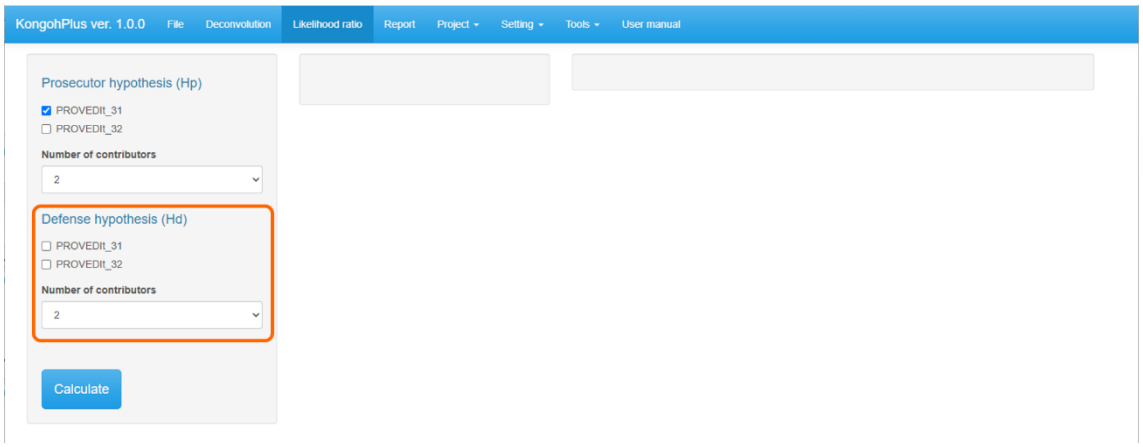
14. Specify the prosecutor hypothesis (the numerator hypothesis of the likelihood ratio).



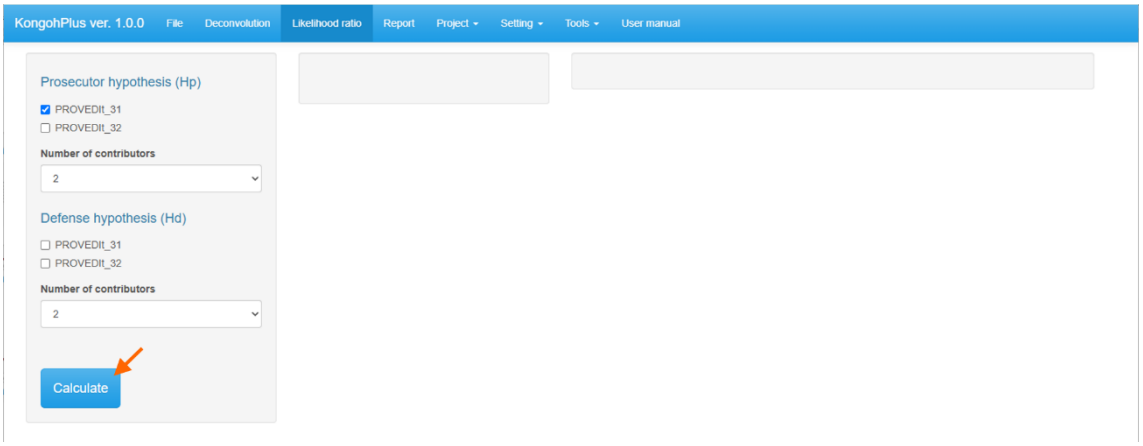
Note:

- Check the individuals assumed to be contributors.
- Select the assumed number of contributors.

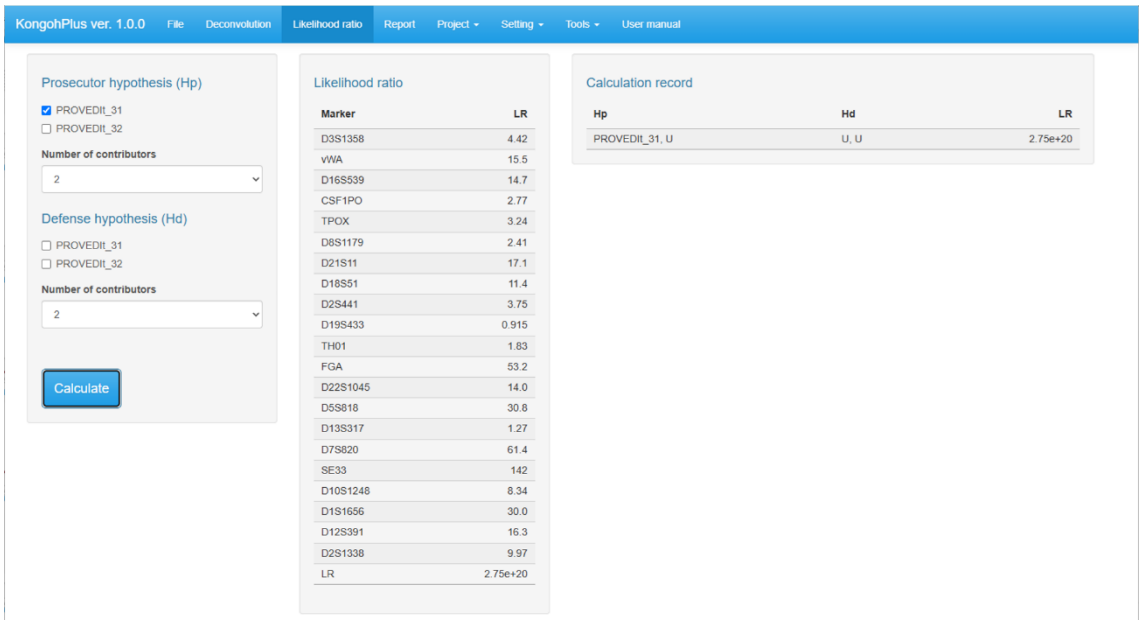
15. Specify the defense hypothesis (the denominator hypothesis of the likelihood ratio).



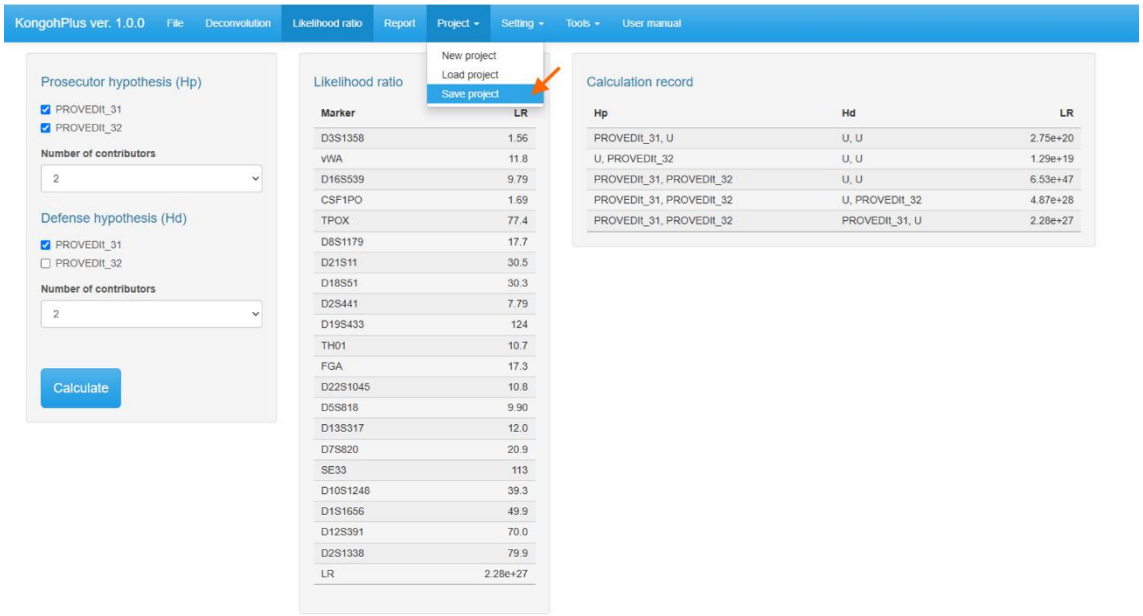
16. Click **“Calculate”**.



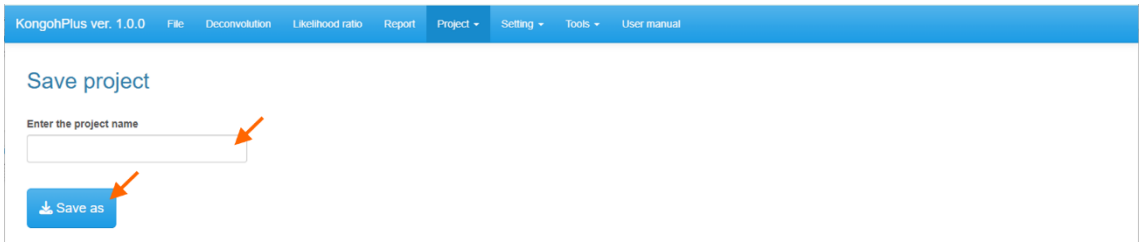
17. After the calculation is complete, the likelihood ratio results are displayed. For details, see Section 7.



18. To save the analysis results, select **"Project" ▶ "Save project"**.



19. Enter a project name, and then click **"Save as"**.



Note:

- You can load a saved project by selecting **"Project" ▶ "Load project"**.

4. Files

This section describes the input file formats required for deconvolution and likelihood ratio analysis.

4.1. Crime stain profile

Prepare the crime stain profile as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2	Allele3	Allele4	…	Height1	Height2	Height3	Height4	…
example	D3S1358	14	15	16		…	305	3865	2317		…
example	vWA	14	15	16	17	…	101	1552	1086	178	…
example	D16S539	9	10	11	12	…	66	1454	782	1372	…
…	…	…	…	…	…	…	…	…	…	…	…

Note:

- An example input file can be downloaded from **“Tools” ▶ “Example files”**.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele and Height values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers. Alternatively, if they are considered pull-up artifacts or noise, remove them from the input file.
- If multiple PCR amplifications have been performed, assign a different SampleName to each profile and combine all profiles into a single CSV file by stacking them vertically.

4.2. Allele frequency

Prepare the allele frequency data as a CSV file in the format shown in the table below.

Allele	D3S1358	vWA	D16S539	CSF1PO	TPOX	…
10			602	648	100	…
11			562	620	1072	…
12	6		517	1267	114	…
13	3	1	209	208	3	…
14	79	586	26	52	2	…
15	1192	78	3	14		…
…	…	…	…	…	…	…

Note:

- An example input file can be downloaded from **“Tools” ▶ “Example files”**.
- For each marker column, enter the observed count for each allele in the population database. Leave the cell blank if the observed count is zero.

4.3. Reference profile

Prepare the reference profile as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2
victim	D3S1358	15	16
victim	vWA	15	18
victim	D16S539	10	10
…	…	…	…
suspect	D3S1358	15	16
suspect	vWA	16	18
suspect	D16S539	11	13
…	…	…	…

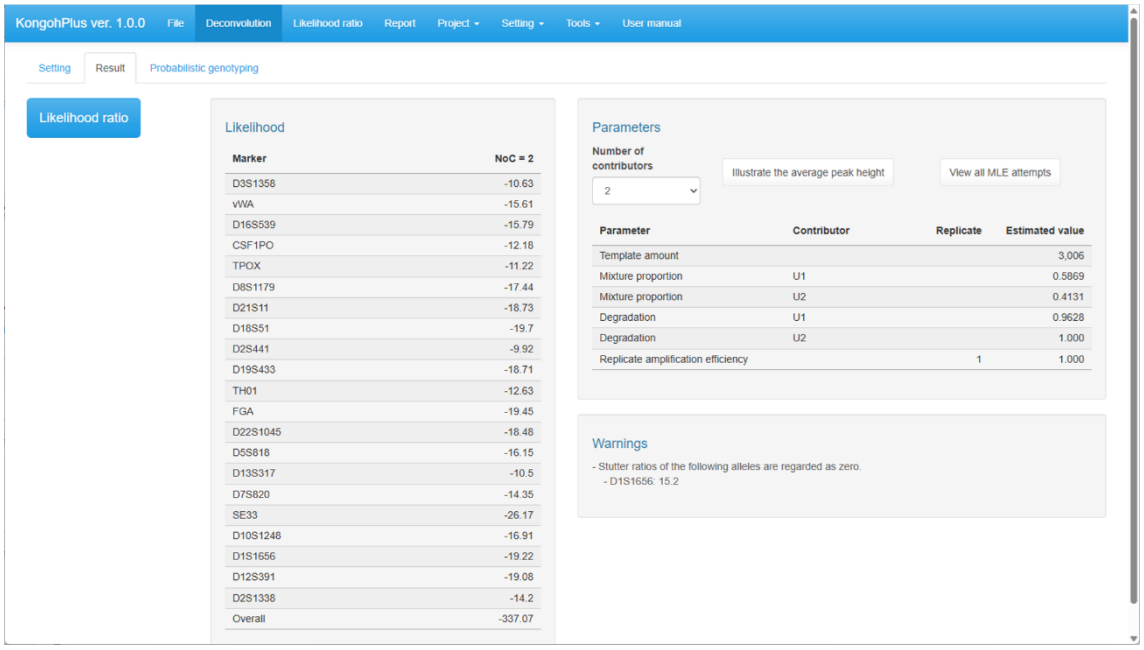
Note:

- An example input file can be downloaded from **“Tools” ▶ “Example files”**.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers.
- For homozygous genotypes, enter the same allele number twice.

5. Deconvolution

This section describes the deconvolution results screen.

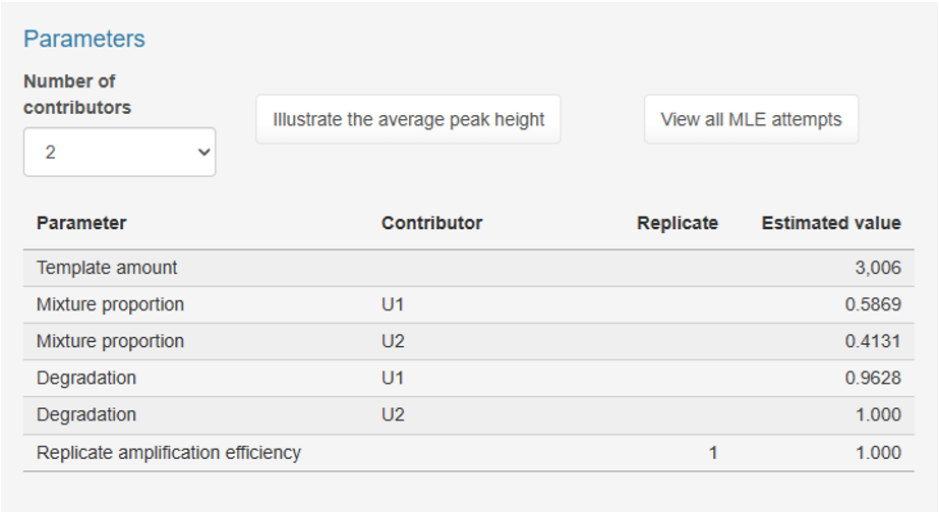
After the deconvolution is complete, the results screen shown below is displayed. The likelihood for each locus and the overall likelihood are shown in the center of the screen. The estimated parameter values are displayed in the upper-right corner, and any warning messages are displayed in the lower-right corner.



Note:

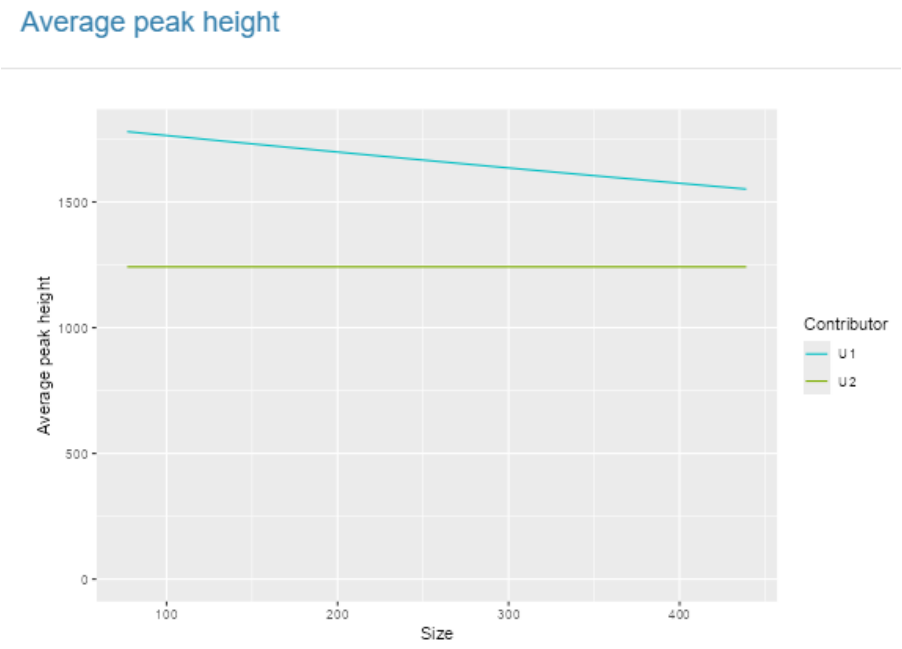
- If an allele for which no stutter ratio has been registered is detected, a warning is displayed indicating that the stutter ratio for that allele was assumed to be zero during the analysis.
- A warning is displayed if the required number of consistent runs is not obtained (see Section 10.4 for details).

The enlarged view below shows the estimated parameter values displayed in the upper-right corner of the screen. For a description of the parameters estimated by deconvolution, see the case-specific parameters (parameters estimated for each profile) described in Section 10.3.1.



Parameter	Contributor	Replicate	Estimated value
Template amount			3,006
Mixture proportion	U1		0.5869
Mixture proportion	U2		0.4131
Degradation	U1		0.9628
Degradation	U2		1.000
Replicate amplification efficiency		1	1.000

Click **"Illustrate the average peak height"** to display the relationship between fragment size and the average peak height per heterozygous allele for each contributor. This provides a visual representation of the estimated parameter values.



Click **“View all MLE attempts”** to assess the stability of the deconvolution results. KongohPlus performs multiple deconvolution runs (up to 10 by default) and terminates the analysis once the minimum number of consistent runs (5 by default) has been obtained (see Section 10.4 for details).

The figure below shows an example in which five consecutive runs were obtained. Columns corresponding to consistent runs are highlighted in blue. The column with the largest log10(likelihood) value is highlighted in light blue, and this result is selected as the final deconvolution result.

All MLE attempts

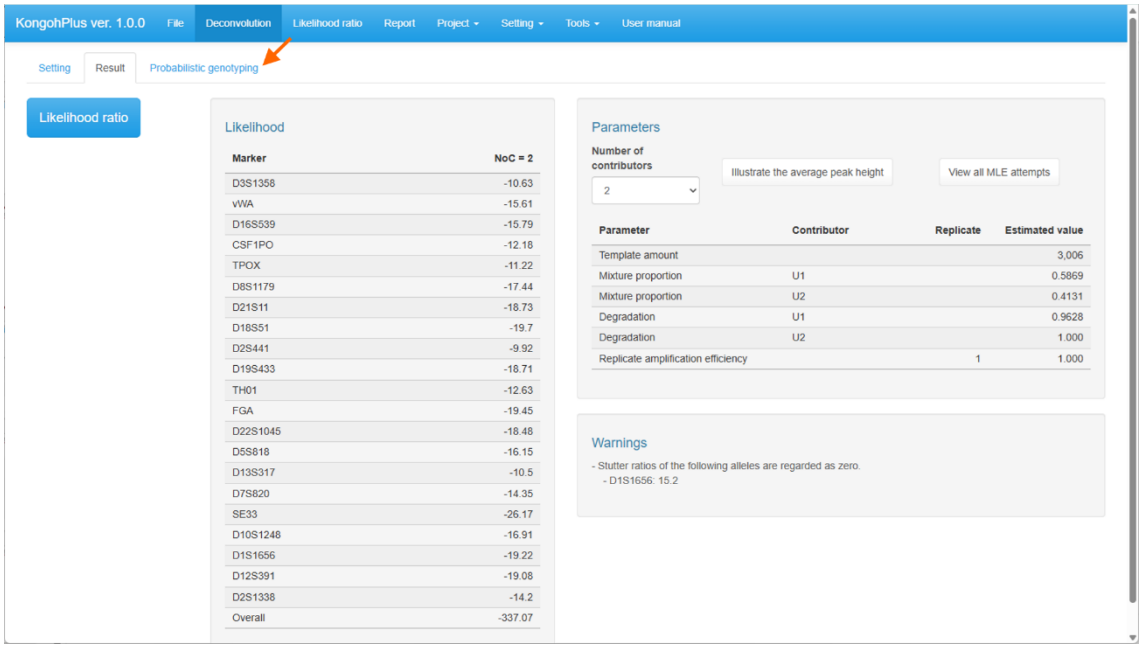
: Selected maximum likelihood : Close to the maximum likelihood

Parameter	MLE 1	MLE 2	MLE 3	MLE 4	MLE 5
Log10 (likelihood)	-337.070	-337.070	-337.070	-337.070	-337.070
Template amount	3,006.00	3,006.06	3,006.16	3,006.25	3,006.18
Mixture proportion	0.586856	0.586896	0.586892	0.586905	0.586886
Mixture proportion	0.413144	0.413104	0.413108	0.413095	0.413114
Degradation	0.962845	0.962766	0.962770	0.962738	0.962788
Degradation	0.999999	0.999999	1.00000	0.999999	1.00000
Replicate amplification efficiency	1.00000	1.00000	1.00000	1.00000	1.00000

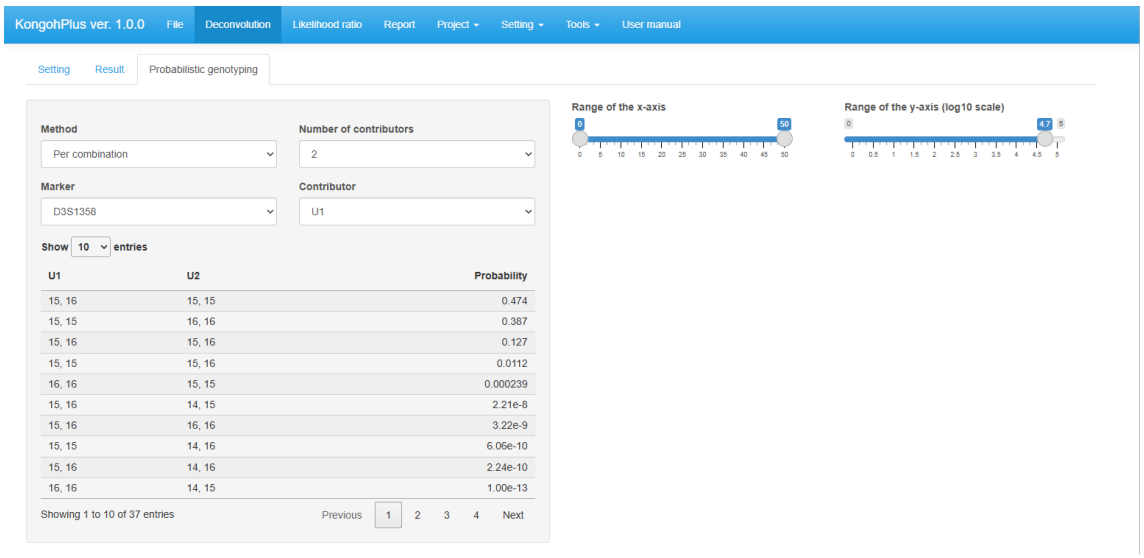
6. Probabilistic Genotyping

This section describes the probabilistic genotyping results screen.

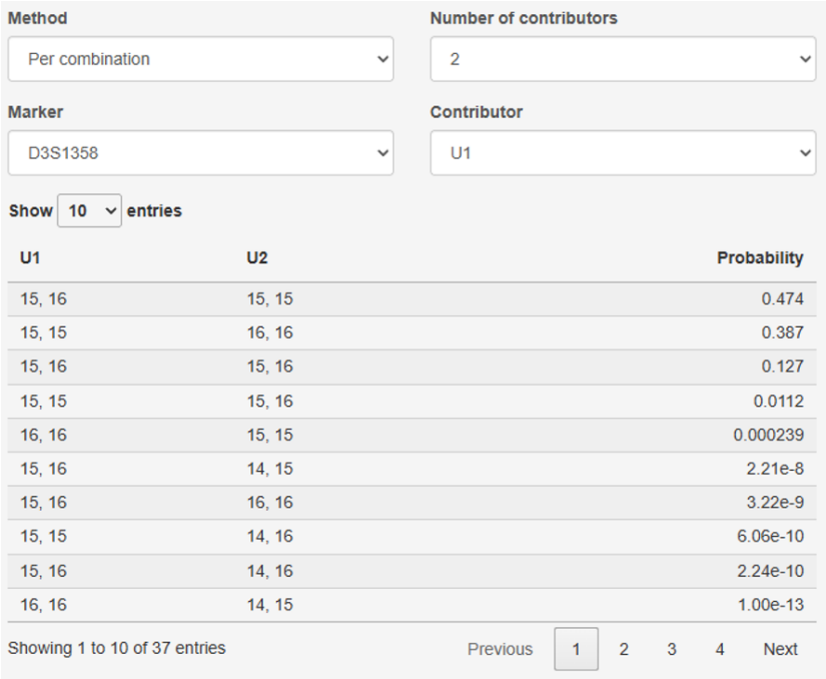
After the deconvolution is complete, you can view the results of probabilistic genotyping, which assigns probabilistic weights to the possible genotypes of each contributor. Click the **“Probabilistic genotyping”** tab, as shown below.



The following screen is then displayed.



The left side of the screen lists the possible genotypes for each contributor together with their corresponding probabilities.



You can change the displayed results using the following options.

Method

- **Per combination:** Displays the probability of each combination of contributor genotypes.
- **Per contributor:** Displays the probability of each genotype for a contributor.

Number of contributors

- Select the assumed number of contributors.

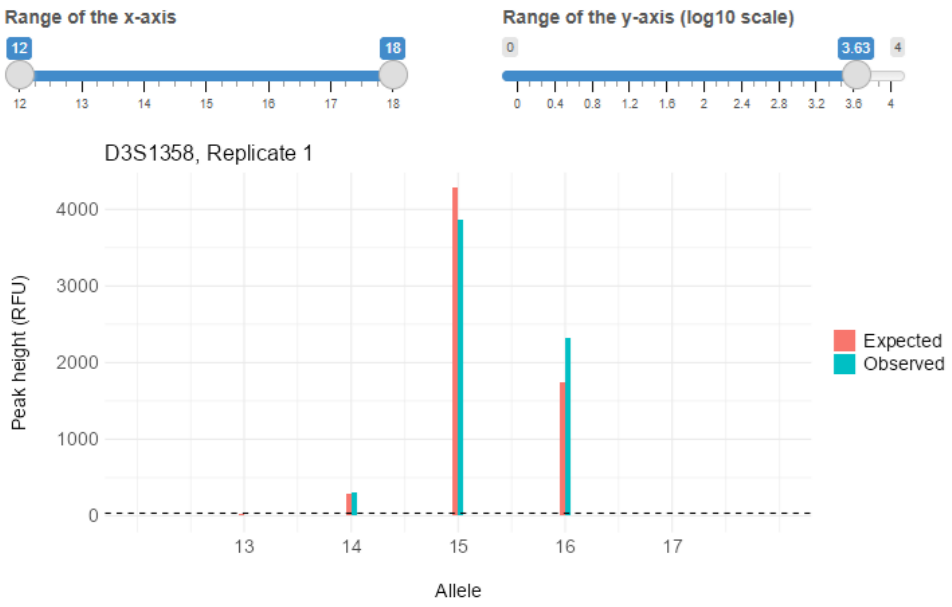
Marker

- Select the marker for which the results are to be displayed.

Contributor

- Select the contributor for whom the results are to be displayed. This option is available only when **Per contributor** is selected as the **Method**.

In addition, selecting a row in the probabilistic genotyping results table displays, on the right side of the screen, a comparison between the expected peak heights under the selected genotype and the observed peak heights.

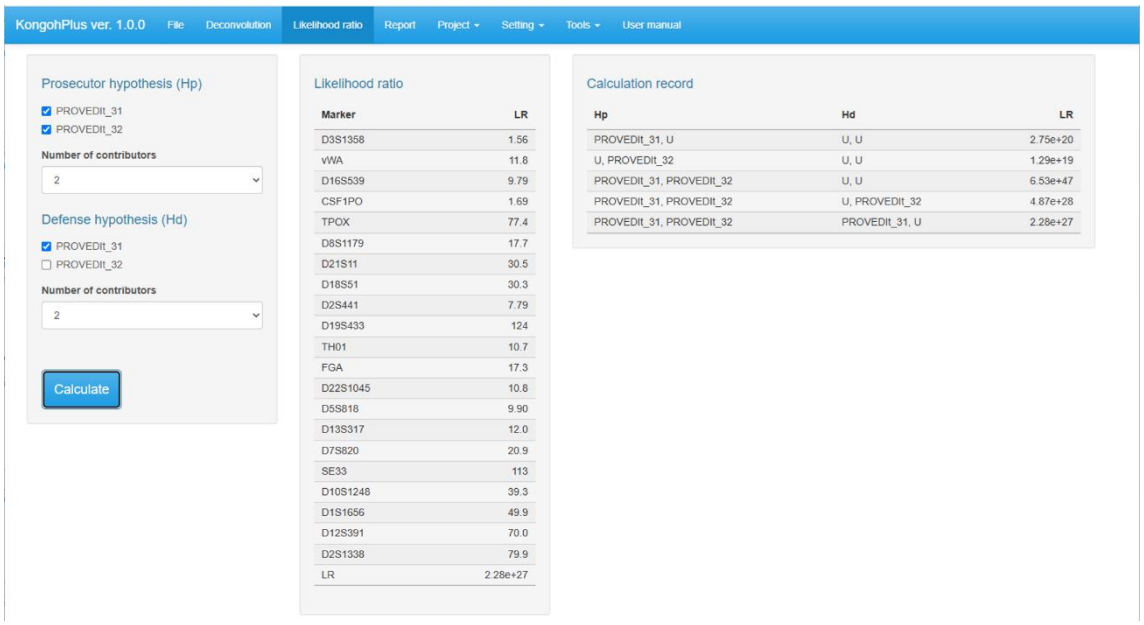


Note:

- When **Per contributor** is selected as the **Method**, the expected peak heights shown are those for the selected contributor only.
- If multiple PCR amplifications have been performed, a comparison between the expected and observed peak heights is displayed for each profile.

7. Likelihood Ratio

This section describes the likelihood ratio results screen.



Note:

- The likelihood ratio can be recalculated as many times as required using different hypotheses.
- All calculation results are recorded in the **“Calculation record”** panel on the right side of the screen.
- The center of the screen displays the likelihood ratio corresponding to the row selected in the **“Calculation record”** panel.
- The individuals listed under **Hp** and **Hd** in the **“Calculation record”** panel are ordered according to their estimated mixture proportions, from largest to smallest. For example, **"PROVEDIt_31, U"** under **Hp** in the figure above indicates that the “PROVEDIt_31” was estimated to be the major contributor, whereas U (an abbreviation for unknown contributor) was estimated to be the minor contributor.

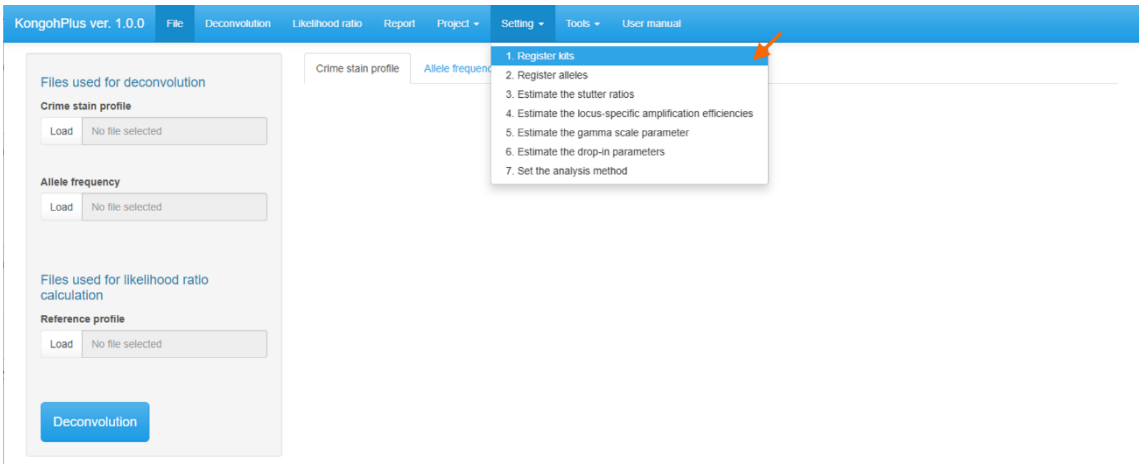
8. Settings

This section describes the settings required for analysis in KongohPlus.

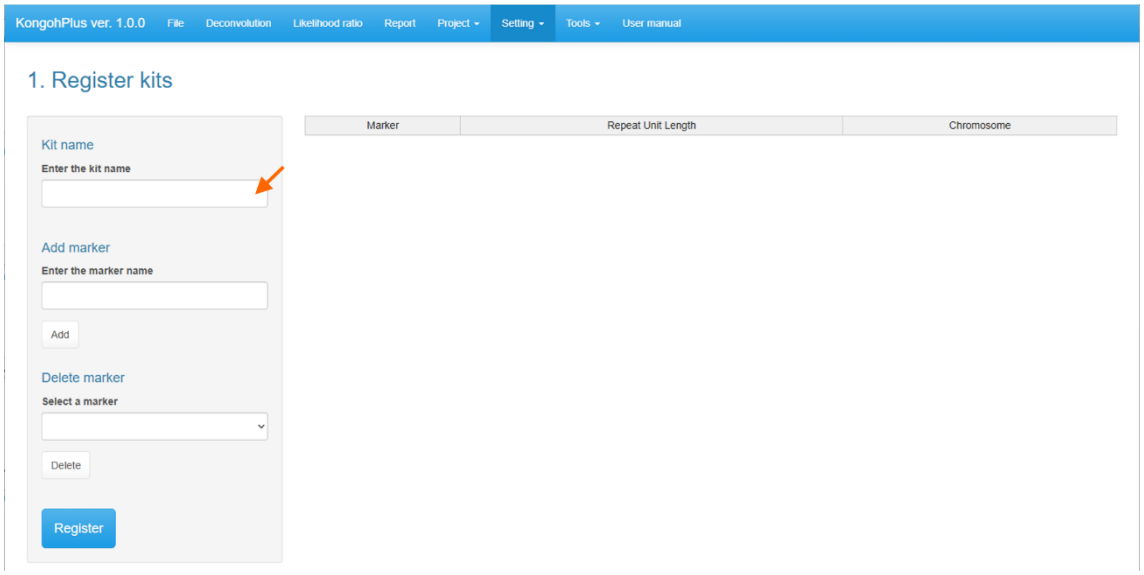
8.1. Registering a DNA Typing Kit

If the DNA typing kit used to generate the profile has not yet been registered, you must first register the marker information included in the kit. The following procedure explains how to do this.

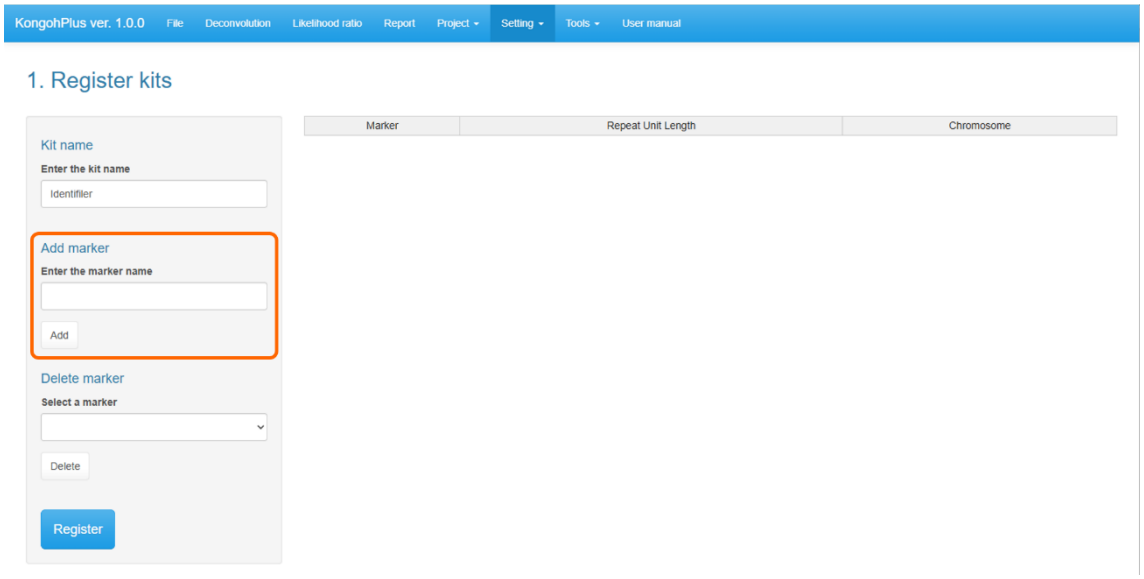
1. Click **“Setting” ▶ “1. Register kits”**.



2. Enter the kit name.



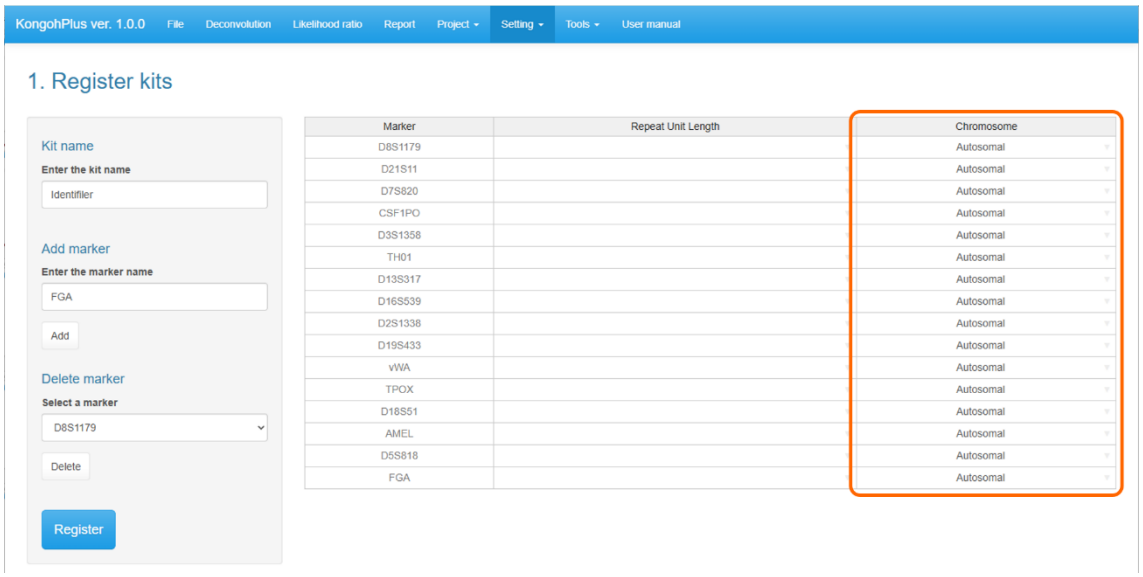
3. Enter a marker name in **”Add marker”**, and then click **”Add”**.



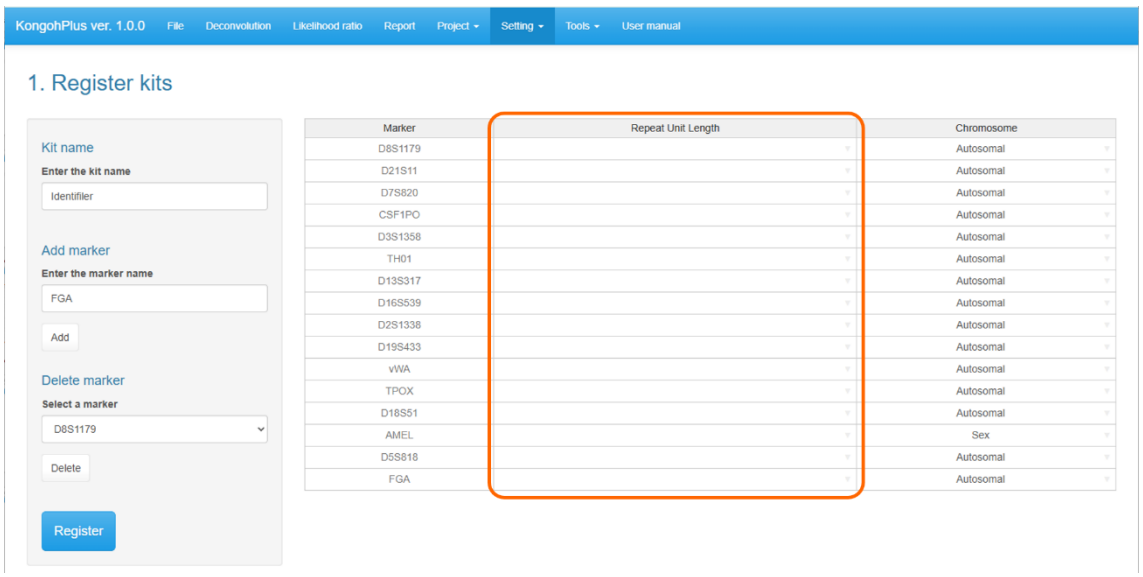
Note:

- Repeat this procedure until all markers have been registered.
- Markers that have been registered incorrectly can be removed using **”Delete marker”**.

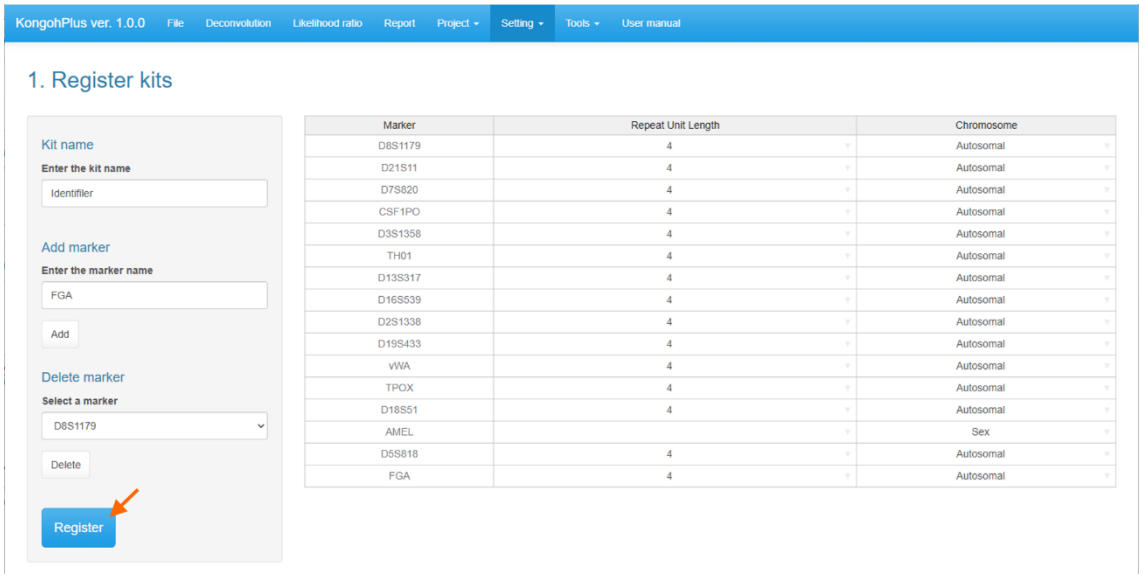
4. Select whether the marker is an autosomal STR (**"Autosomal"**) or a sex chromosome marker (**"Sex"**).



5. For each autosomal STR marker, select the repeat unit length (number of bases per repeat unit).



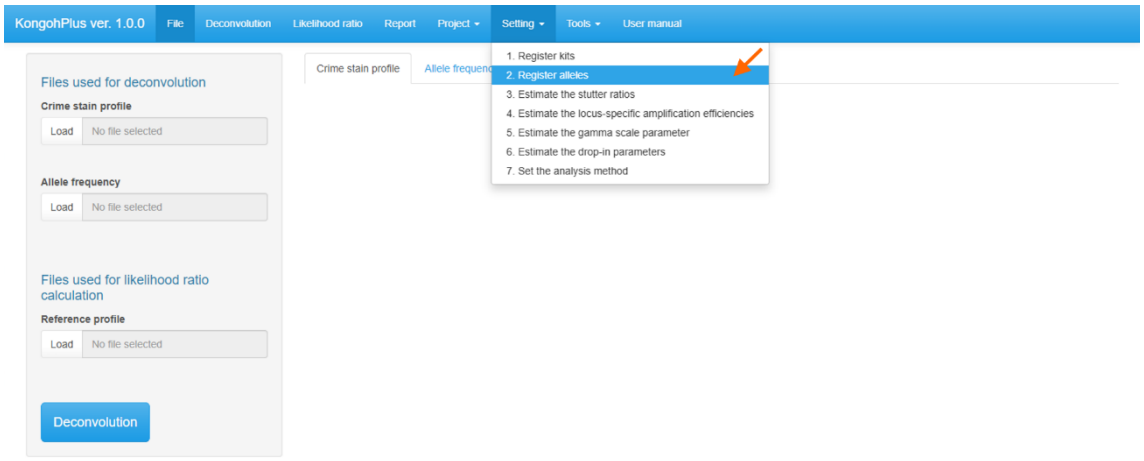
6. Click **“Register”**.



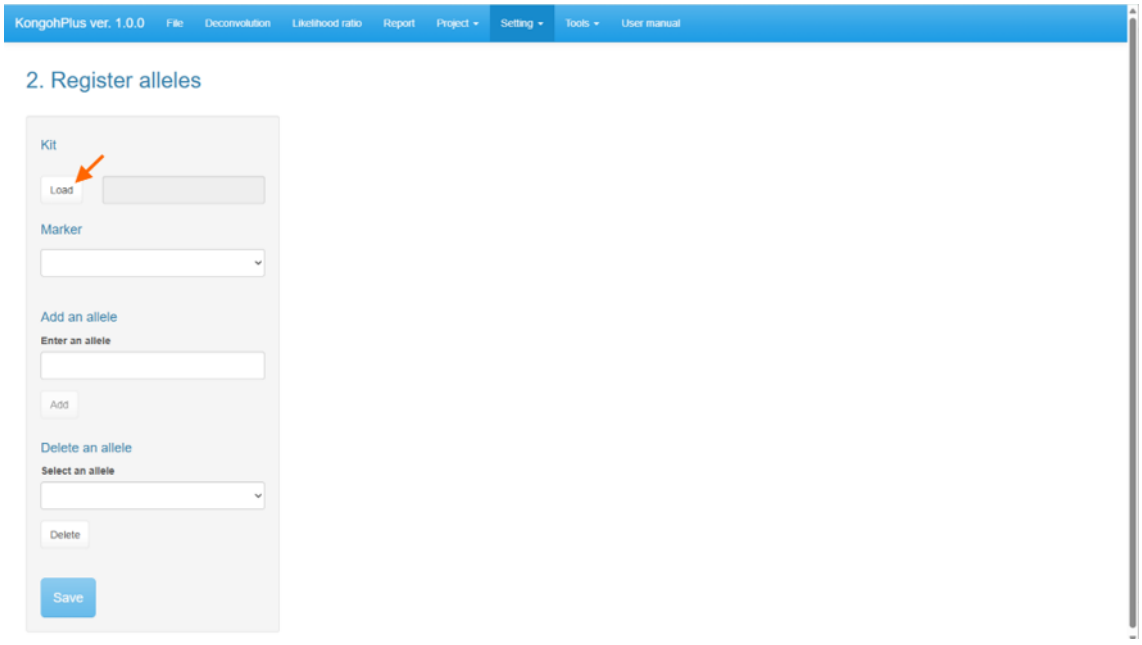
8.2. Registering Alleles

This section describes the procedure for registering the allele information for each marker included in a DNA typing kit. The allele information includes the fragment sizes used to estimate the degree of DNA degradation. In addition, a stutter ratio can be assigned to each registered allele. Therefore, it is recommended that as many alleles as possible that are expected to be encountered in routine casework be registered in advance.

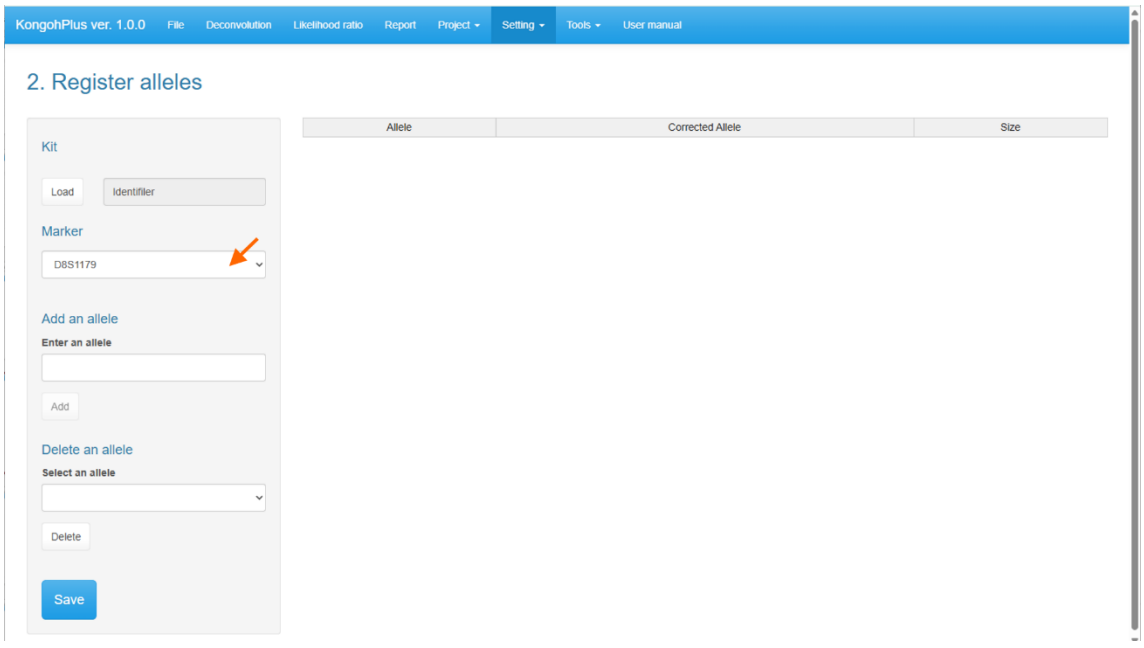
1. Click **“Setting” ▶ “2. Register alleles”**.



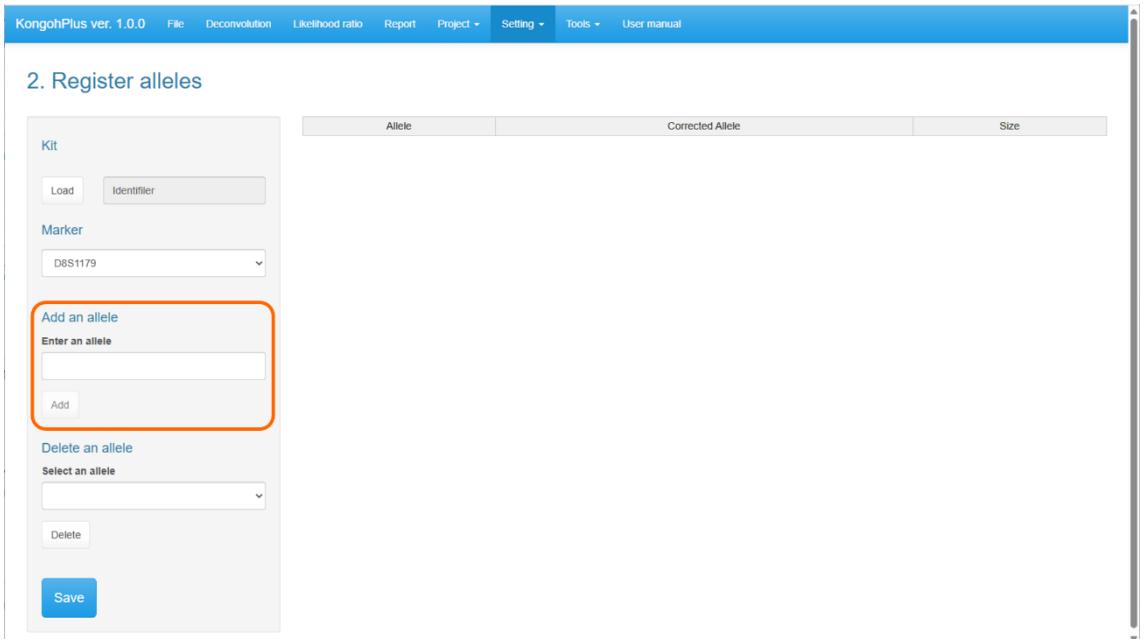
2. Load the registered kit information using the **“Load”** button.



3. Select a marker.



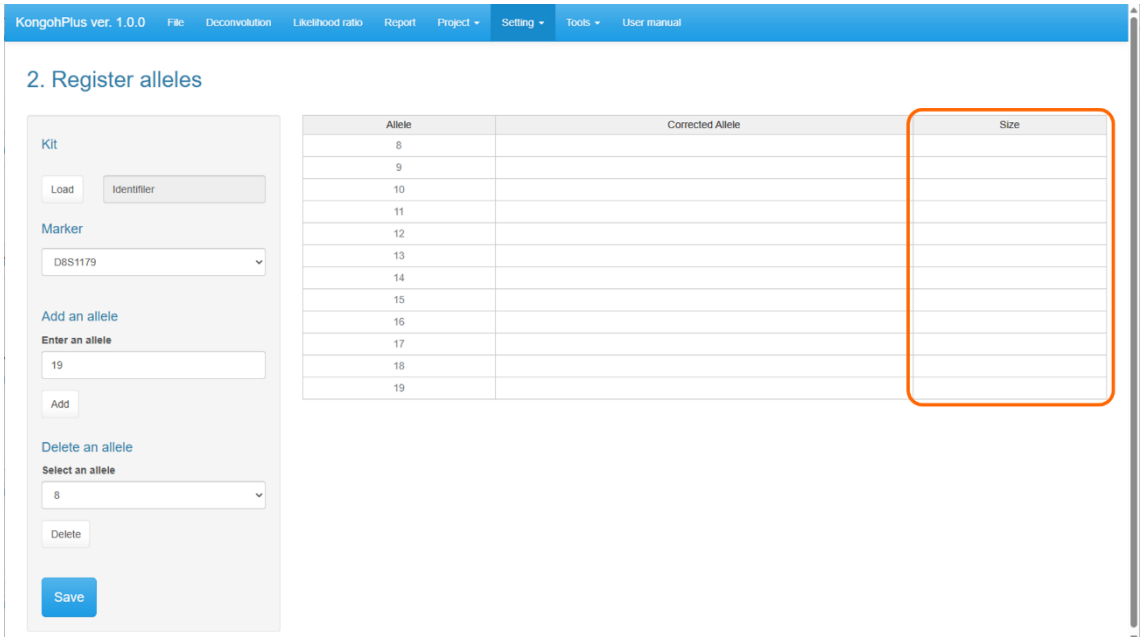
4. Enter an allele number in **“Add an allele”**, and then click **”Add”**.



Note:

- Repeat Steps 3 and 4 until all alleles that may be observed have been registered for every marker.
- Alleles that have been registered incorrectly can be removed using **”Delete an allele”**.

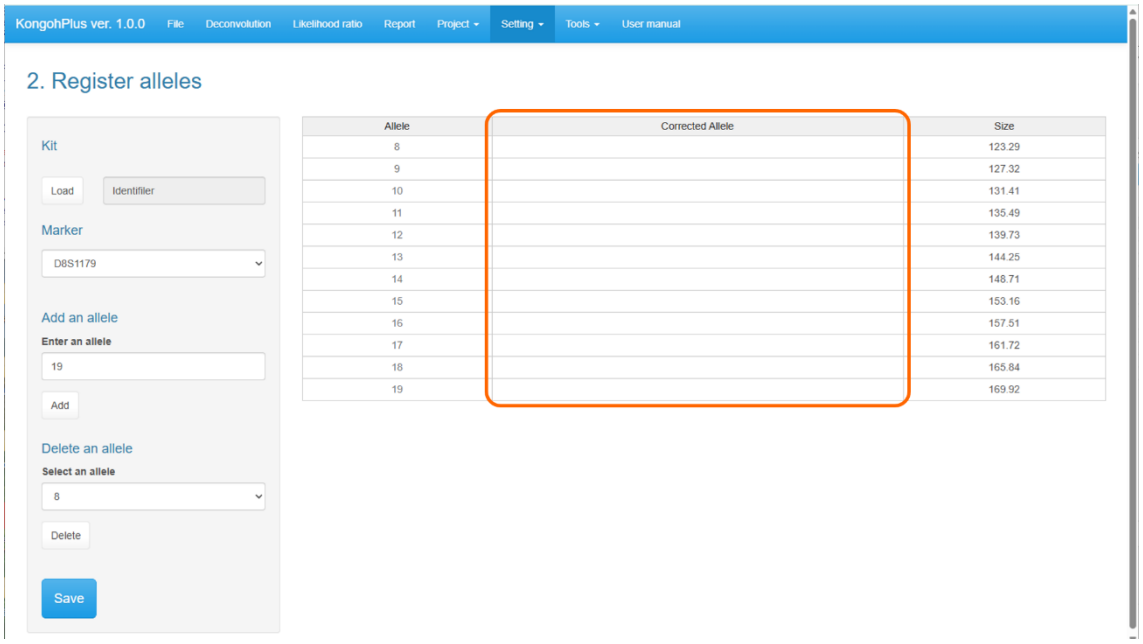
5. Enter the fragment size.



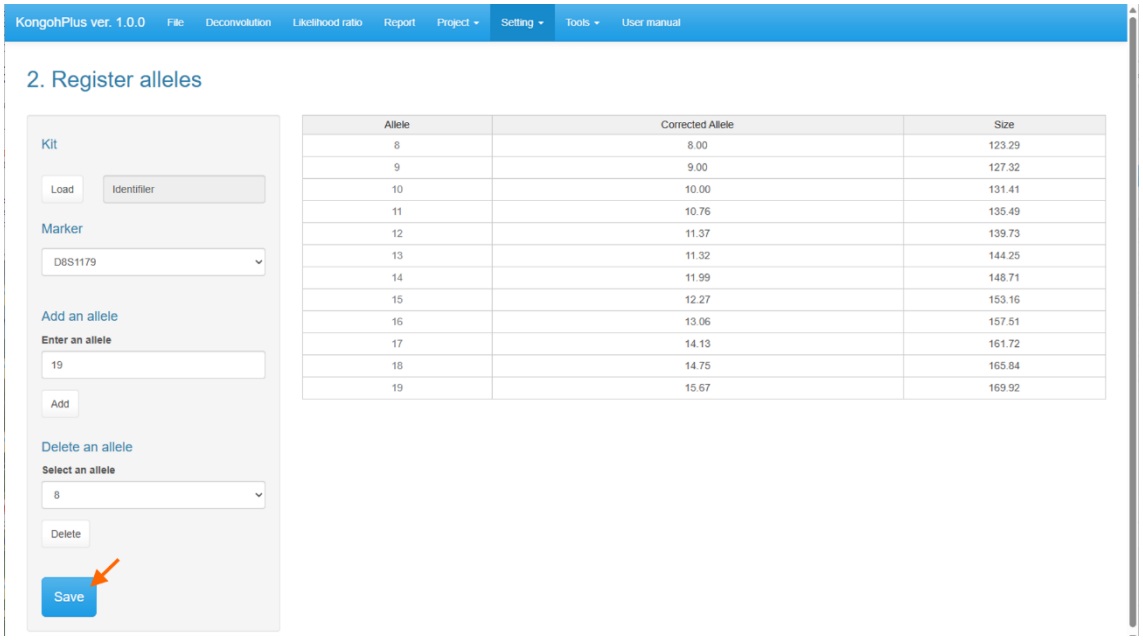
Note:

- Set the fragment size to the average value based on a representative allelic ladder or similar reference data.

6. For markers that use the Linear (Corrected Allele) model as the statistical model for the stutter ratio, enter the Corrected Allele value. For details of the statistical models used for stutter ratios, see Section 10.7.1.



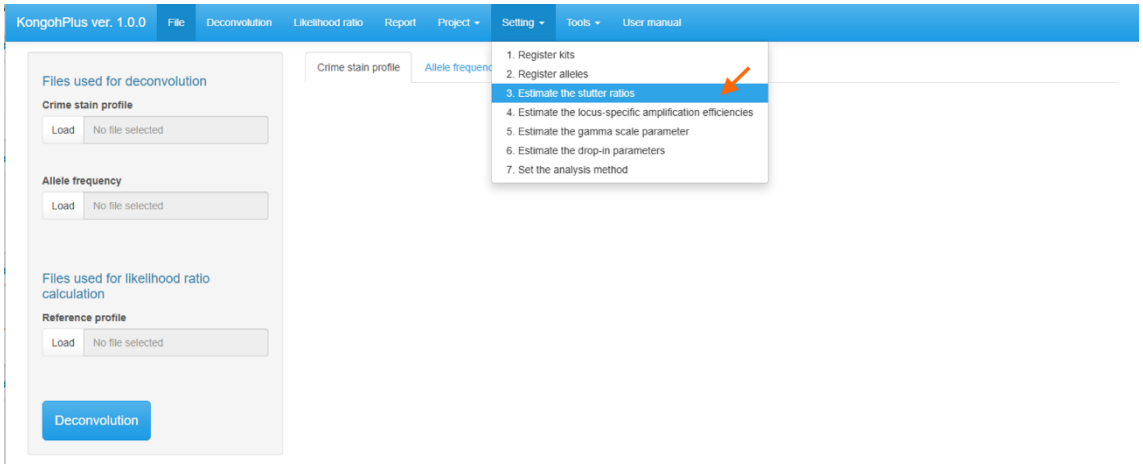
7. After all markers have been configured, click **"Save"**.



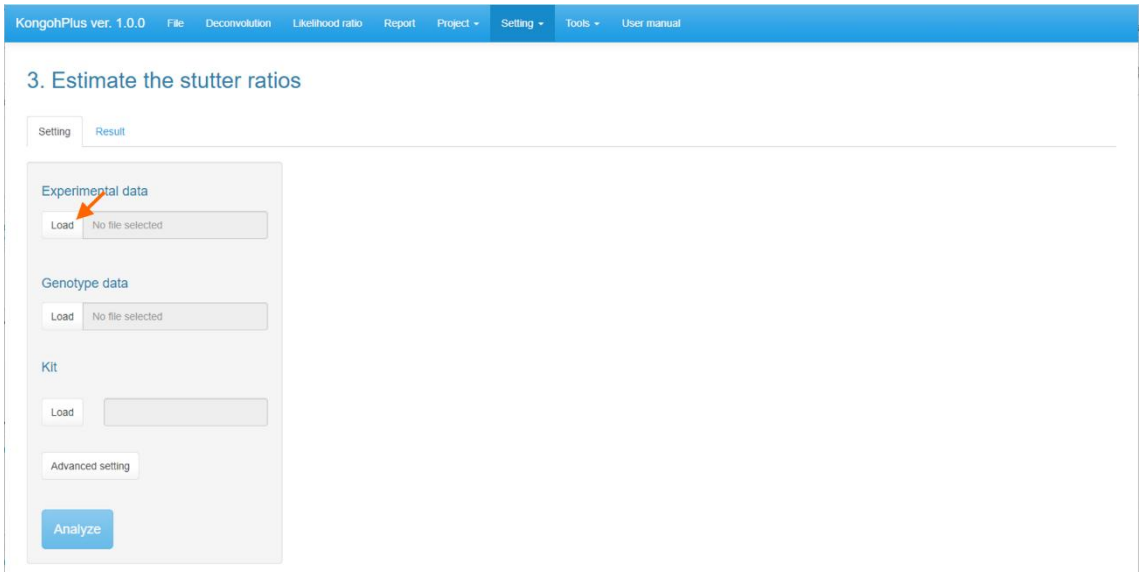
8.3. Estimating Stutter Ratios

This section describes the procedure for estimating allele-specific stutter ratios from experimentally obtained DNA profiles.

1. Click **“Setting” ▶ “3. Estimate the stutter ratios”**.



2. Load the experimental dataset using the **“Load”** button, as shown below.



Requirements for the Experimental Dataset

The experimental dataset should satisfy the following requirements:

- Each profile must be a single-source profile.
- The DNA must not be degraded.
- The dataset should contain profiles from a variety of individuals. Because stutter ratios are estimated separately for each allele, it is desirable that the dataset includes as many different alleles as possible.

File Format of the Experimental Dataset

Prepare the experimental dataset as a CSV file in the format shown in the table below.

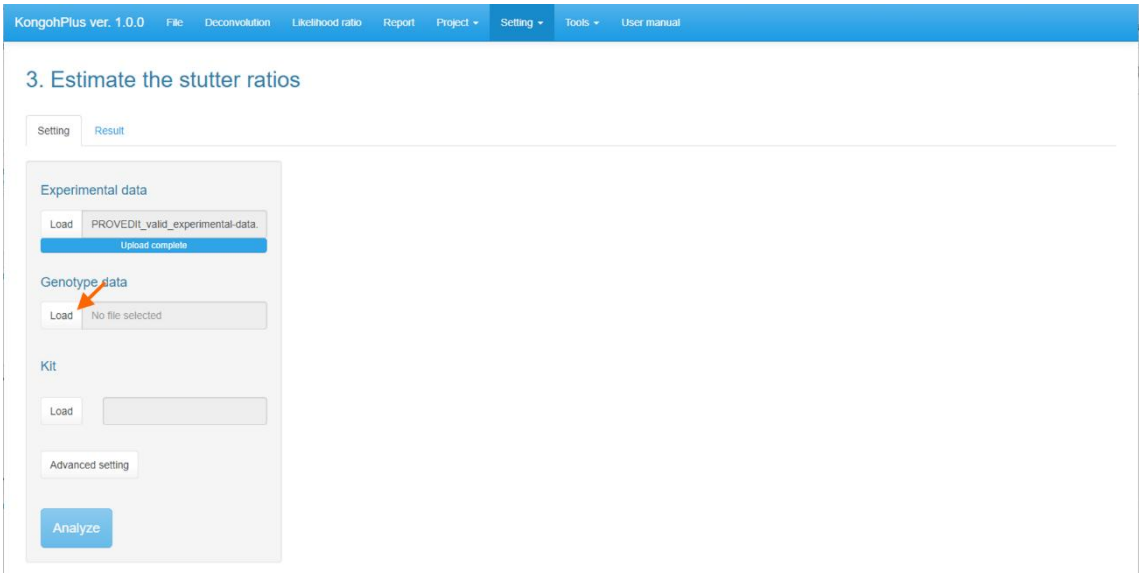
SampleName	Marker	Allele1	Allele2	Allele3	Allele4	…	Height1	Height2	Height3	Height4	…
example1	D3S1358	13	14	17	18	…	103	2130	152	1701	…
example1	vWA	16	17	18	19	…	93	1394	104	1328	…
example1	D16S539	11	12	13		…	84	1610	1367		…
…	…	…	…	…	…	…	…	…	…	…	…

Note:

- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele and Height values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers. Alternatively, if they are considered pull-up artifacts or noise, remove them from the input file.

- Assign a unique SampleName to each profile and combine all profiles into a single CSV file by stacking them vertically.

3. Load the genotype dataset for the experimental profiles using the **“Load”** button, as shown below.



File Format of the Genotype Dataset

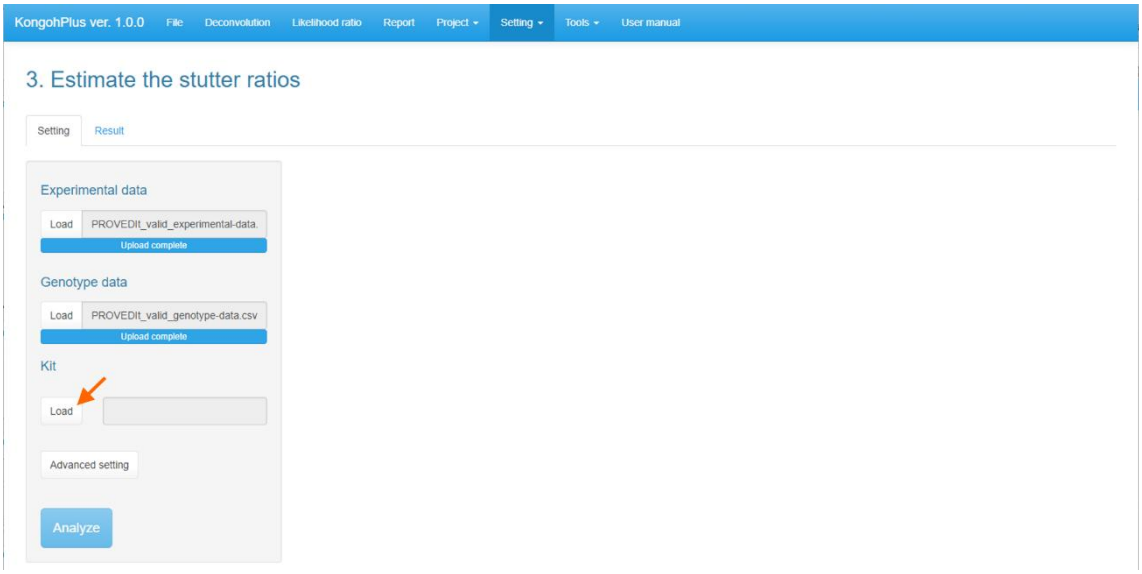
Prepare the genotype dataset as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2
example1	D3S1358	15	16
example1	vWA	15	18
example1	D16S539	10	10
…	…	…	…

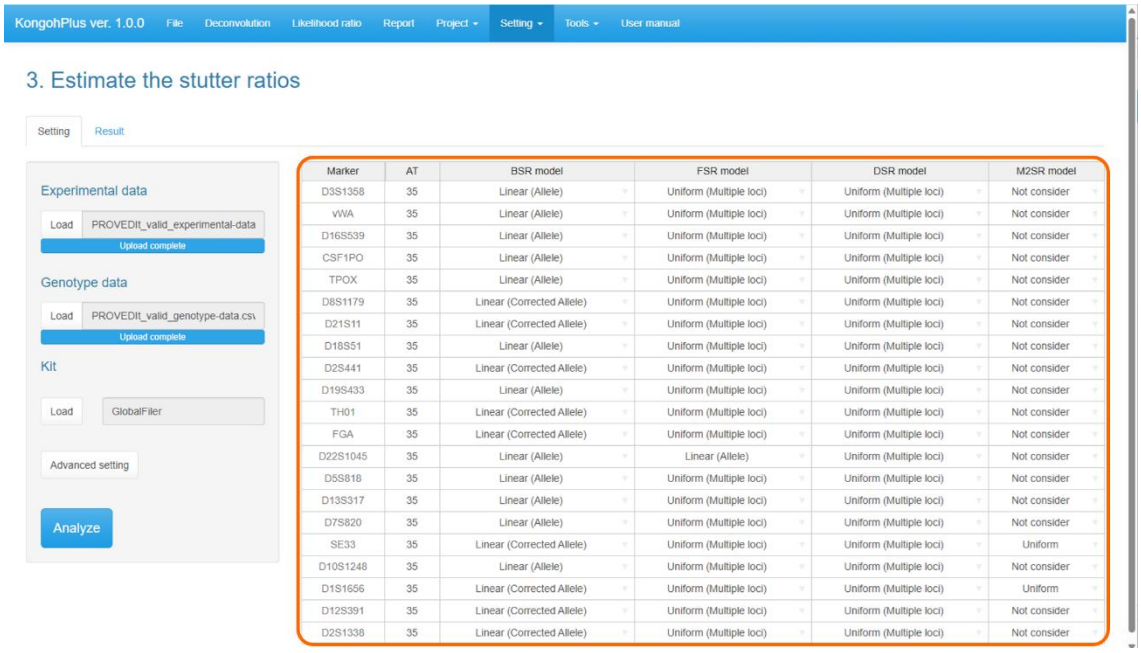
Note:

- The SampleName values must correspond to those in the experimental dataset.
- Assign a unique SampleName to each profile and combine all genotype data into a single CSV file by stacking them vertically.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers.
- For homozygous genotypes, enter the same allele number twice.

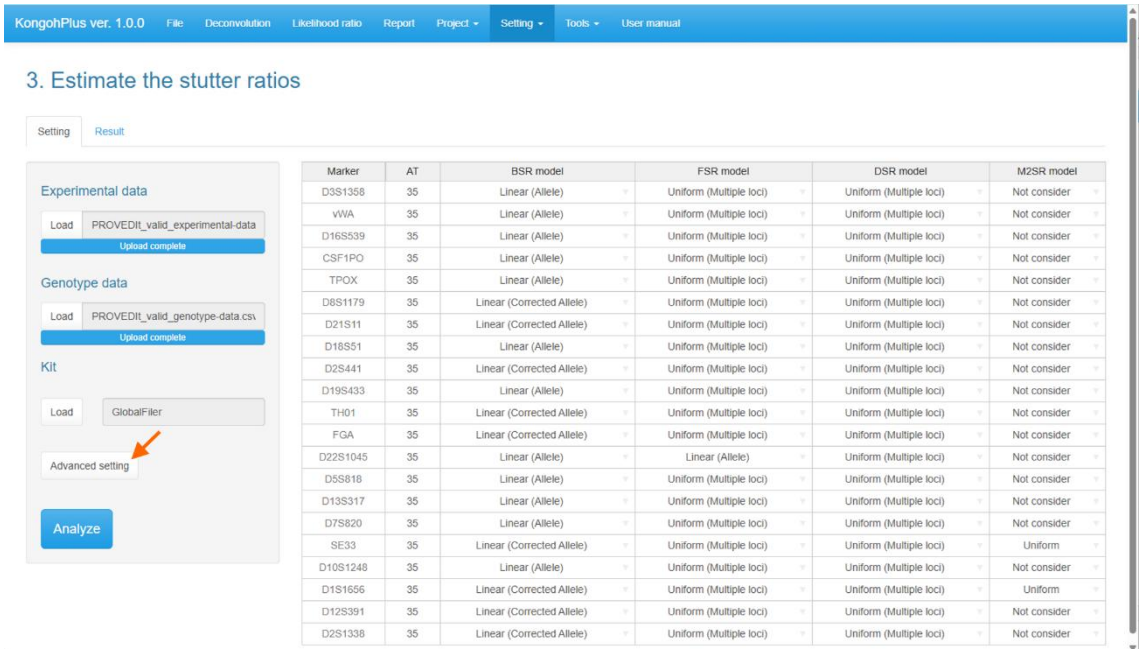
4. Load the registered kit information by clicking the **"Load"** button, as shown below.



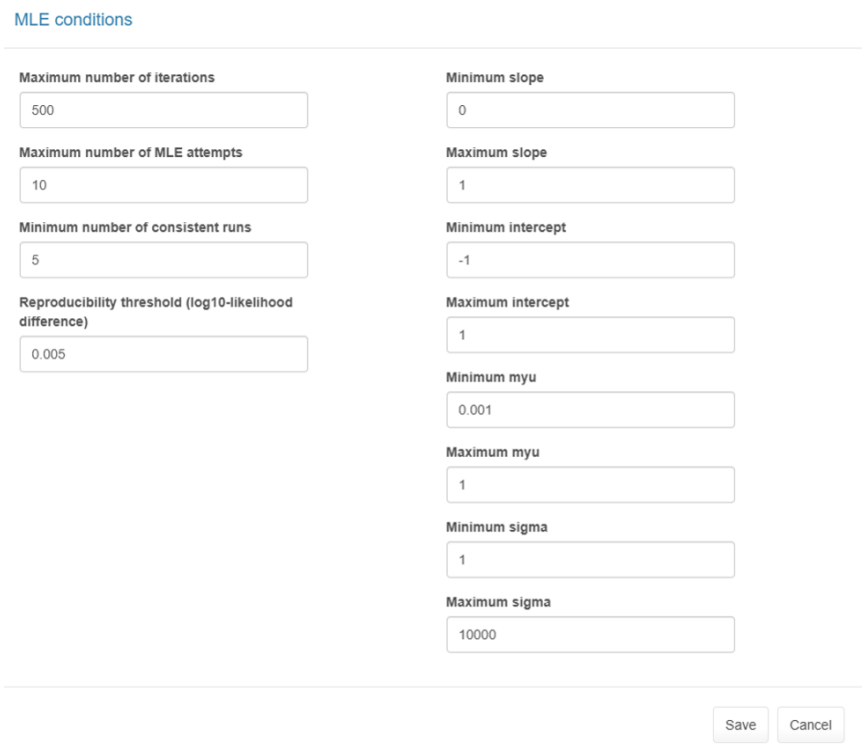
5. Set the analytical threshold (AT) for each marker and select the statistical model for each type of stutter ratio. For details of the statistical models used for stutter ratios, see Section 10.7.1.



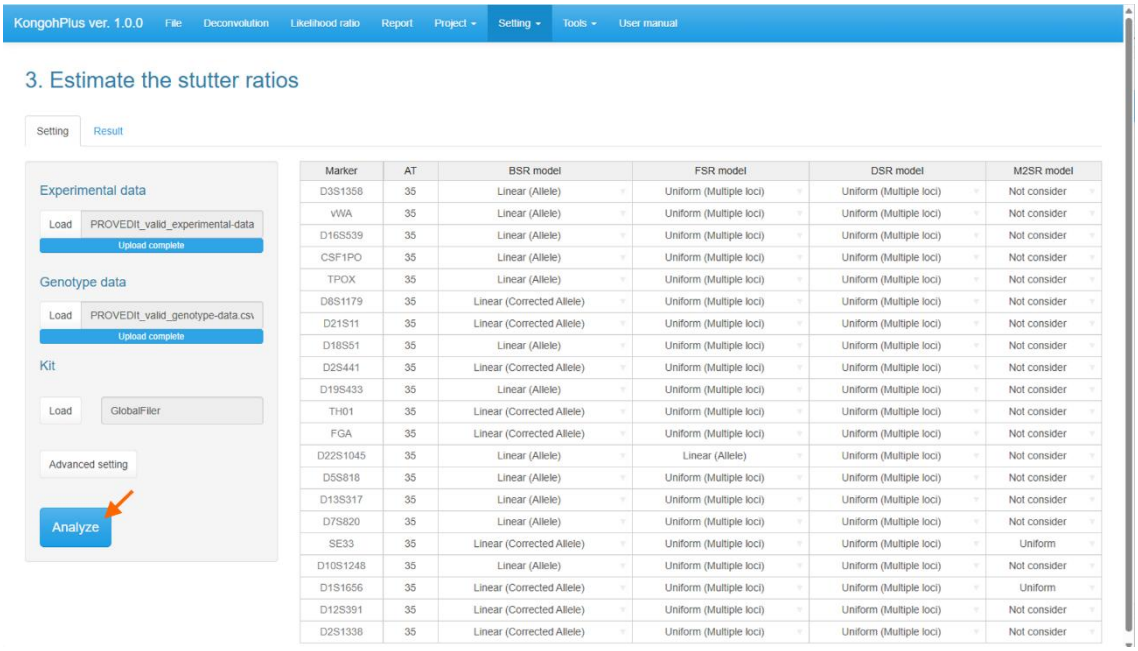
6. If necessary, click **”Advanced setting”** to configure the advanced settings.



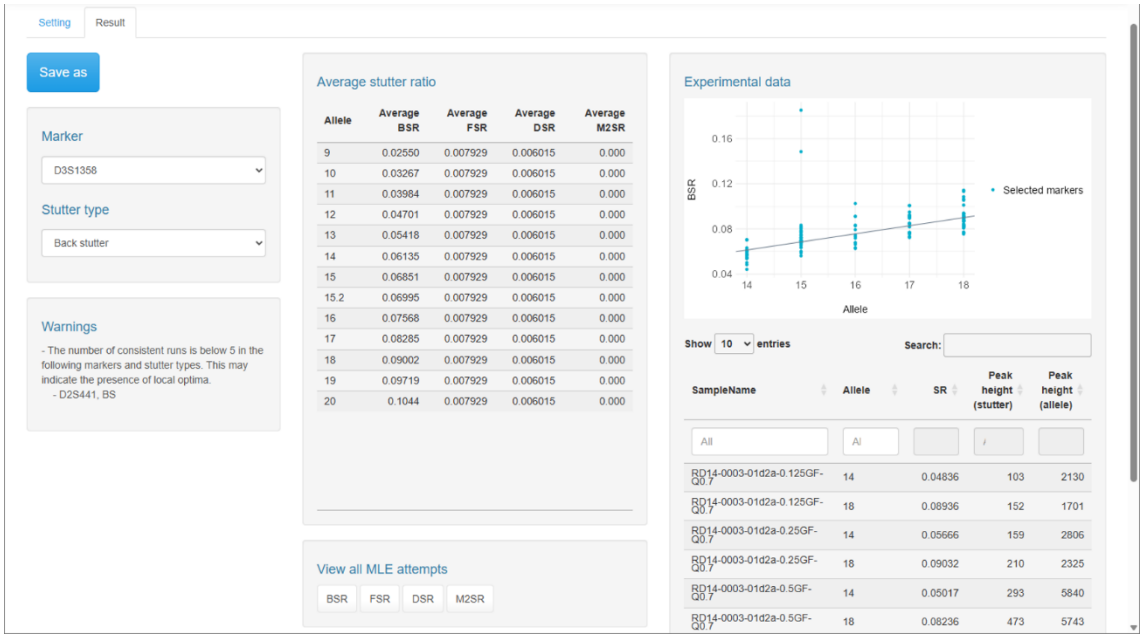
7. Configure the maximum likelihood estimation settings used during stutter ratio estimation. After confirming the values for each setting, click **"Save"**. The maximum likelihood estimation approach is the same as that used for deconvolution, as described in detail in Section 10.4. The parameters to be estimated are described in Section 10.7.1.



8. Click **“Analyze”**.



9. Review the analysis results.



Note:

- You can select the marker and stutter type to be displayed from the controls in

the upper-left corner of the screen.

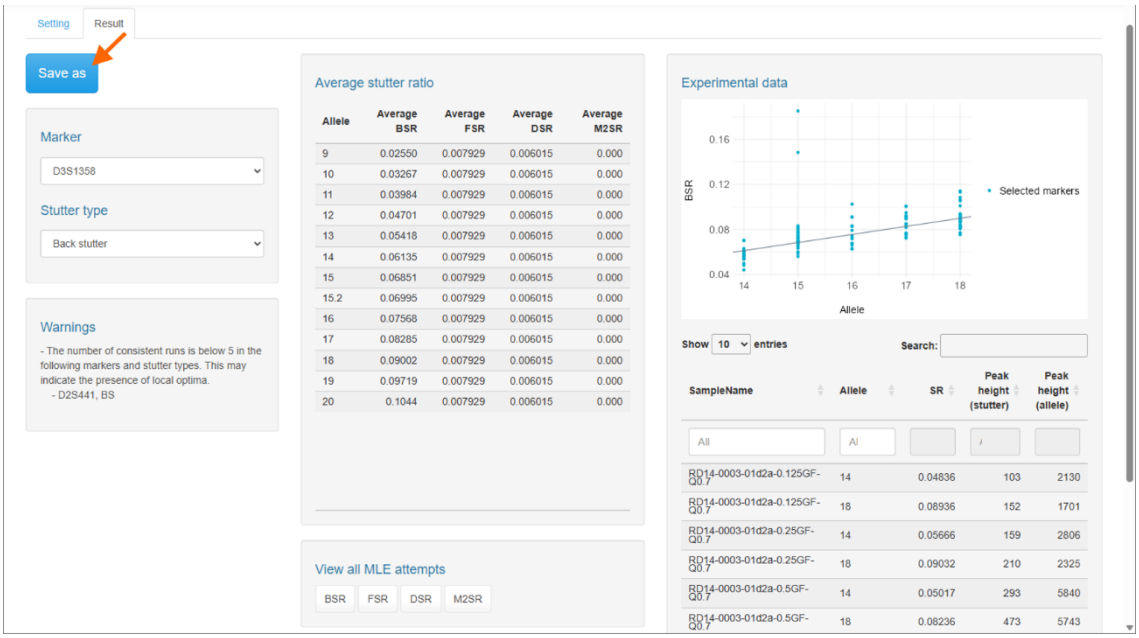
- A warning is displayed if the required number of consistent runs is not obtained.
- **“Average stutter ratio”** shows the estimated average stutter ratio for each allele and each stutter type. These values are used in the deconvolution analysis.
- **“Experimental data”** illustrates the relationship between the stutter ratio and the allele number (or the corrected allele value). The table below summarizes the stutter ratios and related values obtained from the experimental data. Selecting a row in the table also displays the corresponding data in the plot.
- Click **“View all MLE attempts”** to assess the stability of the estimated stutter ratios for the selected marker. KongohPlus performs multiple independent estimations of the stutter ratios (up to 10 by default) and terminates the analysis once the required number of consistent runs (default: 5) has been obtained.
- The figure below shows an example in which five consecutive runs were obtained. Columns corresponding to consistent runs are highlighted in blue. The column with the largest log10(likelihood) value is highlighted in light blue, and this result is selected as the final deconvolution result.

All MLE attempts (BSR)

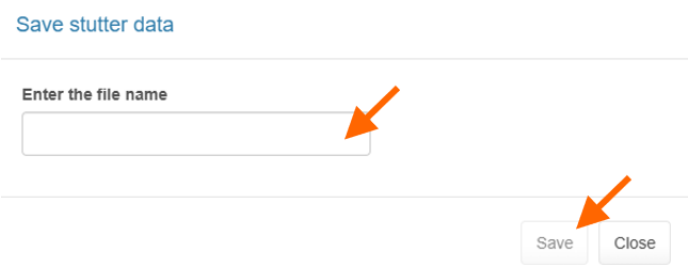
■ : Selected maximum likelihood ■ : Close to the maximum likelihood

Parameter	MLE 1	MLE 2	MLE 3	MLE 4	MLE 5
Log10 (likelihood)	186.670	186.670	186.670	186.670	186.670
Slope	0.00716880	0.00716880	0.00716880	0.00716880	0.00716880
Intercept	-0.0390171	-0.0390171	-0.0390171	-0.0390171	-0.0390171
Sigma	42.1072	42.1072	42.1072	42.1072	42.1072

10. Click **”Save as”**.



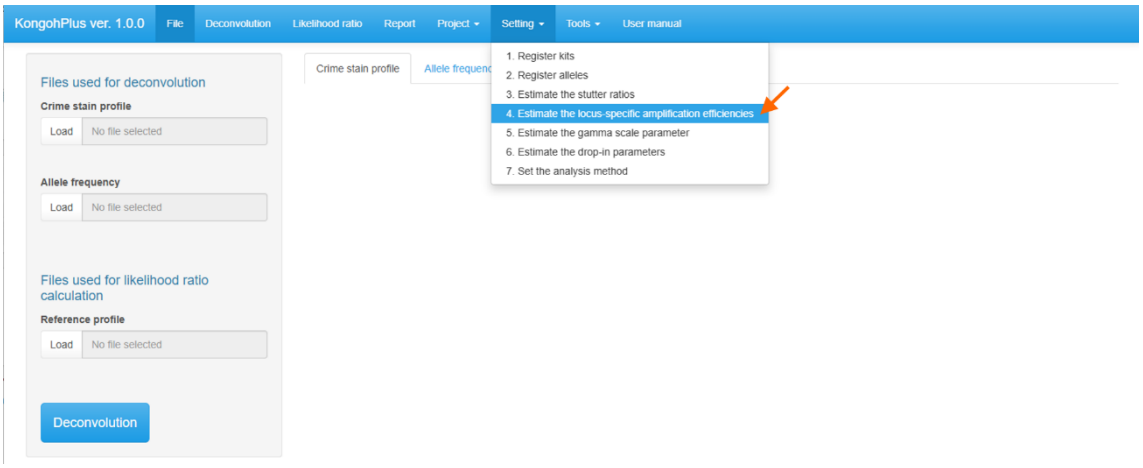
11. Enter a file name, and then click **”Save”**.



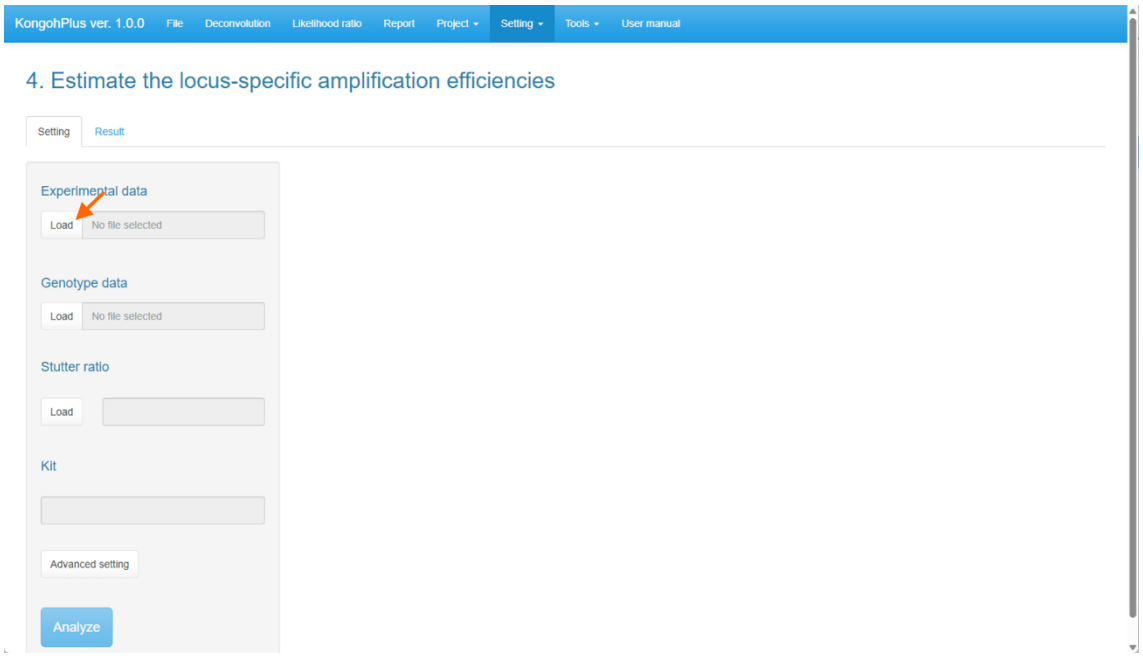
8.4. Estimating Locus-Specific Amplification Efficiencies

This section describes the procedure for estimating locus-specific amplification efficiencies from experimentally obtained DNA profiles.

1. Click **“Setting” ▶ “4. Estimate the locus specific amplification efficiencies”**.



2. Load the experimental dataset using the **“Load”** button, as shown below.



Requirements for the Experimental Dataset

The experimental dataset should satisfy the following requirements:

- Each profile must be a single-source profile.
- The DNA must not be degraded.

File Format of the Experimental Dataset

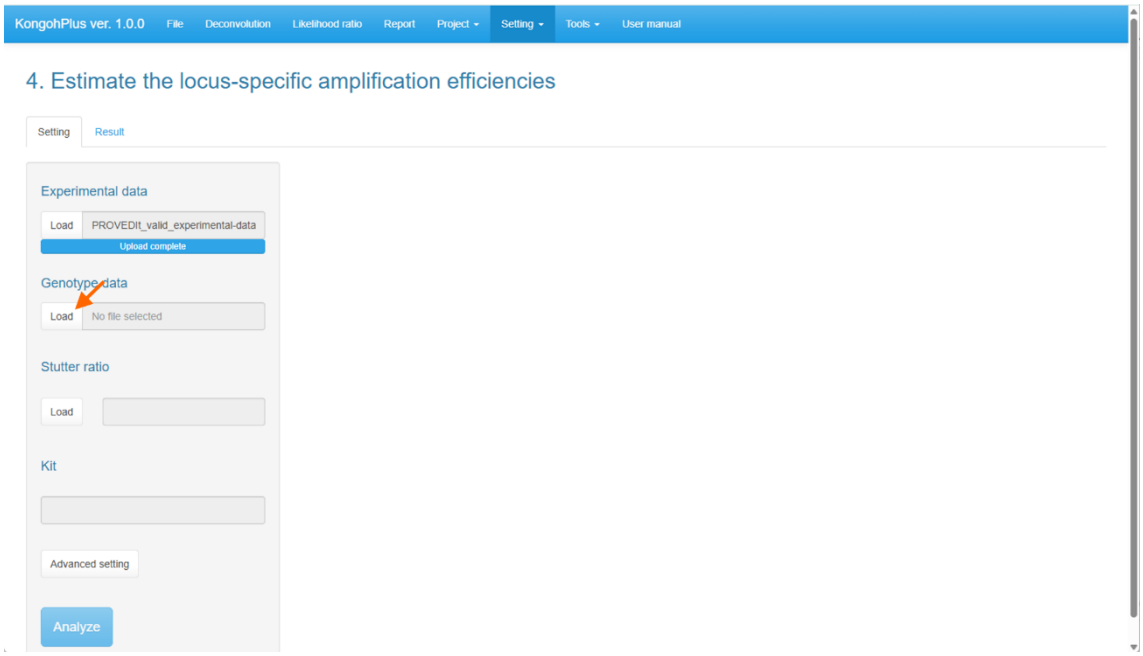
Prepare the experimental dataset as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2	Allele3	Allele4	…	Height1	Height2	Height3	Height4	…
example1	D3S1358	13	14	17	18	…	103	2130	152	1701	…
example1	vWA	16	17	18	19	…	93	1394	104	1328	…
example1	D16S539	11	12	13		…	84	1610	1367		…
…	…	…	…	…	…	…	…	…	…	…	…

Note:

- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele and Height values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers. Alternatively, if they are considered pull-up artifacts or noise, remove them from the input file.
- Assign a unique SampleName to each profile and combine all profiles into a single CSV file by stacking them vertically.

3. Load the genotype dataset for the experimental profiles using the **“Load”** button, as shown below.



File Format of the Genotype Dataset

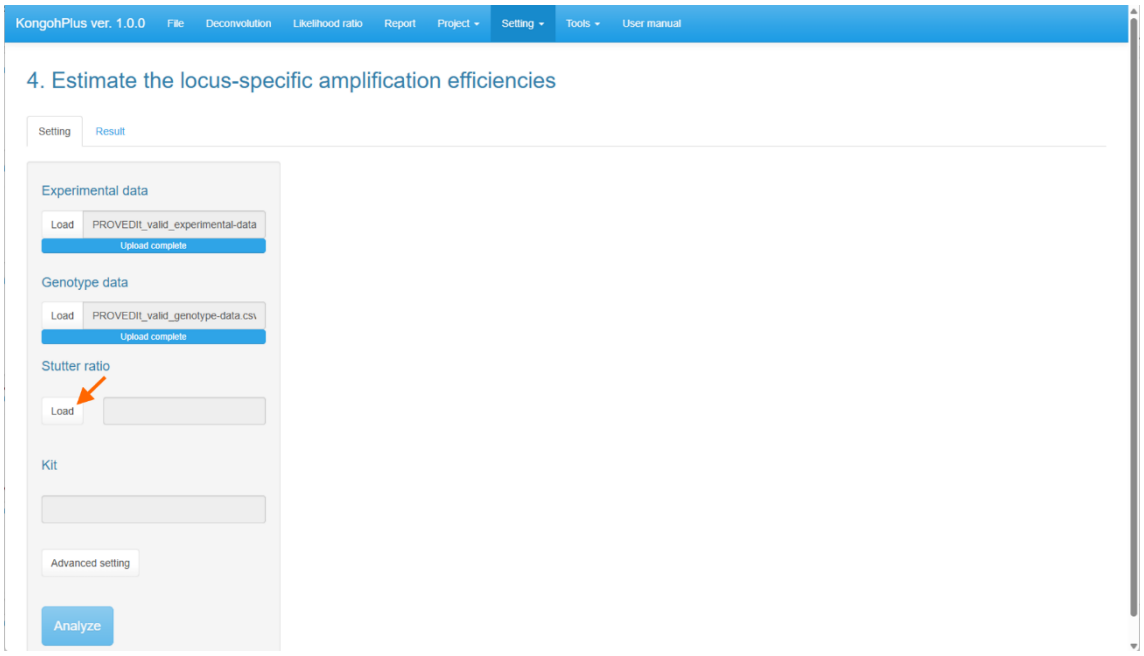
Prepare the genotype dataset as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2
example1	D3S1358	15	16
example1	vWA	15	18
example1	D16S539	10	10
…	…	…	…

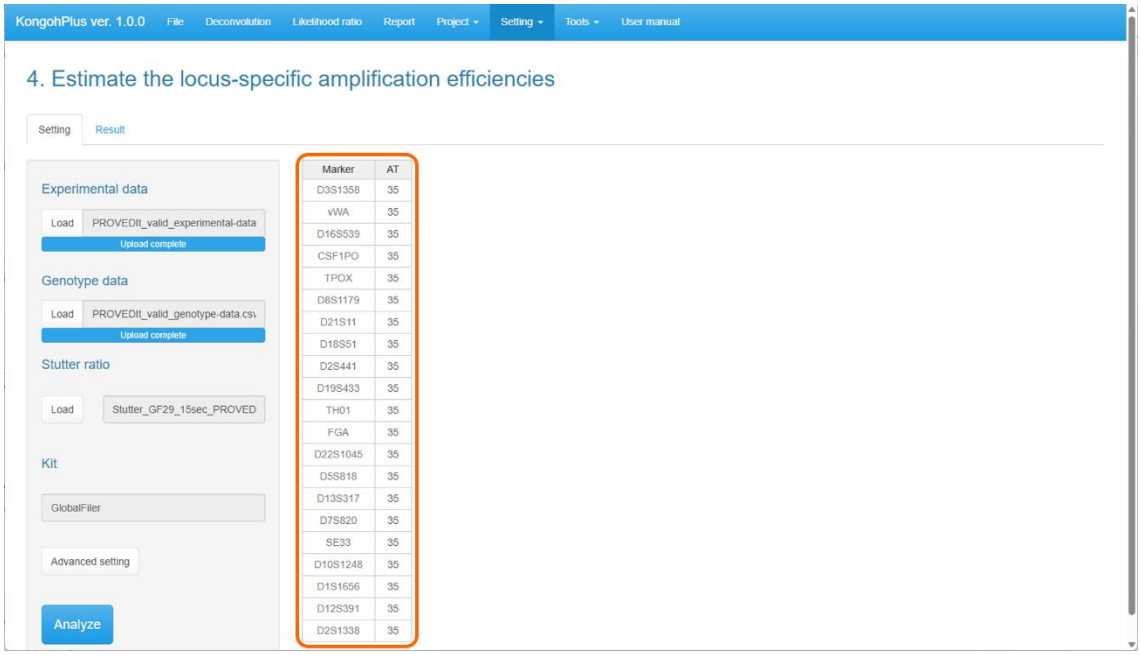
Note:

- The SampleName values must correspond to those in the experimental dataset.
- Assign a unique SampleName to each profile and combine all genotype data into a single CSV file by stacking them vertically.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers.
- For homozygous genotypes, enter the same allele number twice.

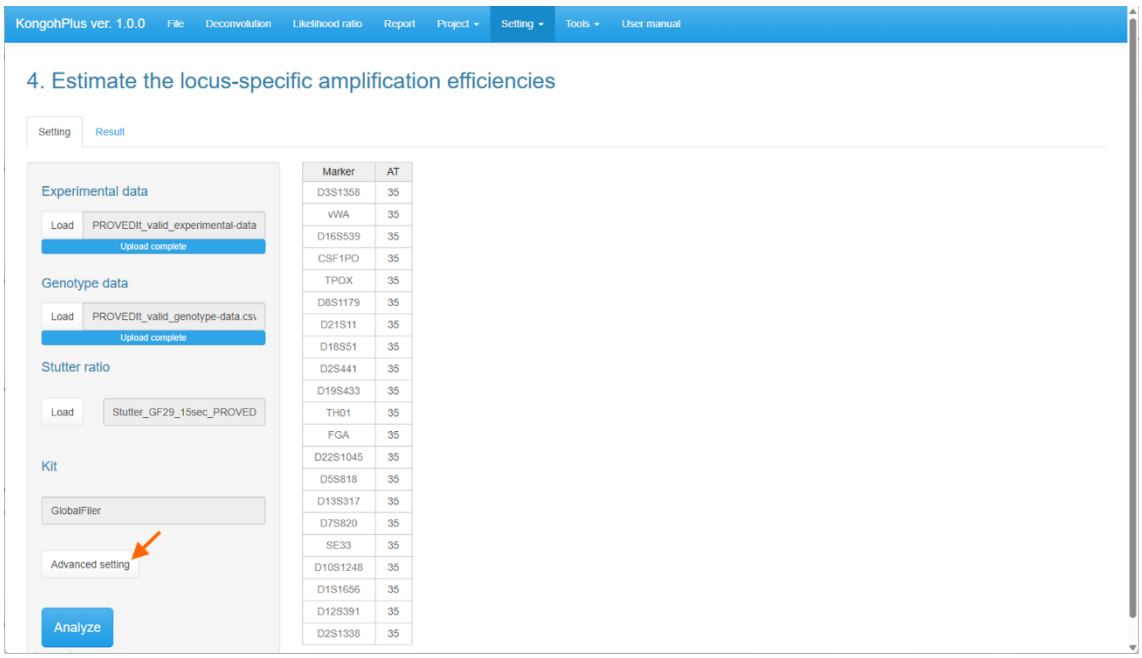
4. Load the stutter ratio data estimated according to the procedure described in Section 8.3 by clicking the **“Load”** button, as shown below.



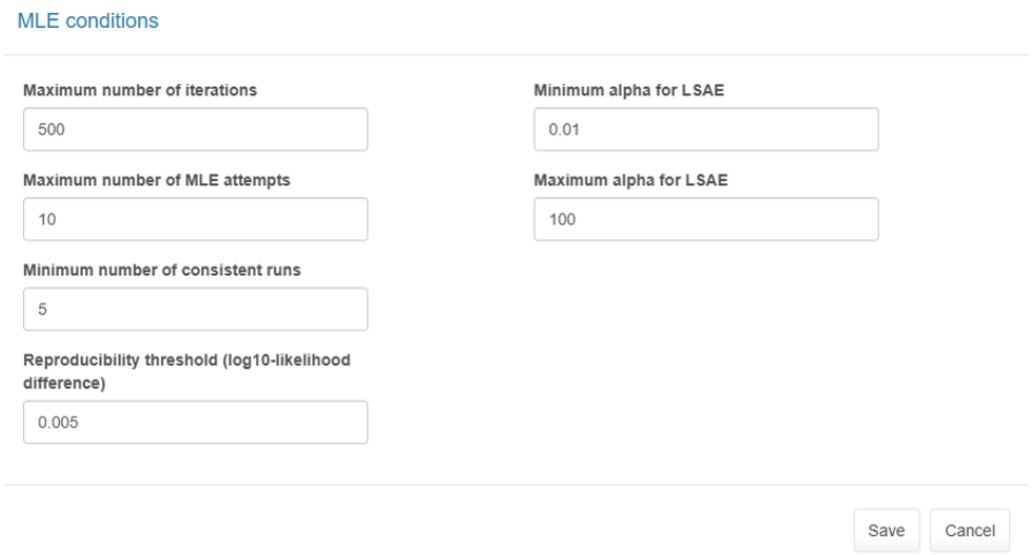
5. Set the analytical threshold (AT) for each marker.



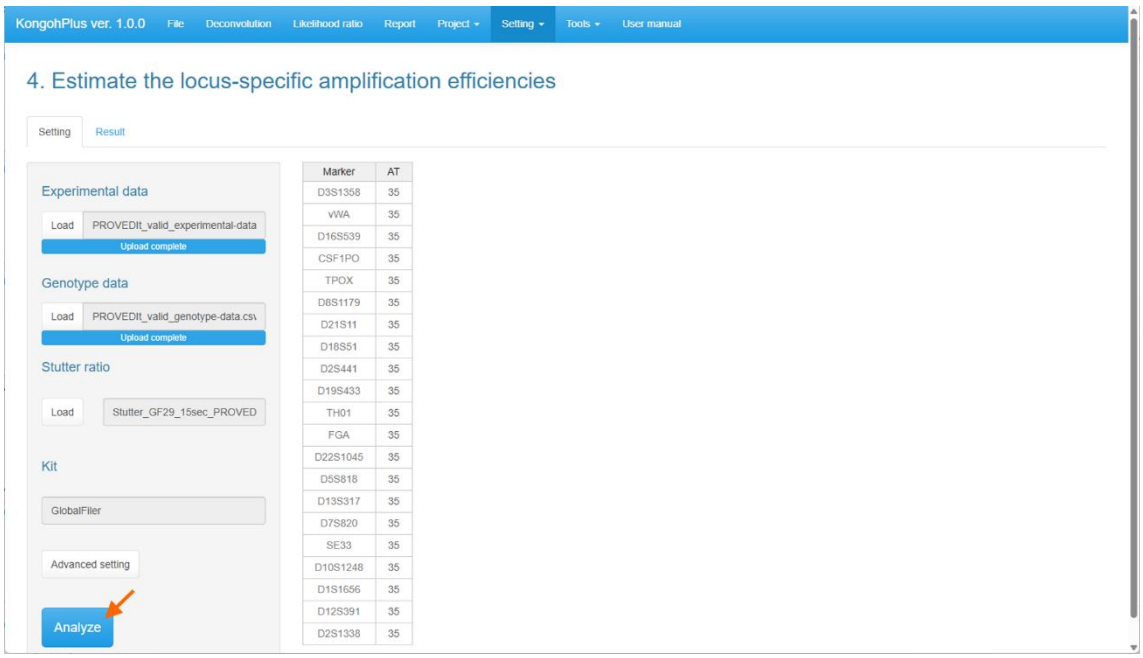
6. If necessary, click **”Advanced setting”** to configure the advanced settings.



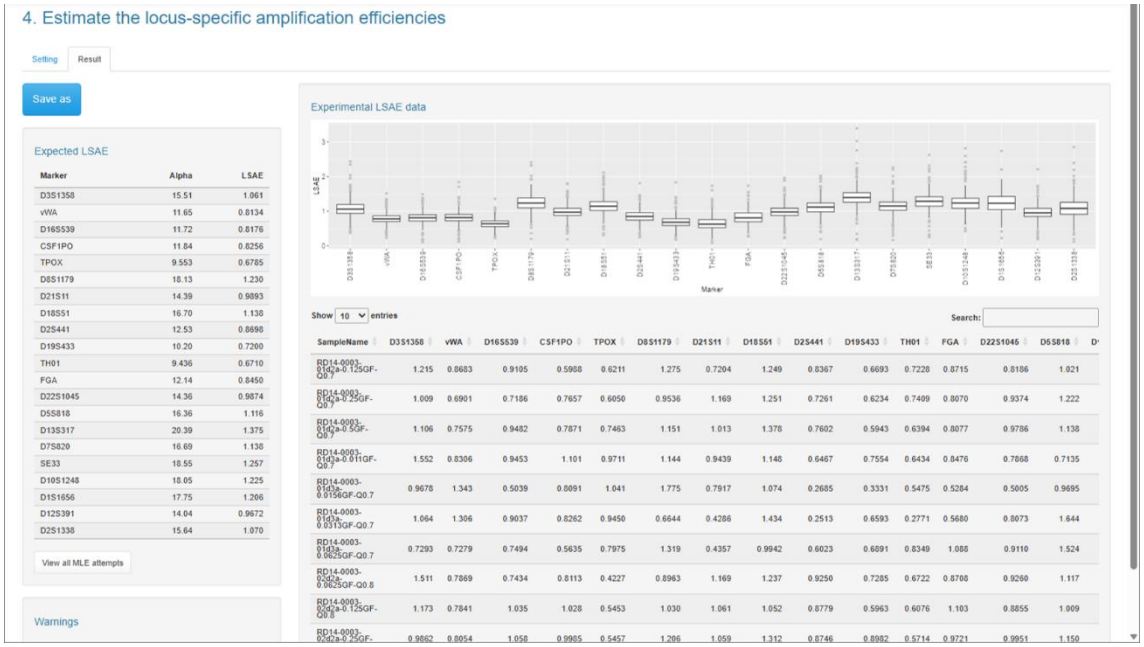
7. Configure the maximum likelihood estimation settings used during the estimation of locus-specific amplification efficiencies. After confirming the values for each setting, click **"Save"**. The maximum likelihood estimation approach is the same as that used for deconvolution, as described in detail in Section 10.4. The parameters to be estimated are described in Section 10.7.2.



8. Click **"Analyze"**.



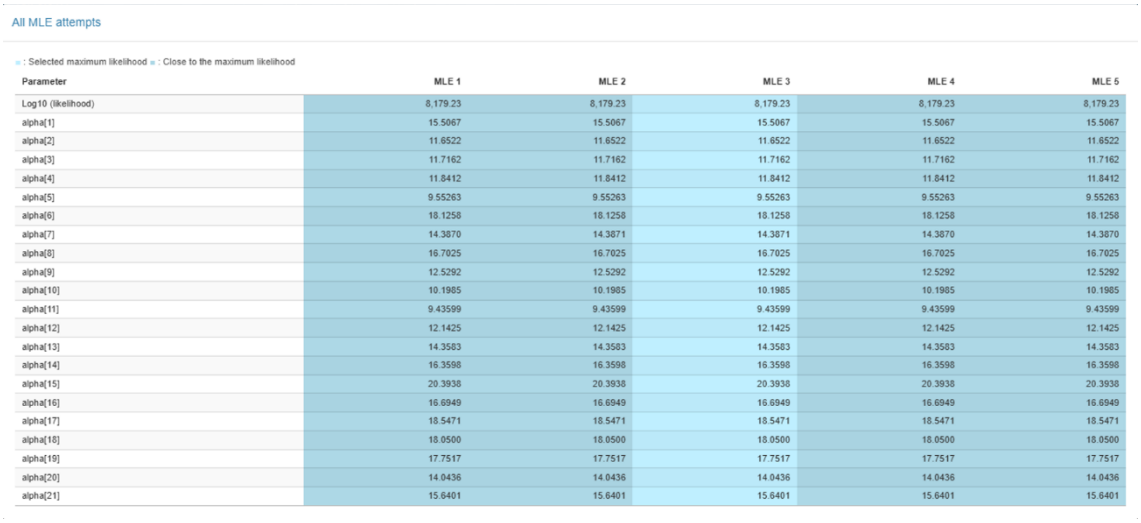
9. Review the analysis results.



Note:

- A warning is displayed if the required number of consistent runs is not obtained.
- **"Expected LSAE"** displays the estimated values of **Alpha** and **LSAE** (locus-specific amplification efficiency) for each locus. The estimated **LSAE** values are used in the deconvolution analysis. **Alpha** is a parameter estimated under the assumption that the normalized locus-specific amplification efficiencies follow a Dirichlet distribution (see Section 10.7.2 for details).
- **"Experimental data"** illustrates the amplification efficiencies estimated from the experimental data for each locus. The table below summarizes the amplification efficiency estimated from each experimental profile. Selecting a row in the table also displays the corresponding data in the plot.
- Click **"View all MLE attempts"** to assess the stability of the estimated locus-specific amplification efficiencies. KongohPlus performs multiple independent estimations (up to 10 by default) and terminates the analysis once the required number of consistent runs (default: 5) has been obtained.

- The figure below shows an example in which the required number of consistent runs was obtained. Columns corresponding to consistent runs are highlighted in blue. The column with the highest log10(likelihood) value is highlighted in light blue, and this result is selected as the final estimate of the locus-specific amplification efficiencies.

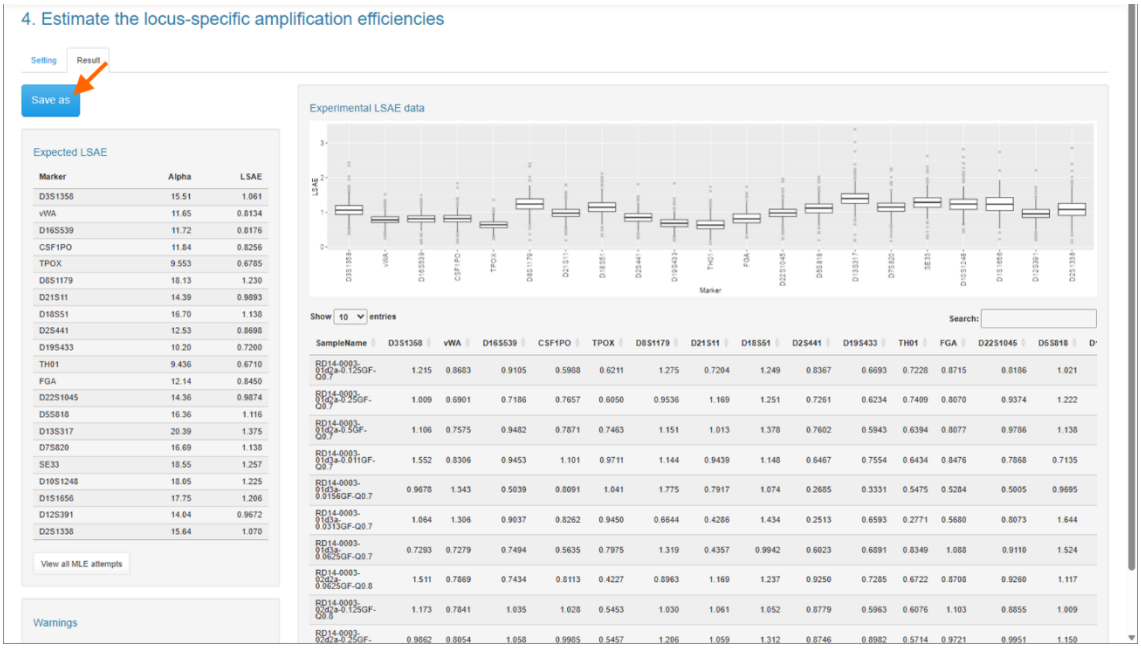


All MLE attempts

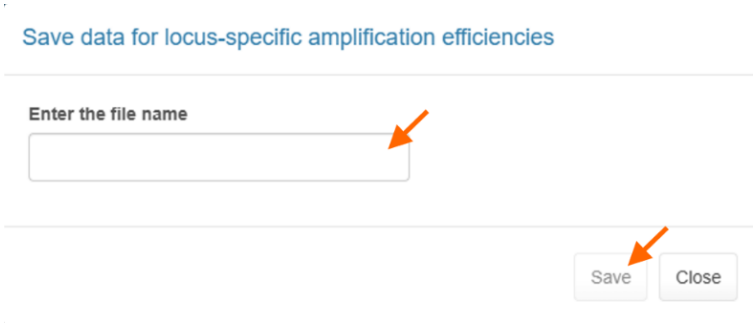
: Selected maximum likelihood : Close to the maximum likelihood

Parameter	MLE 1	MLE 2	MLE 3	MLE 4	MLE 5
Log10 (likelihood)	8,179.23	8,179.23	8,179.23	8,179.23	8,179.23
alpha[1]	15.5067	15.5067	15.5067	15.5067	15.5067
alpha[2]	11.6522	11.6522	11.6522	11.6522	11.6522
alpha[3]	11.7162	11.7162	11.7162	11.7162	11.7162
alpha[4]	11.8412	11.8412	11.8412	11.8412	11.8412
alpha[5]	9.55263	9.55263	9.55263	9.55263	9.55263
alpha[6]	18.1258	18.1258	18.1258	18.1258	18.1258
alpha[7]	14.3870	14.3871	14.3871	14.3870	14.3870
alpha[8]	16.7025	16.7025	16.7025	16.7025	16.7025
alpha[9]	12.5292	12.5292	12.5292	12.5292	12.5292
alpha[10]	10.1985	10.1985	10.1985	10.1985	10.1985
alpha[11]	9.43599	9.43599	9.43599	9.43599	9.43599
alpha[12]	12.1425	12.1425	12.1425	12.1425	12.1425
alpha[13]	14.3583	14.3583	14.3583	14.3583	14.3583
alpha[14]	16.3598	16.3598	16.3598	16.3598	16.3598
alpha[15]	20.3938	20.3938	20.3938	20.3938	20.3938
alpha[16]	16.6949	16.6949	16.6949	16.6949	16.6949
alpha[17]	18.5471	18.5471	18.5471	18.5471	18.5471
alpha[18]	18.0500	18.0500	18.0500	18.0500	18.0500
alpha[19]	17.7517	17.7517	17.7517	17.7517	17.7517
alpha[20]	14.0436	14.0436	14.0436	14.0436	14.0436
alpha[21]	15.6401	15.6401	15.6401	15.6401	15.6401

10. Click **"Save as"**.



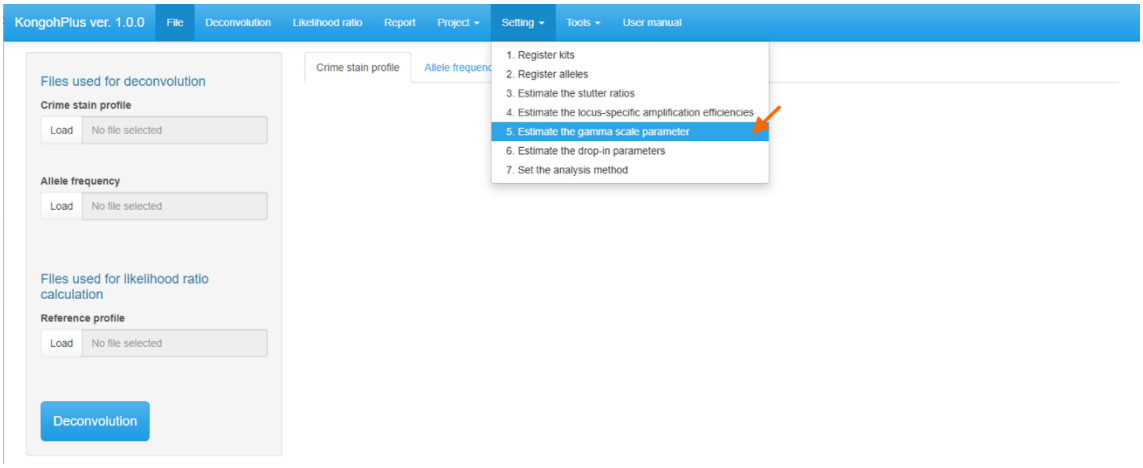
11. Enter a file name, and then click **”Save”**.



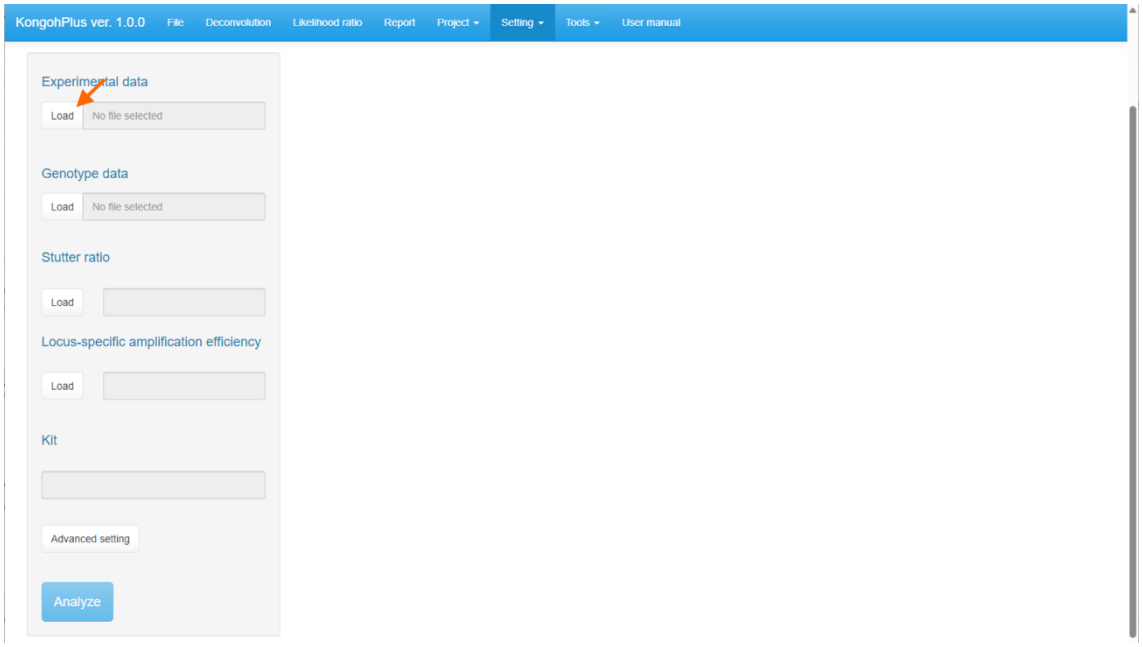
8.5. Estimating the Gamma Scale Parameter

This section describes the procedure for estimating the scale parameters of the gamma distributions from experimentally obtained DNA profiles.

1. Click **“Setting” ▶ “5. Estimate the gamma scale parameter”**.



2. Load the experimental dataset using the **“Load”** button, as shown below.



Requirements for the Experimental Dataset

The experimental dataset should satisfy the following requirements:

- Each profile must be a single-source profile.
- The DNA must not be degraded.
- The dataset should contain profiles with a wide range of PCR template DNA quantities.

File Format of the Experimental Dataset

Prepare the experimental dataset as a CSV file in the format shown in the table below.

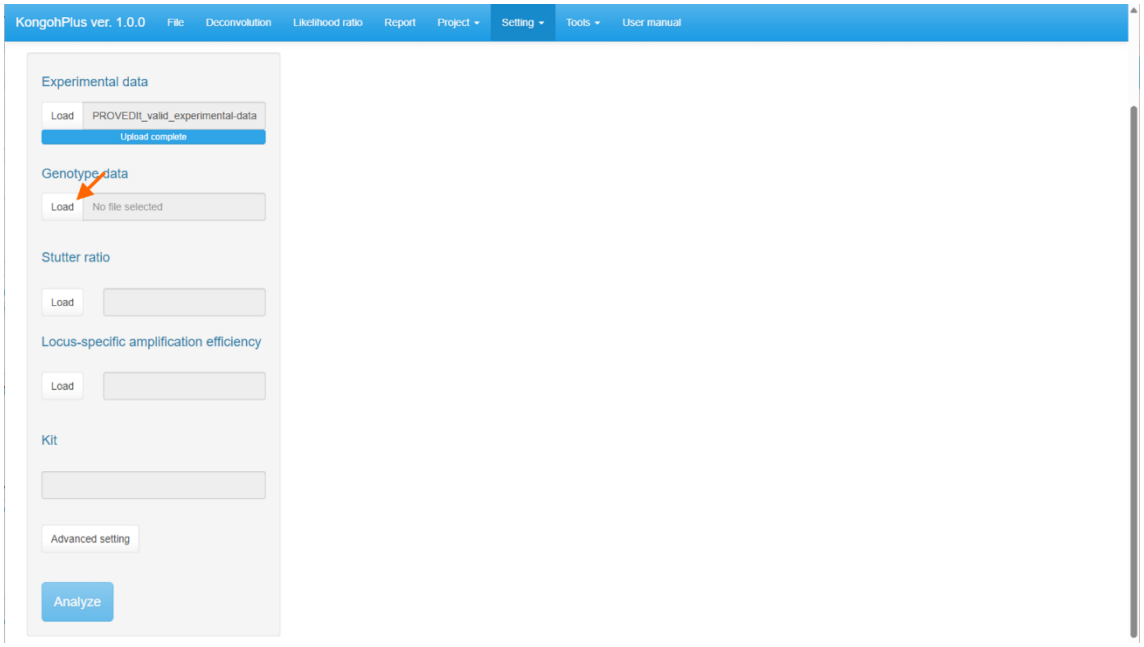
SampleName	Marker	Allele1	Allele2	Allele3	Allele4	…	Height1	Height2	Height3	Height4	…
example1	D3S1358	13	14	17	18	…	103	2130	152	1701	…
example1	vWA	16	17	18	19	…	93	1394	104	1328	…
example1	D16S539	11	12	13		…	84	1610	1367		…
…	…	…	…	…	…	…	…	…	…	…	…

Note:

- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele and Height values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers. Alternatively, if they are considered pull-up artifacts or noise, remove them from the input file.

- Assign a unique SampleName to each profile and combine all profiles into a single CSV file by stacking them vertically.

3. Load the genotype dataset for the experimental profiles using the **“Load”** button, as shown below.



File Format of the Genotype Dataset

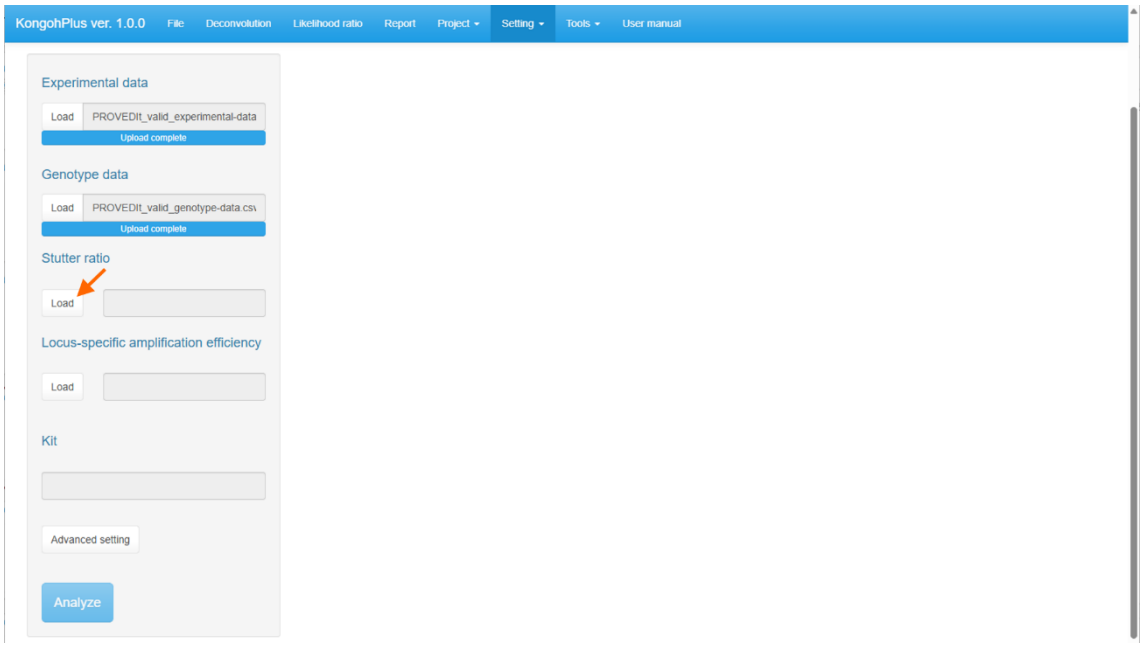
Prepare the genotype dataset as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2
example1	D3S1358	15	16
example1	vWA	15	18
example1	D16S539	10	10
…	…	…	…

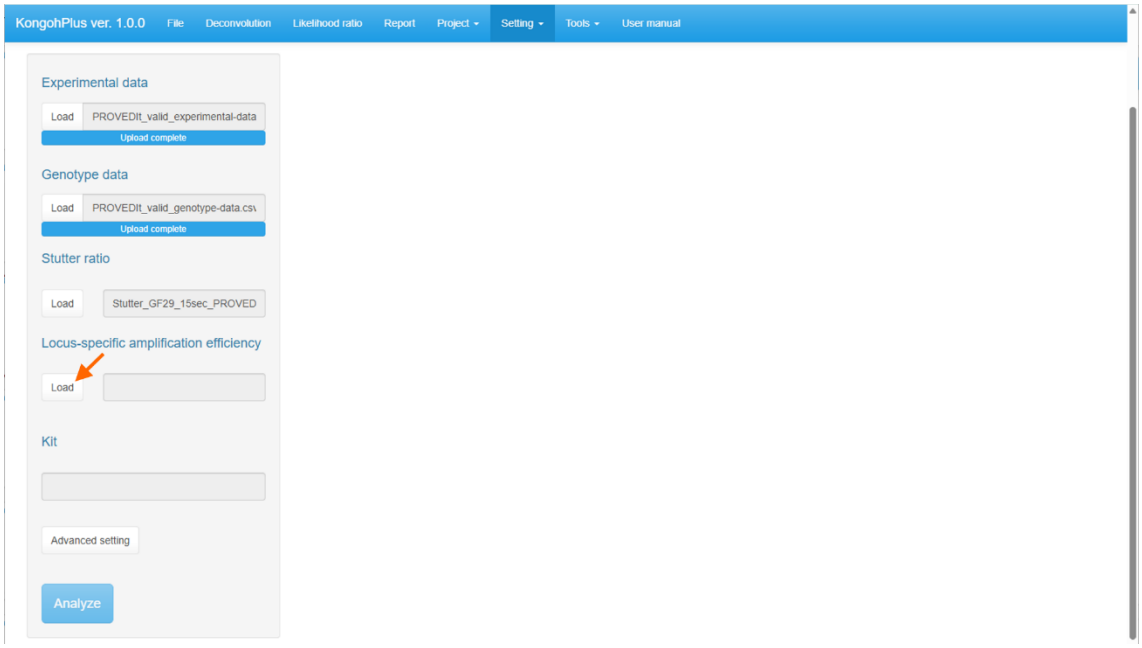
Note:

- The SampleName values must correspond to those in the experimental dataset.
- Assign a unique SampleName to each profile and combine all genotype data into a single CSV file by stacking them vertically.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- The Allele values for autosomal STR markers must be numeric. For Off-ladder (OL) and Outside Marker Range (OMR) alleles, assign the appropriate allele numbers.
- For homozygous genotypes, enter the same allele number twice.

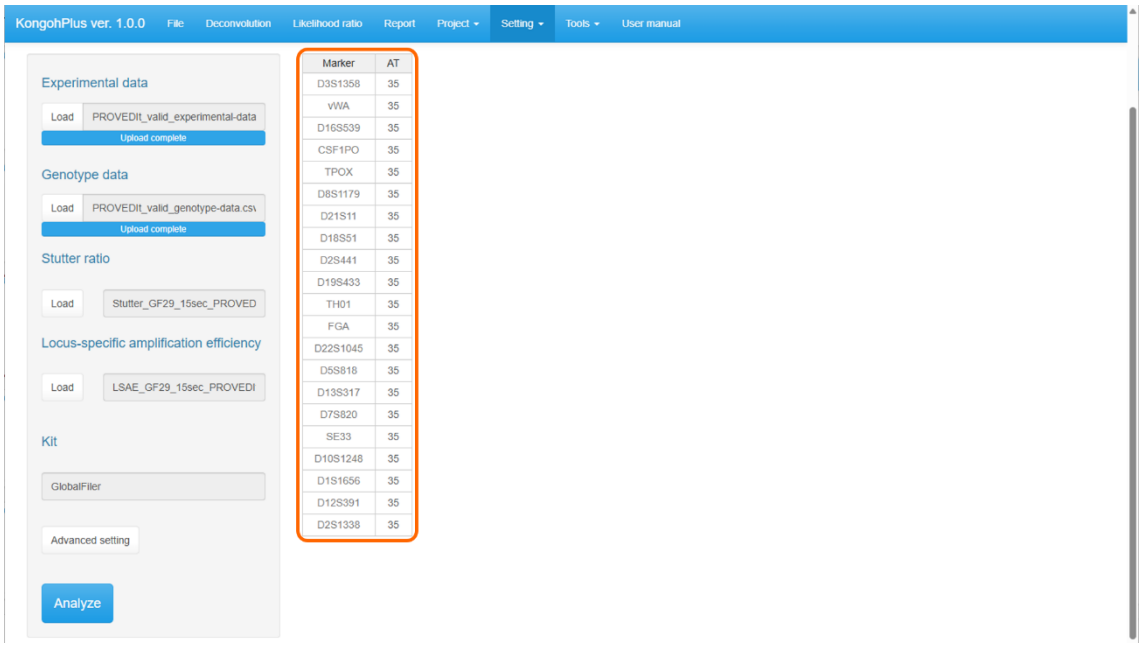
4. Load the stutter ratio data estimated according to the procedure described in Section 8.3 by clicking the **“Load”** button.



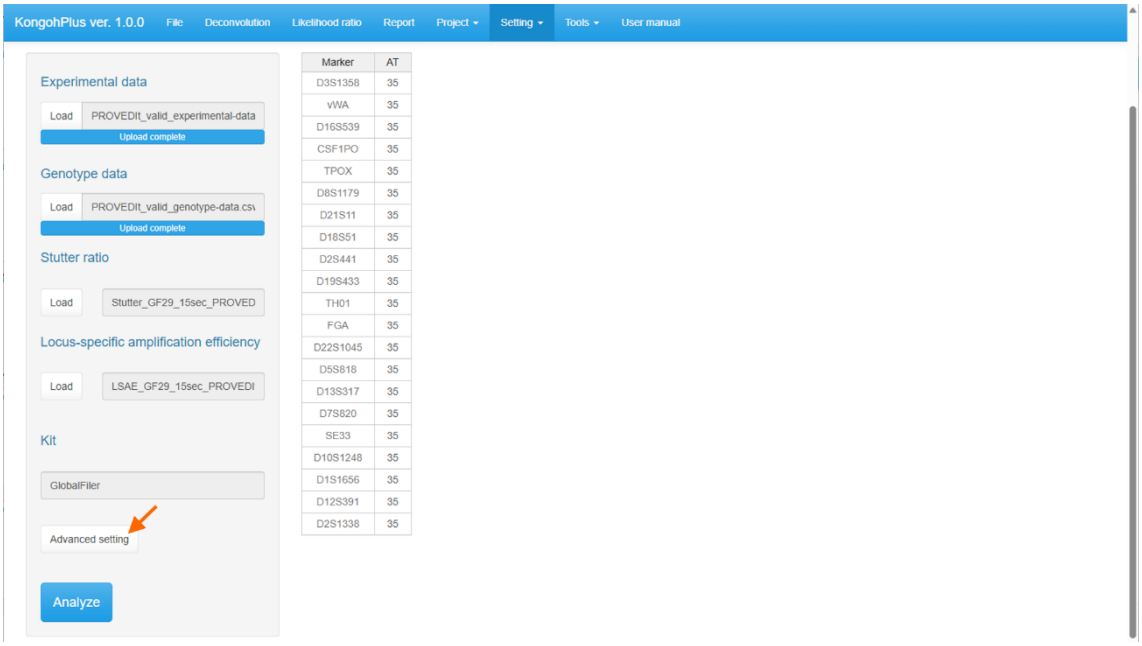
5. Load the locus-specific amplification efficiency data estimated according to the procedure described in Section 8.4 by clicking the **“Load”** button.



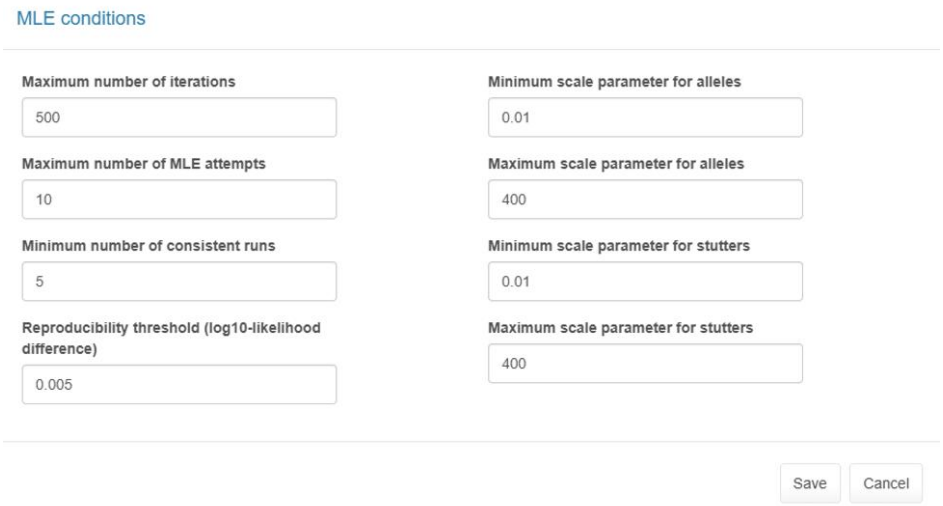
6. Set the analytical threshold (AT) for each marker.



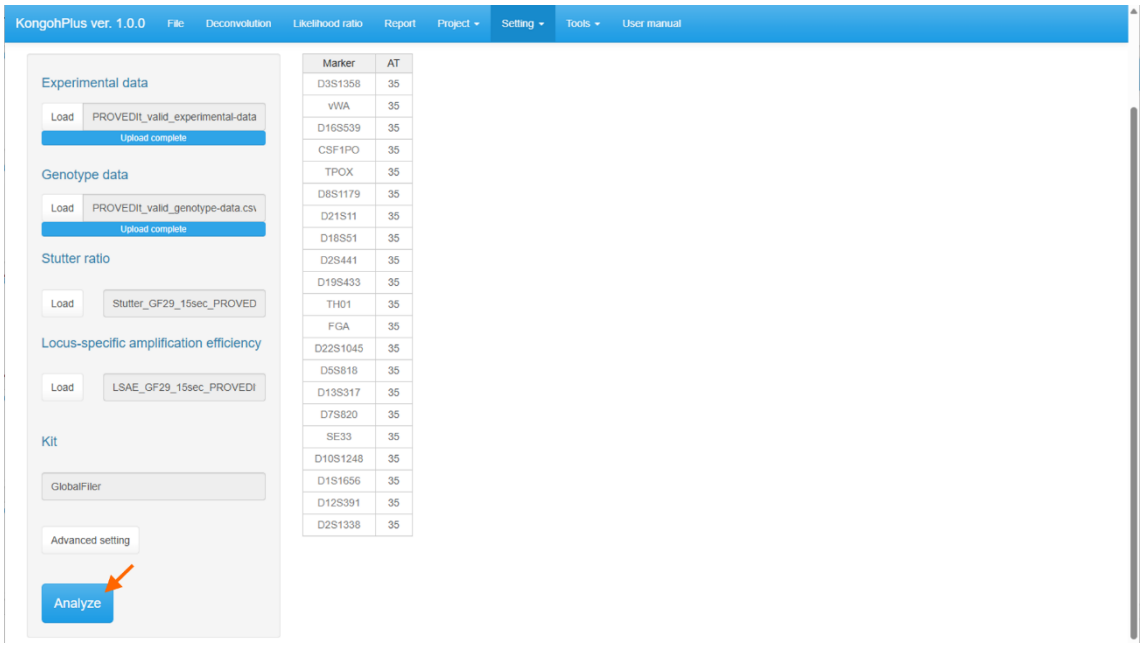
7. If necessary, click **”Advanced setting”** to configure the advanced settings.



8. Configure the maximum likelihood estimation settings used during estimation of the gamma scale parameters. After confirming the values for each setting, click **”Save”**. The maximum likelihood estimation approach is the same as that used for deconvolution, as described in detail in Section 10.4. The parameters to be estimated are described in Section 10.7.3.



9. Click **“Analyze”**.



10. Review the analysis results.



Note:

- A warning is displayed if the required number of consistent runs is not obtained.
- **"Estimated parameters"** displays the estimated scale parameters of the gamma distributions for the allele and stutter peak heights.
- **"Experimental peak height data"** shows scatter plots comparing the observed and expected peak heights. The left plot shows the allelic products only, the center plot shows the stutter products only, and the right plot shows data containing both allelic and stutter products. The table below summarizes the values obtained from the experimental data. Selecting a row in the table also displays the corresponding data in the plots.
- Click **"View all MLE attempts"** to assess the stability of the estimated gamma scale parameters. KongohPlus performs multiple independent estimations (up to 10 by default) and terminates the analysis once the required number of consistent runs (default: 5) has been obtained.
- The figure below shows an example in which the required number of consistent runs was obtained. Columns corresponding to consistent runs are highlighted in blue. The column with the highest log10(likelihood) value is highlighted in light blue, and this result is selected as the final estimate of the gamma scale parameters.

All MLE attempts

: Selected maximum likelihood : Close to the maximum likelihood

Parameter	MLE 1	MLE 2	MLE 3	MLE 4	MLE 5
Log10 (likelihood)	-57,415.5	-57,415.5	-57,415.5	-57,415.5	-57,415.5
Scale parameter (allele)	81.7536	81.7536	81.7536	81.7536	81.7536
Scale parameter (stutter)	10.7909	10.7909	10.7909	10.7909	10.7909

11. Click **”Save as”**.



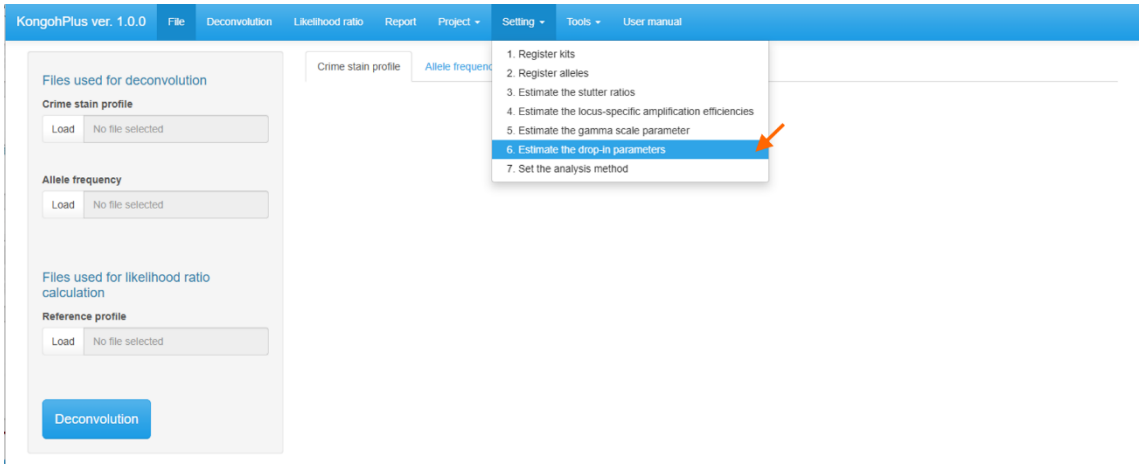
12. Enter a file name, and then click **”Save”**.



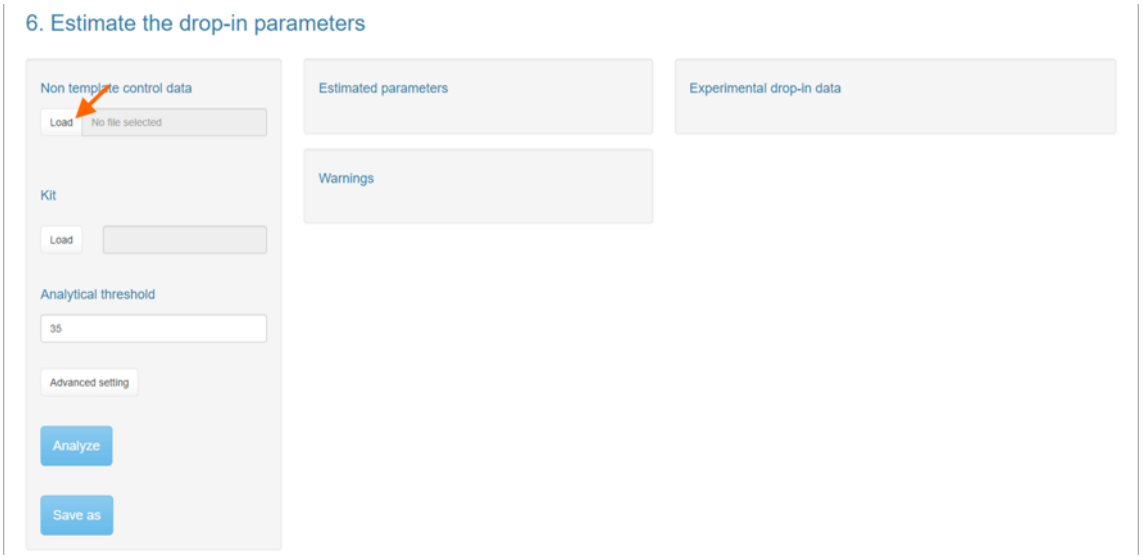
8.6. Estimating Drop-in Parameters

This section describes the procedure for estimating the parameters of the statistical model for drop-in events and peak heights using experimentally obtained non-template control profiles.

1. Click **“Setting” ▶ “6. Estimate the drop-in parameters”**.



2. Load the non-template control dataset using the **“Load”** button, as shown below.



File Format of the Non-template Control Dataset

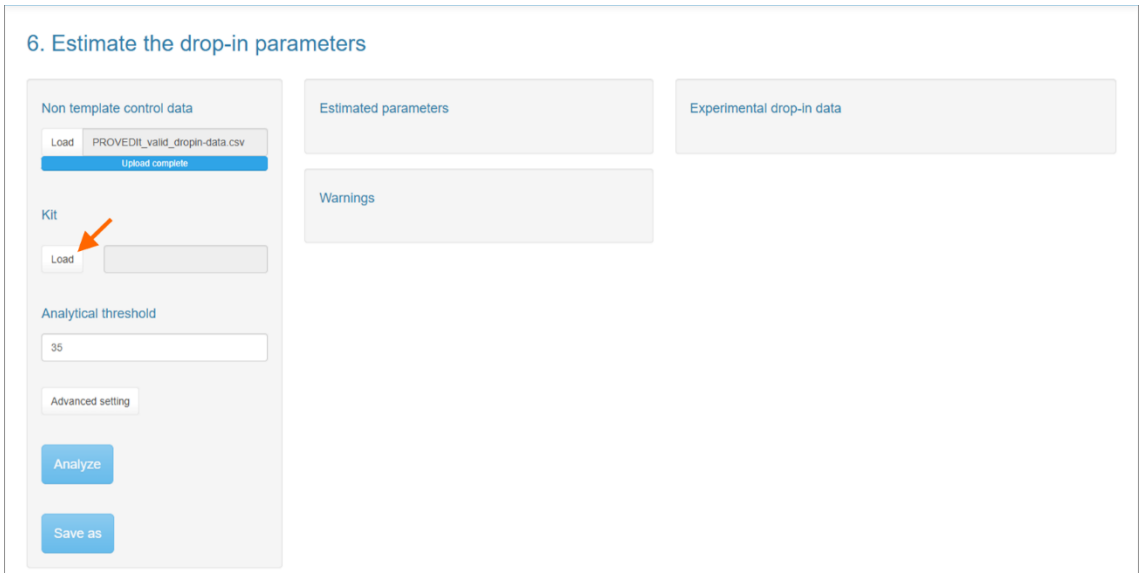
Prepare the non-template control dataset as a CSV file in the format shown in the table below.

SampleName	Marker	Allele1	Allele2	Allele3	Allele4	…	Height1	Height2	Height3	Height4	…
example1	D3S1358					…					…
example1	vWA	16				…	63				…
example1	D16S539					…					…
…	…	…	…	…	…	…	…	…	…	…	…

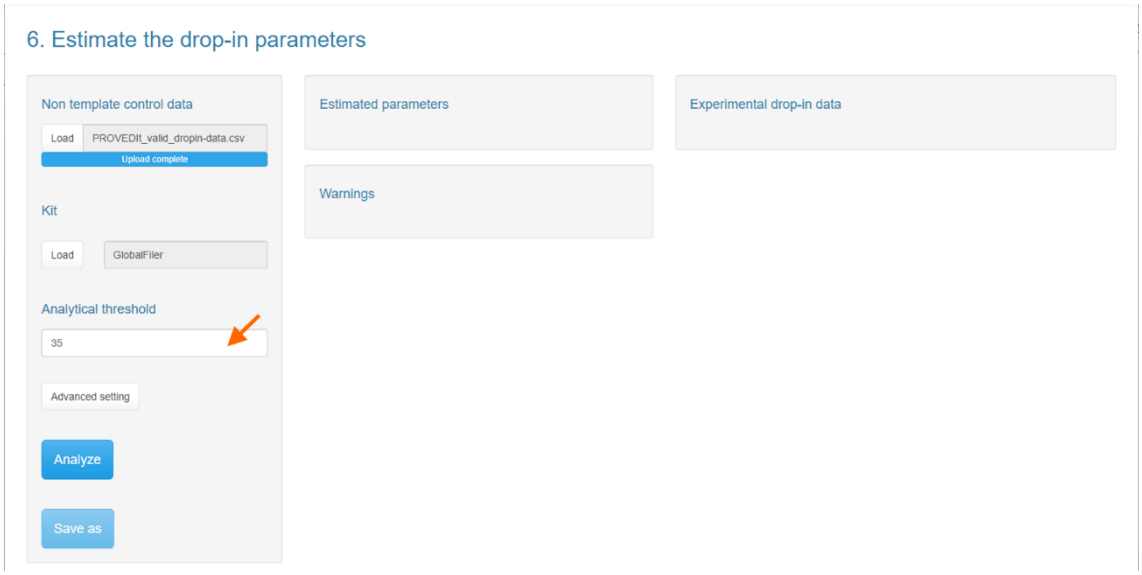
Note:

- Include all analyzed non-template control profiles in the dataset, not only those in which drop-in events were observed.
- Although only autosomal STR data are used for the analysis, the input file may also contain sex chromosome marker data.
- Assign a unique SampleName to each profile and combine all profiles into a single CSV file by stacking them vertically.

3. Load the registered kit information by clicking the **“Load”** button, as shown below.



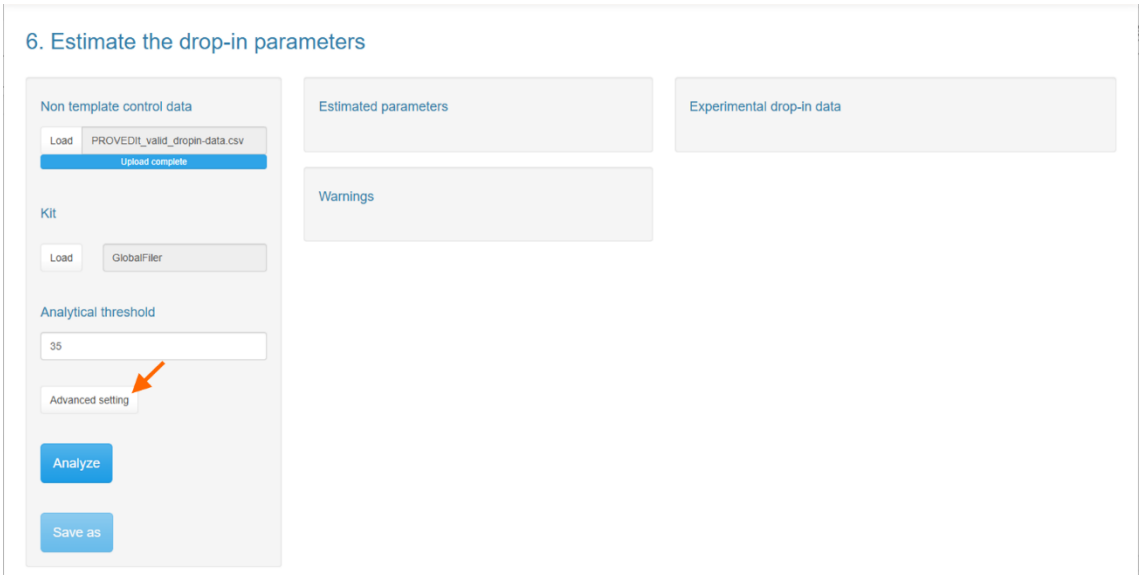
4. Set the analytical threshold (AT).



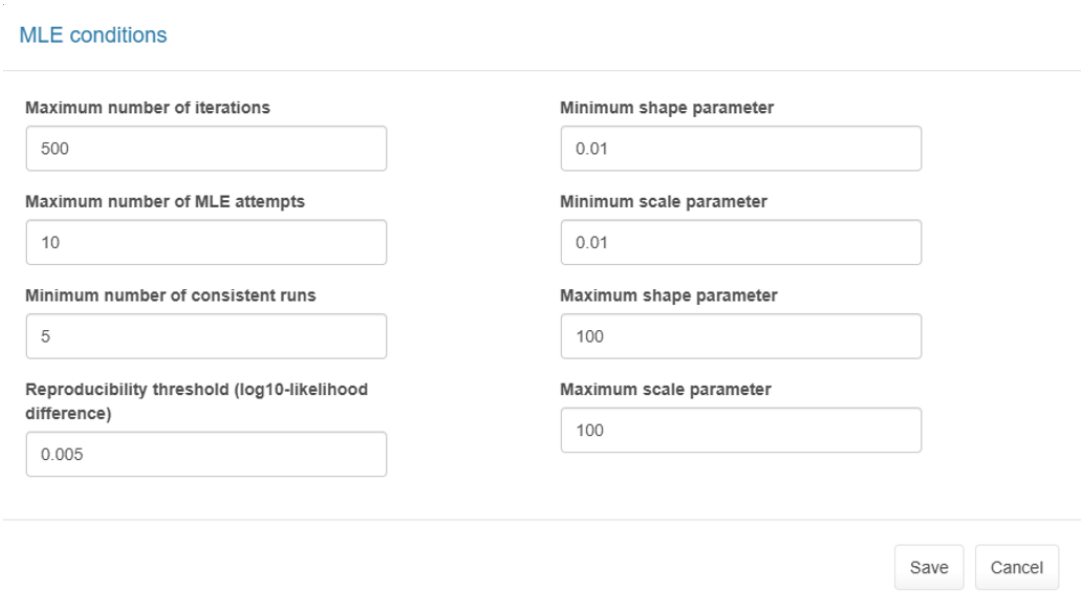
Note:

- A common AT is applied to all markers.
- To maximize the amount of drop-in data available for estimation, the AT should be set as low as practically possible.

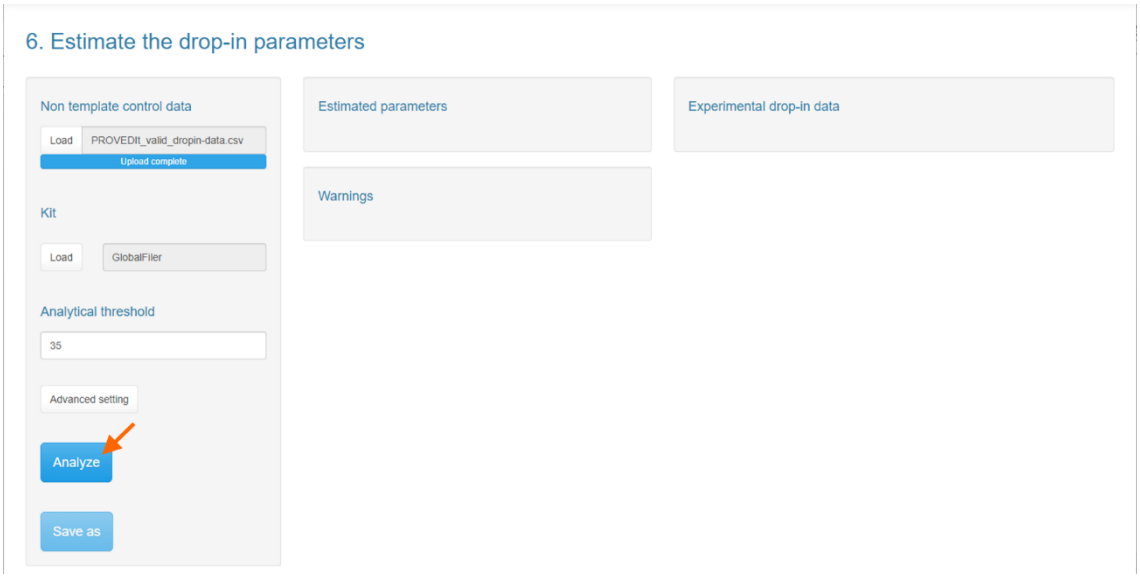
5. If necessary, click **”Advanced setting”** to configure the advanced settings.



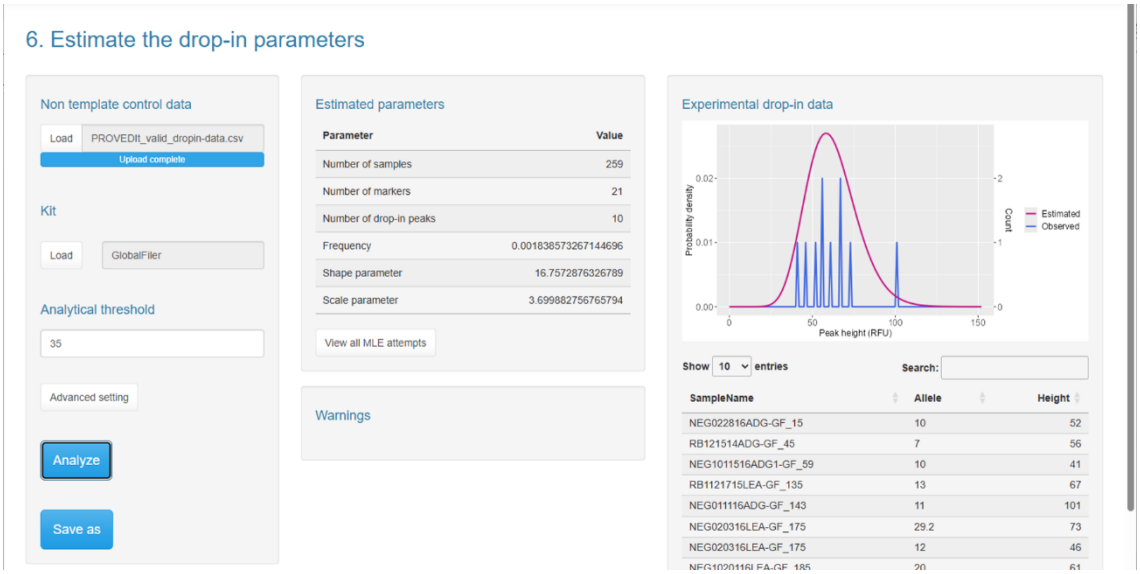
6. Configure the maximum likelihood estimation settings used during estimation of the drop-in parameters. After confirming the values for each setting, click **“Save”**. The maximum likelihood estimation approach is the same as that used for deconvolution, as described in detail in Section 10.4. The parameters to be estimated are described in Section 10.7.4.



7. Click **“Analyze”**.



8. Review the analysis results.



Note:

- A warning is displayed if the required number of consistent runs is not obtained.
- **“Estimated parameters”** displays the observed number and frequency of drop-in events, together with the estimated shape and scale parameters of the

gamma distribution used to model drop-in peak heights.

- **“Experimental drop-in data”** illustrates the observed drop-in peak heights together with the fitted gamma distribution. The table below summarizes the positions and peak heights of the drop-in events observed in the non-template control dataset.

- Click **“View all MLE attempts”** to assess the stability of the estimated gamma distribution parameters for drop-in peak heights. KongohPlus performs multiple independent estimations (up to 10 by default) and terminates the analysis once the required number of consistent runs (default: 5) has been obtained.

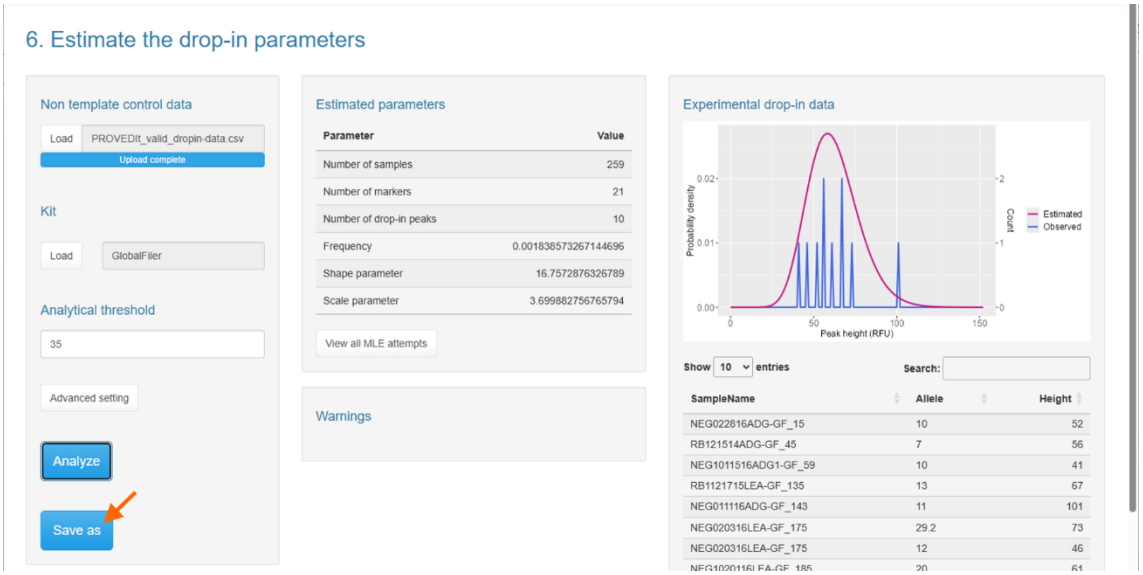
- The figure below shows an example in which the required number of consistent runs was obtained. Columns corresponding to consistent runs are highlighted in blue. The column with the highest log10(likelihood) value is highlighted in light blue, and this result is selected as the final estimate of the drop-in parameters.

All MLE attempts

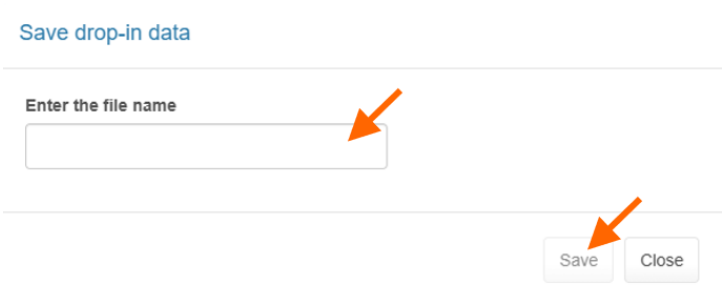
: Selected maximum likelihood : Close to the maximum likelihood

Parameter	MLE 1	MLE 2	MLE 3	MLE 4	MLE 5
Log10 (likelihood)	-17.8776	-17.8776	-17.8776	-17.8776	-17.8776
Shape parameter	16.7573	16.7573	16.7573	16.7573	16.7573
Scale parameter	3.69988	3.69988	3.69988	3.69988	3.69988

9. Click **”Save as”**.



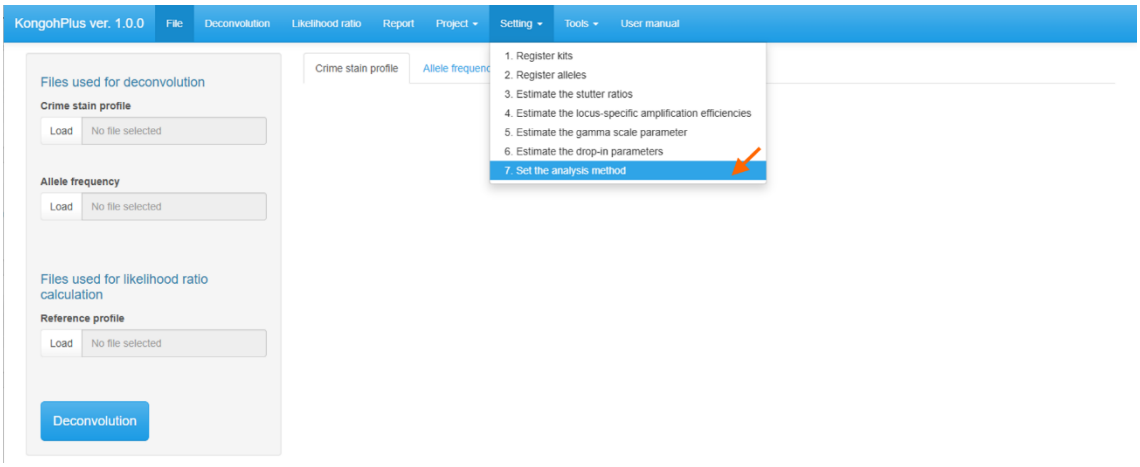
10. Enter a file name, and then click **”Save”**.



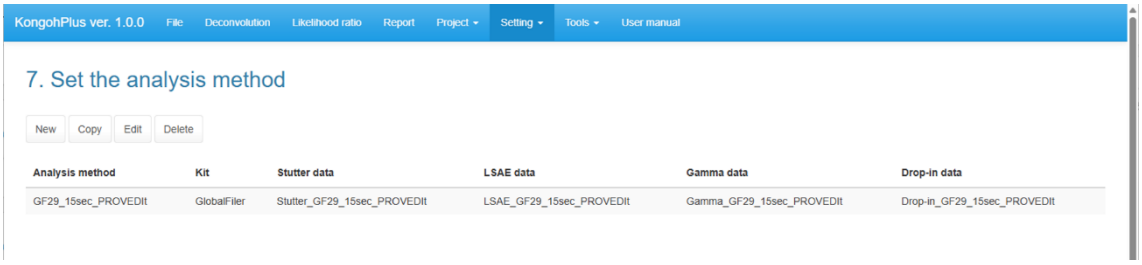
8.7. Setting the Analysis Method

This section describes how to configure an analysis method.

1. Click **“Setting” ▶ “7. Set the analysis method”**.



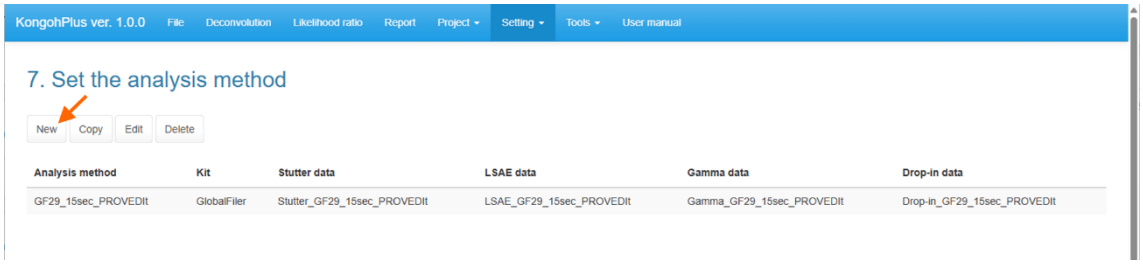
2. The following window is displayed.



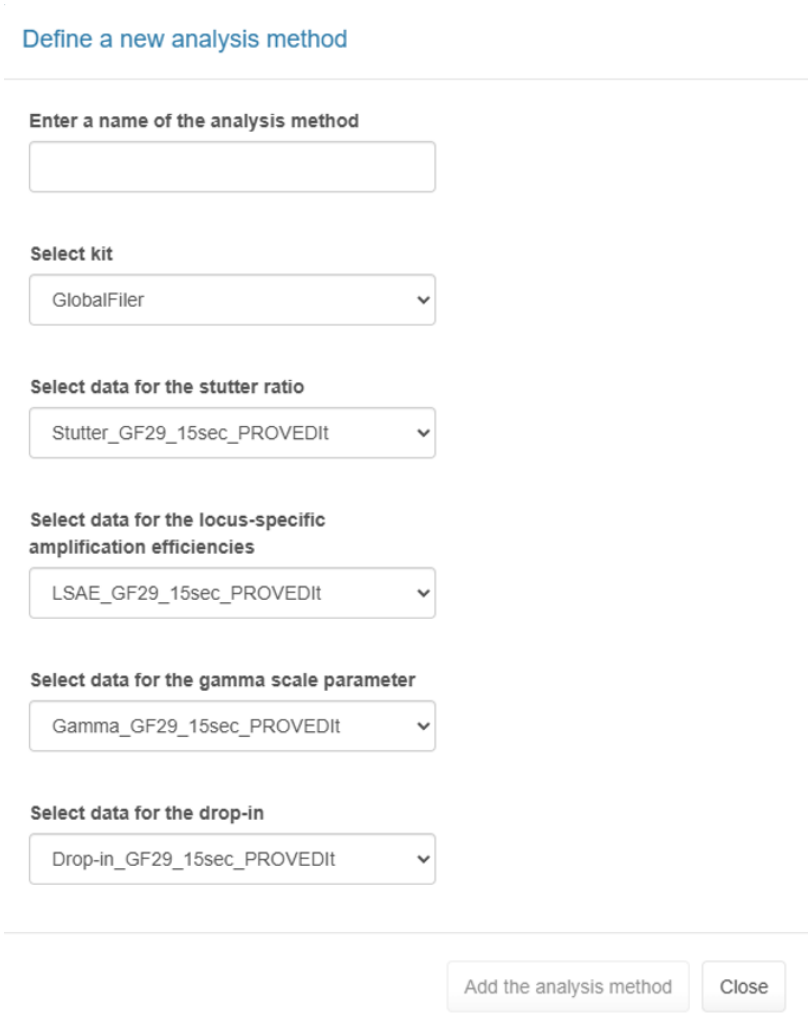
Note:

- **New:** Creates a new analysis method.
- **Copy:** Creates a copy of the selected analysis method.
- **Edit:** Edits the selected analysis method.
- **Delete:** Deletes the selected analysis method.

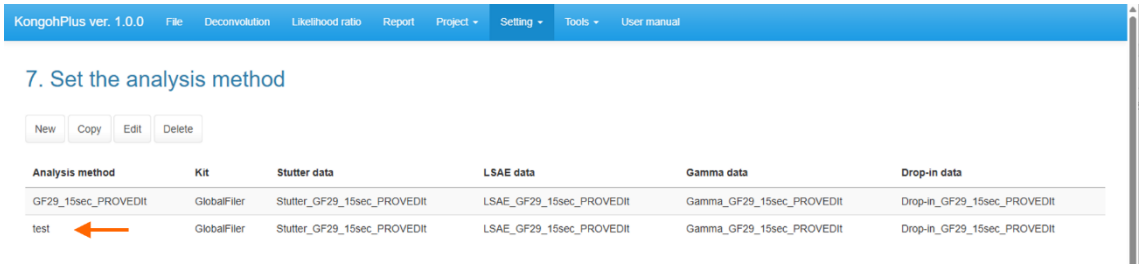
3. The following procedure describes how to create a new analysis method. Click **”New”**.



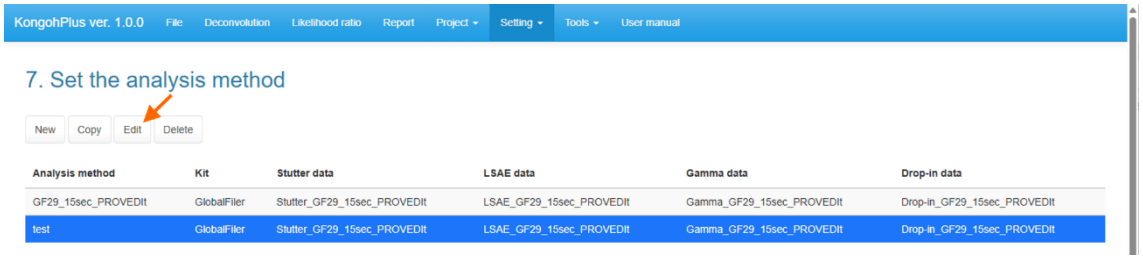
4. Configure the settings shown below, and then click **”Add the analysis method”**. For the third and subsequent items, select the parameter files estimated according to the procedures described in Sections 8.3 – 8.6.



5. The newly created analysis method is added to the list.

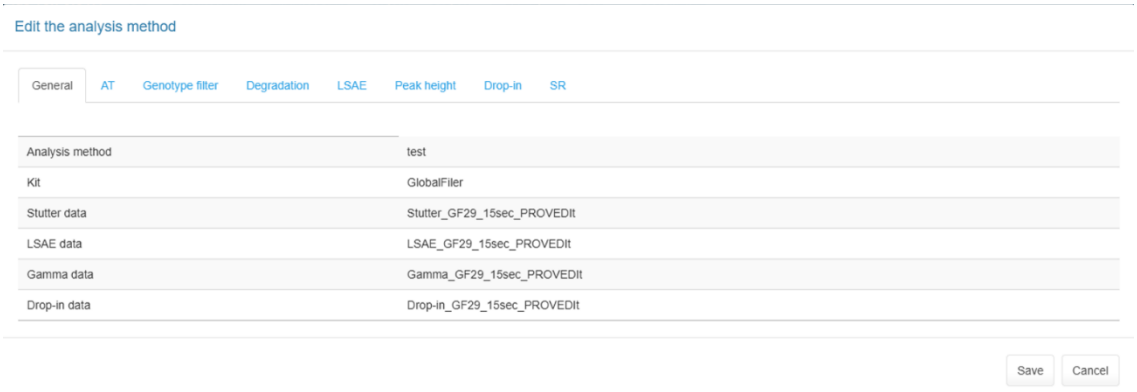


6. Select the newly created analysis method, and then click **"Edit"**.



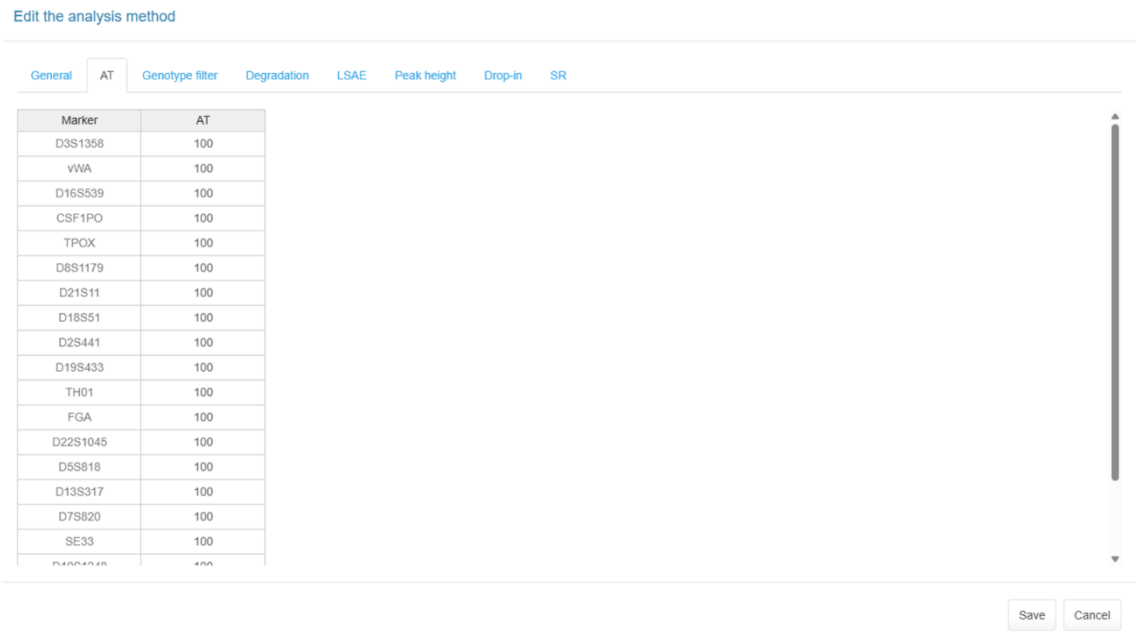
7. A configuration dialog is displayed. The following explains each tab.

General



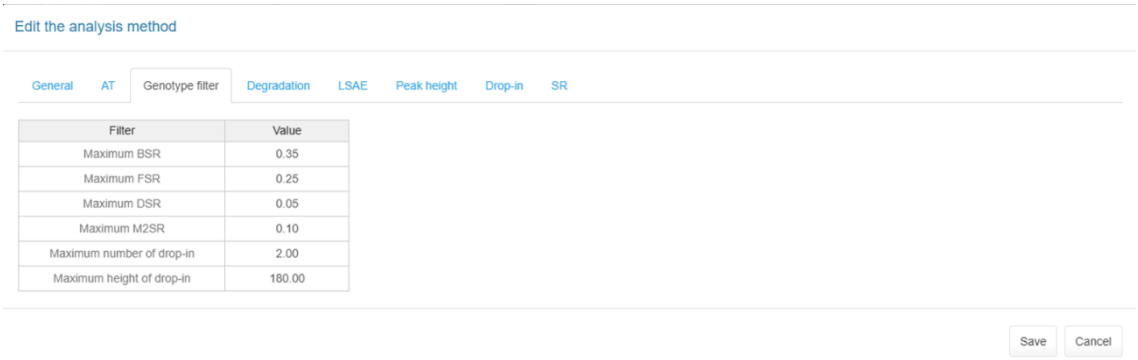
Displays the name of the analysis method, the DNA typing kit information, and the names of the parameter datasets estimated from the experimental data.

AT



Sets the analytical threshold (AT) for each marker.

Genotype filter

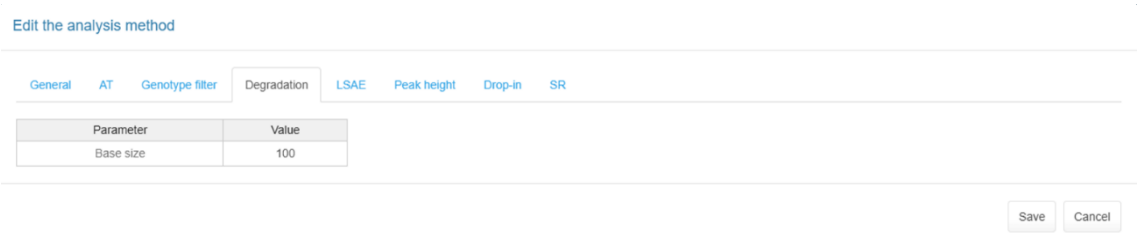


Configures the criteria used to exclude unrealistic genotypes from the analysis.

- **Maximum SR:** Excludes genotypes from the analysis if the specified stutter ratio (SR) exceeds the threshold. This setting can be configured separately for the back stutter ratio (BSR), forward stutter ratio (FSR), double-back stutter ratio (DSR), and minus 2-nt stutter ratio (M2SR).
- **Maximum number of drop-in:** Specifies the maximum allowable number of drop-in events per locus.

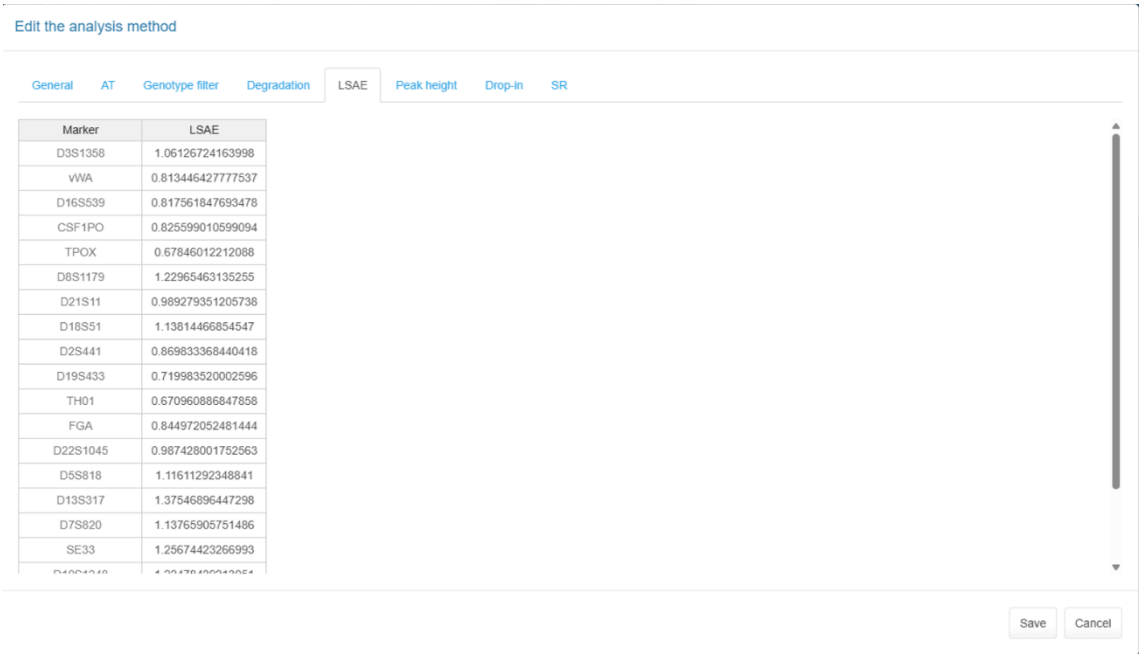
- **Maximum height of drop-in:** Specifies the maximum peak height considered for drop-in events. Peaks exceeding this value are not considered potential drop-in events.

Degradation



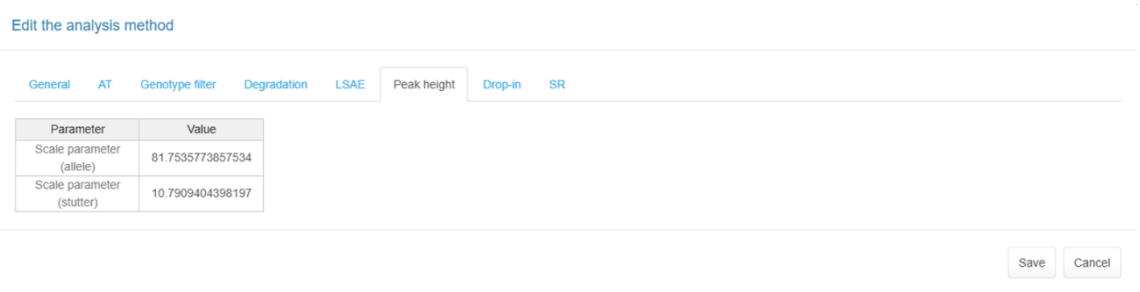
Sets the reference fragment size ($\hat{f}$). For details, see Section 10.3.1.

LSAE



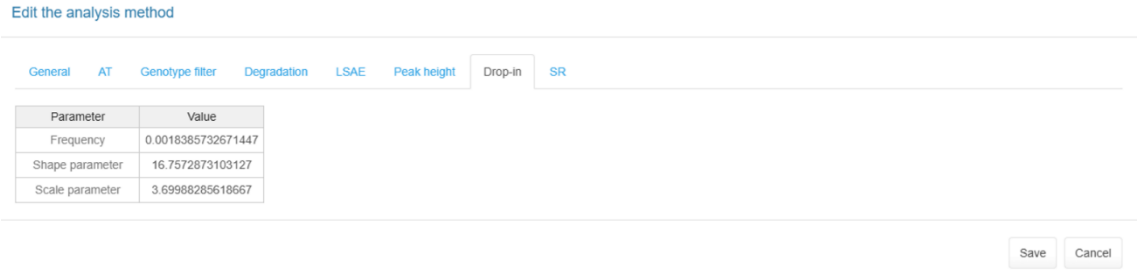
Sets the locus-specific amplification efficiencies. By default, the values estimated according to the procedure described in Section 8.4 are used. These values can also be modified by the user.

Peak height



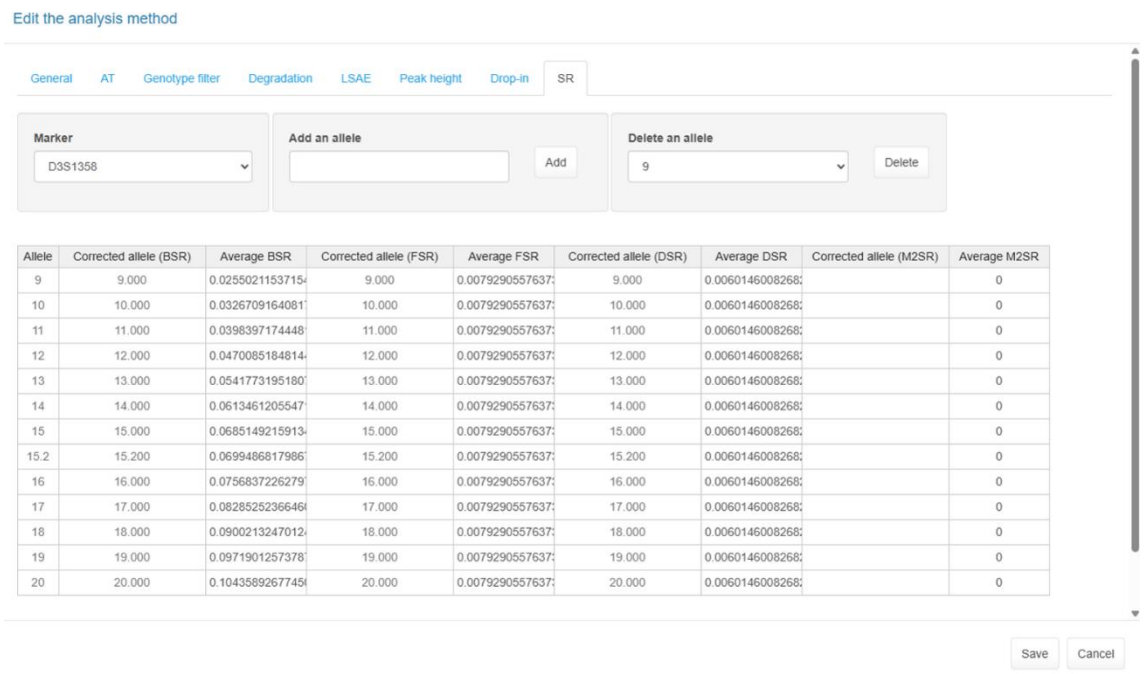
Configures the parameters of the gamma distributions used to model peak heights. By default, the values estimated according to the procedure described in Section 8.5 are used. These values can also be modified by the user.

Drop-in



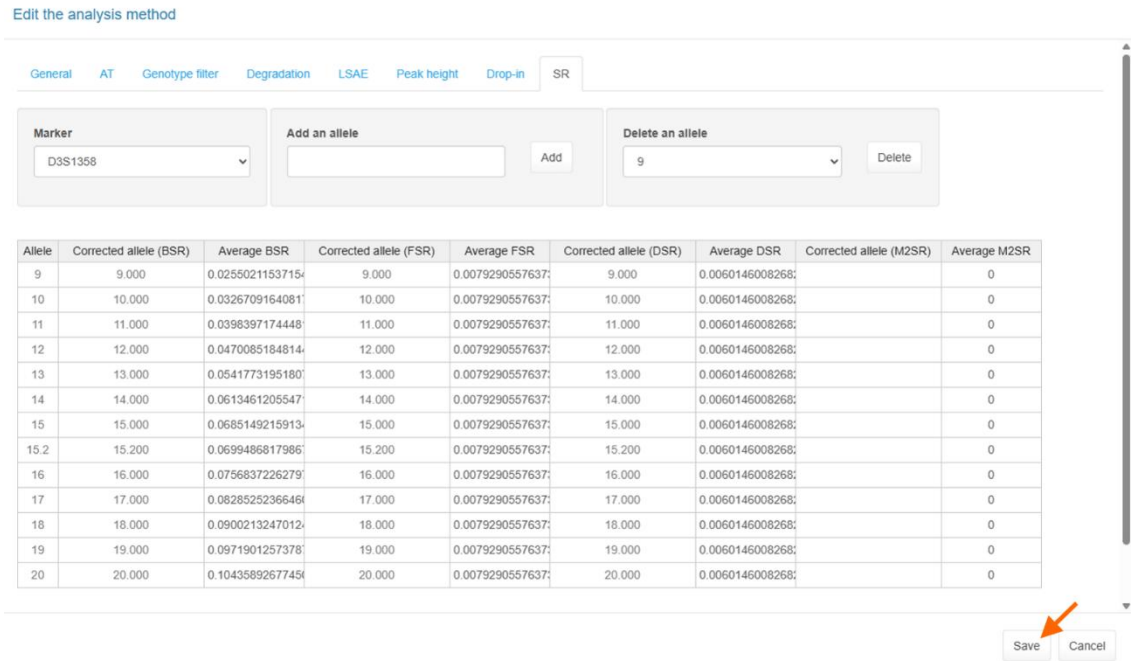
Sets the parameters related to drop-in events. By default, the values estimated according to the procedure described in Section 8.6 are used. These values can also be modified by the user.

SR



Sets the allele-specific stutter ratios for each locus. By default, the values estimated according to the procedure described in Section 8.3 are used. These values can also be modified by the user. New alleles can also be added.

8. After all settings have been configured, click **"Save"**.

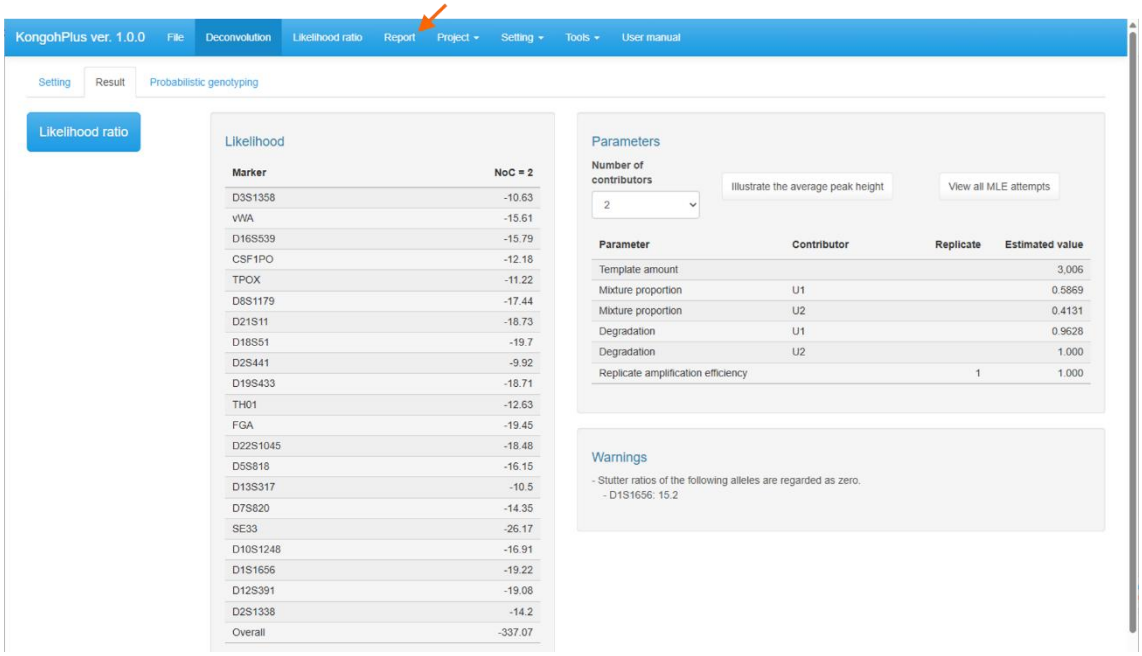


9. Other Functions

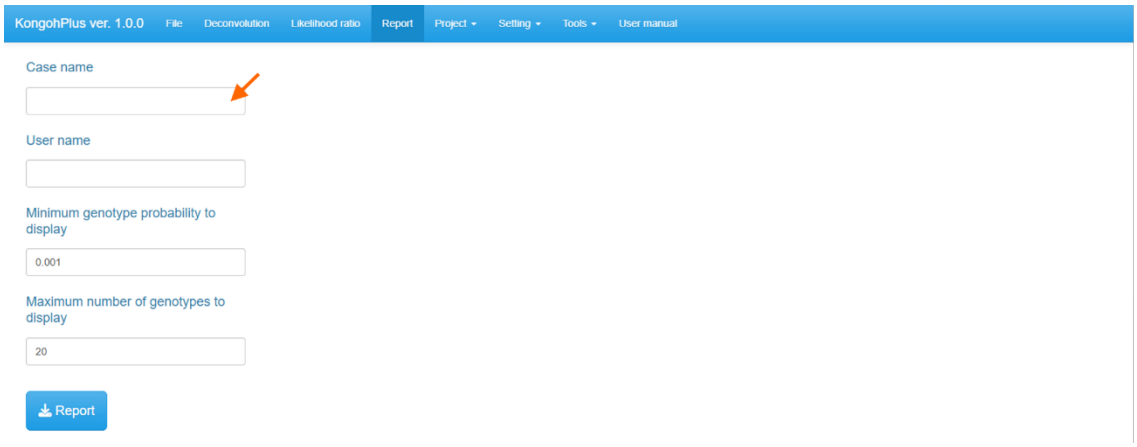
9.1. Report Output

This section describes the procedure for generating a report of the analysis results.

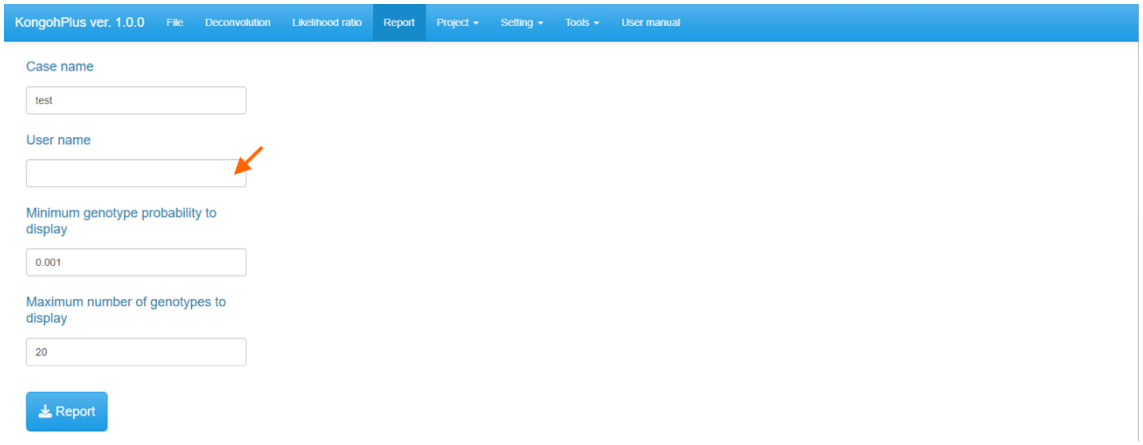
1. Click the **“Report”** tab.



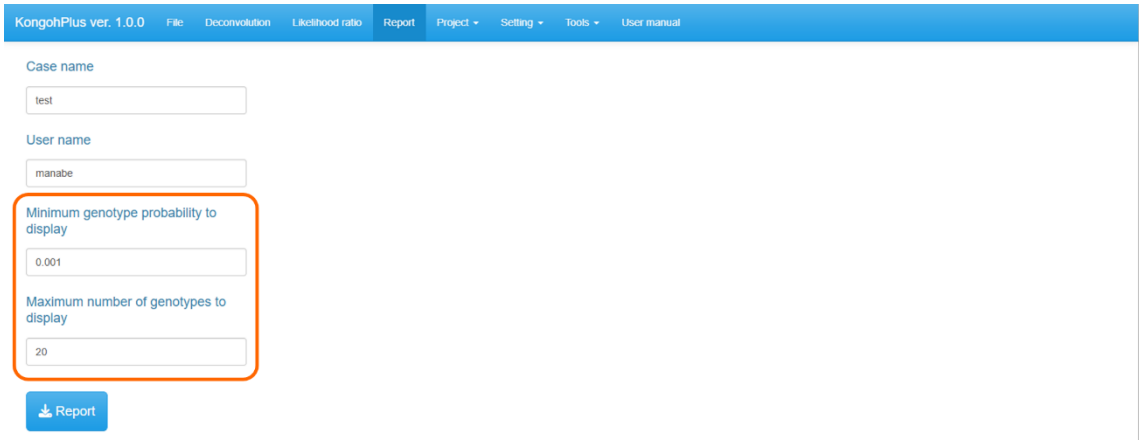
2. Enter the **“Case name”**.



3. Enter the **“User name”**.



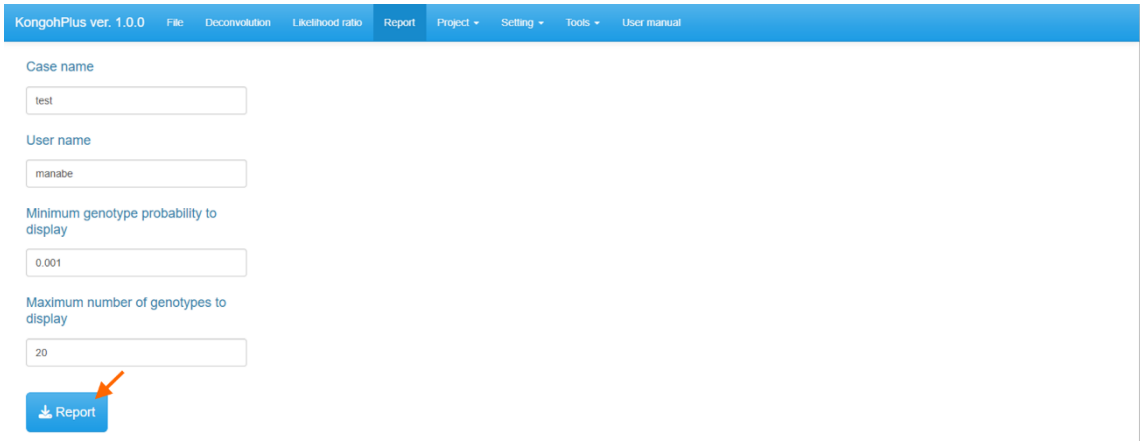
4. Configure the settings for the genotype candidates to be included in the report.



Note:

- **Minimum genotype probability to display:** Specifies the minimum genotype probability to be included in the report. Only genotypes with probabilities greater than or equal to this threshold are included.
- **Maximum number of genotypes to display:** Specifies the maximum number of genotype candidates to be included in the report.

5. Click **“Report”**.

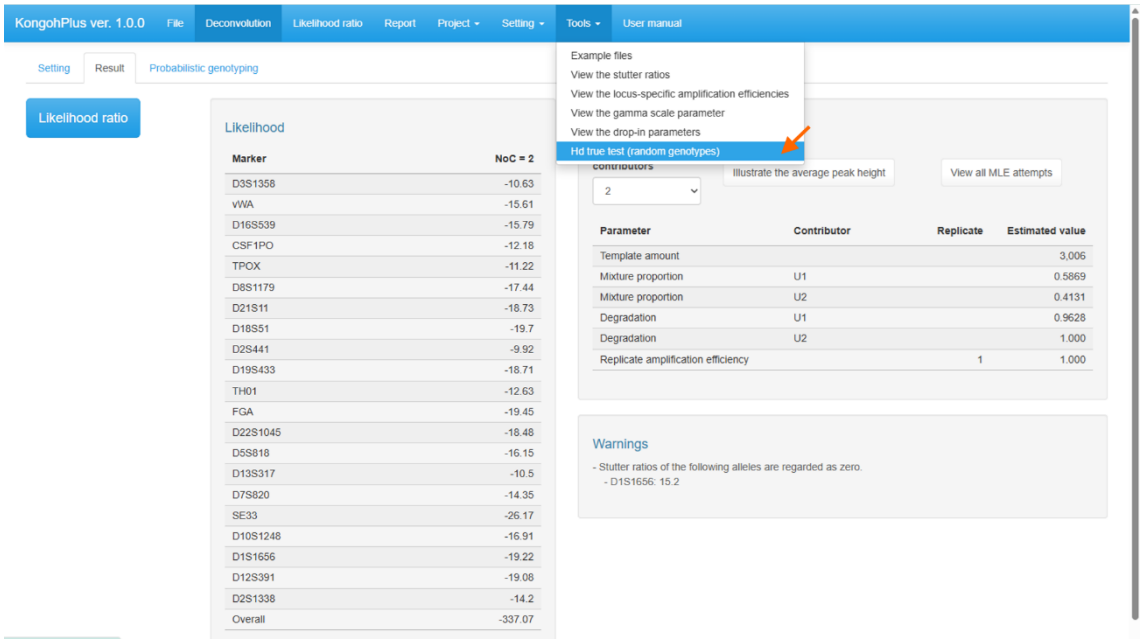


9.2. Hd True Test

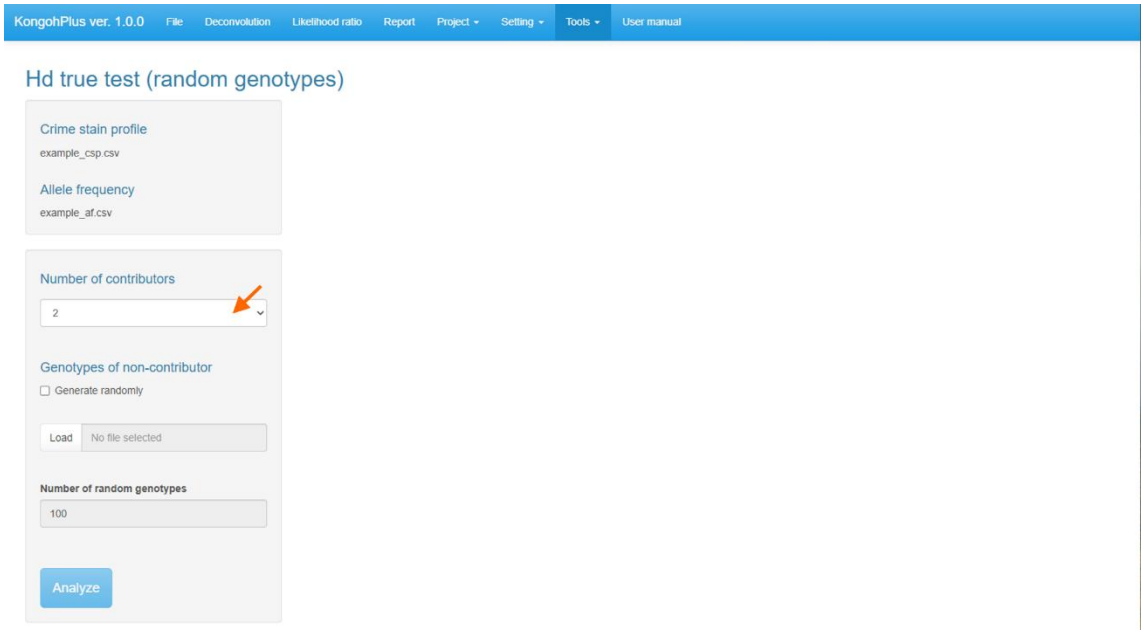
This section describes the Hd true test (random genotypes) function, which simulates the distribution of likelihood ratios using the genotypes of individuals who did not contribute DNA to the crime stain profile. This corresponds to the scenario in which the defense hypothesis (Hd) is true—that is, the person of interest (POI) is not an actual contributor to the crime stain profile.

1. Perform deconvolution of the crime stain profile. For details, see Section 3.

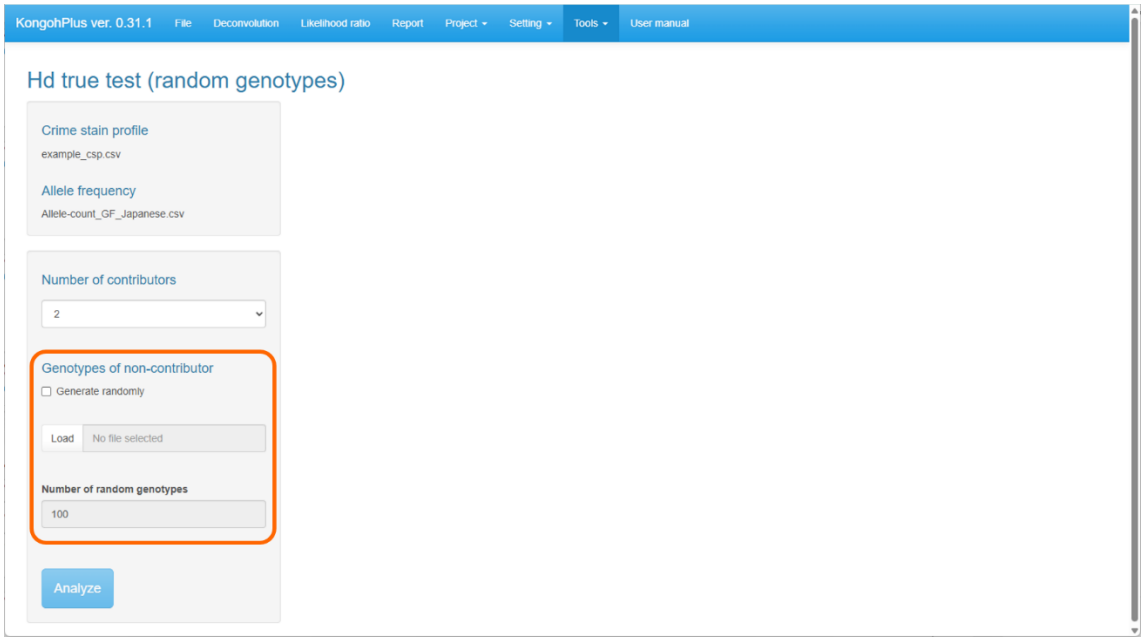
2. Click **“Tools” ▶ “Hd true test (random genotypes)”**.



3. Select the assumed number of contributors.



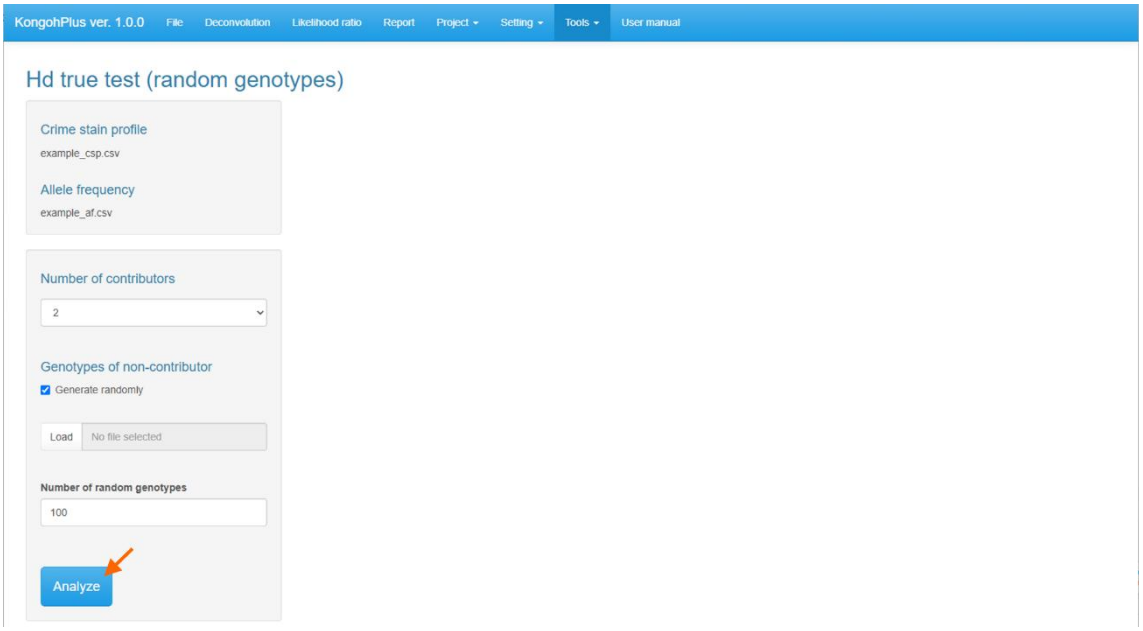
4. Configure the settings for the non-contributor genotypes.



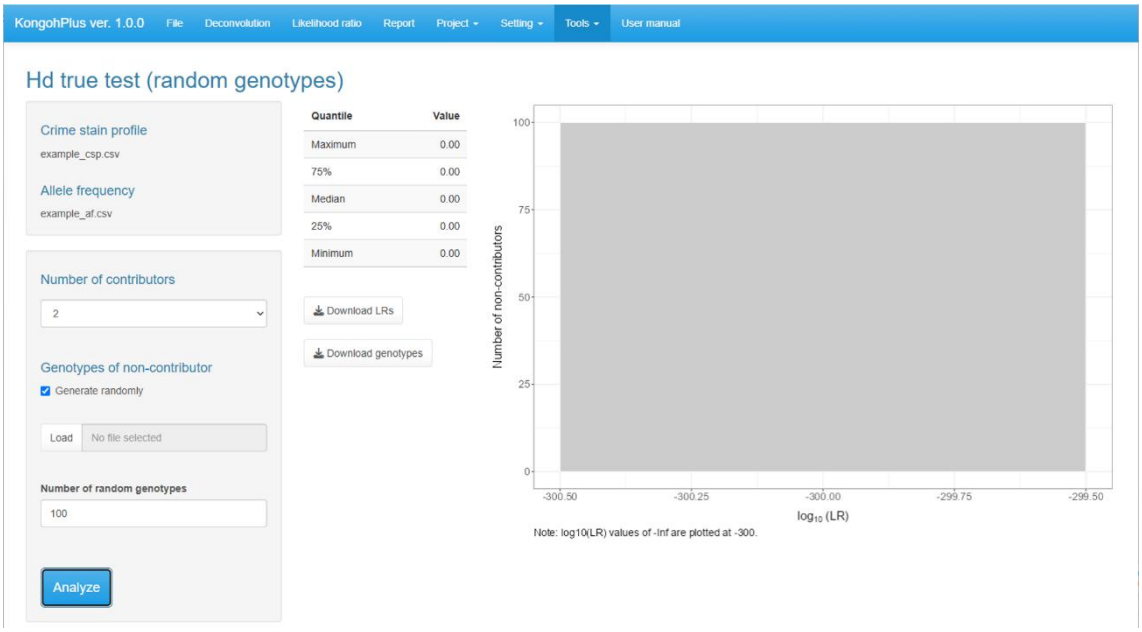
Note:

- If **“Generate randomly”** is selected, non-contributor genotypes are generated randomly according to the loaded allele frequency data. When using this option, specify the number of genotypes to generate in **”Number of random genotypes”**.
- Alternatively, you can use a set of non-contributor genotypes prepared by the user. In this case, clear the **“Generate randomly”** checkbox, click **”Load”**, and load the genotype file. The required file format is identical to that of the **reference profile** described in **Section** 4.3.

5. Click **“Analyze”**.



6. After the analysis is complete, summary statistics and a histogram of the likelihood ratios are displayed. The likelihood ratio results and the genotypes used in the analysis can be downloaded.



10. Calculation Principle

The analysis performed by KongohPlus consists of two steps: deconvolution and likelihood ratio (LR) calculation (Figure 1).

In the deconvolution step, KongohPlus estimates the parameters that best explain the observed DNA profile, such as mixture proportions and DNA degradation. Based on these estimated parameters, probabilistic weights are assigned to all possible genotype combinations that could explain the observed data. Before performing deconvolution, the forensic scientist must specify the number of contributors to the DNA profile.

In the second step, KongohPlus calculates the likelihood ratio (LR) based on the deconvolution results and the two competing hypotheses specified by the forensic scientist. The LR quantitatively measures the extent to which the observed DNA profile supports one hypothesis over the other.

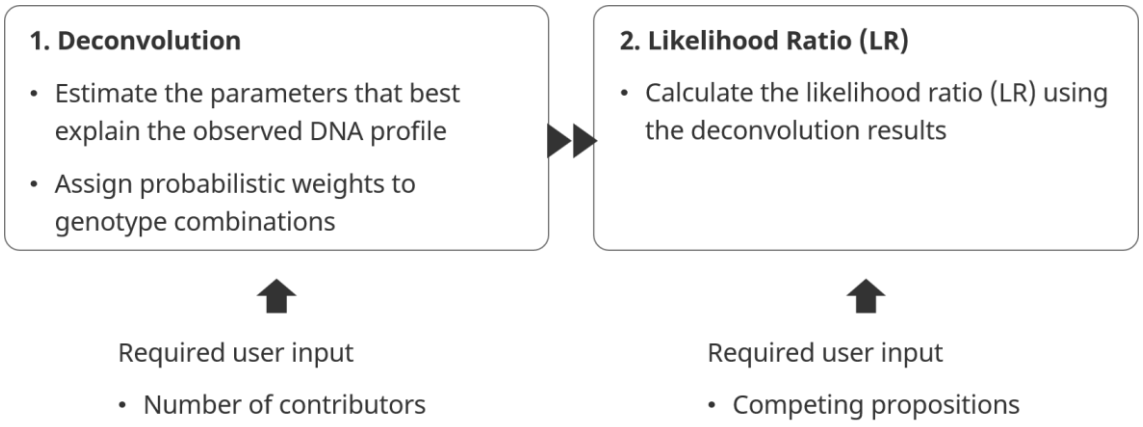


Figure 1. Analysis workflow in KongohPlus

10.1. Deconvolution

The objective of deconvolution is to estimate the parameters that best explain the observed crime stain profile under the assumed number of unknown

contributors. Following the general framework of fully continuous model described by Taylor et al. [3], the likelihood of the observed CSP, conditional on the model parameters, is expressed as:

$$f(CSP \mid H, M) = \prod_{l} \sum_{i} f\left(CSP_l \mid G_{l,i}, M\right) \Pr\left(G_{l,i} \mid H\right) \quad (1)$$

CSP_l : crime stain profile (observed peak heights) at locus l

$G_{l,i}$: the i-th genotype combination at locus l

M : parameter set

H : hypothesis (the assumed number of unknown contributors)

In KongohPlus, the parameter set M consists of all parameters required to model the observed peak heights. These parameters are classified into two categories:

- **Case specific parameters**, which vary from profile to profile (e.g., DNA quantity and mixture proportions).
- **Laboratory-specific parameters**, which are determined experimentally in advance (e.g., locus-specific amplification efficiencies and stutter ratios) (see Section 10.7 for details).

During deconvolution, the laboratory-specific parameters are treated as fixed constants, whereas the case-specific parameters are optimized by repeatedly calculating the likelihood. The case-specific parameter set that best explains the observed crime stain profile is selected as the maximum likelihood estimate.

The deconvolution procedure consists of the following steps:

1. Enumerate all possible genotype combinations $G_{l,i}$ $(i = 1, 2, \ldots, I)$.
2. Calculate the probability of each genotype combination under the hypothesis $(\Pr\left(G_{l,i} \mid H\right))$.

3. Evaluate how well each genotype combination explains the observed peak heights ($f(CSP_l \mid G_{l,i}, M)$).
4. Calculate the weighted sum of these values.
5. Multiply the likelihoods across all loci.

These calculations are repeated while varying the case-specific parameters, and the parameter set that maximizes the likelihood is selected as the final estimate.

10.2. Genotype Probability Calculation

This section describes the calculation of $\Pr(G_{l,i} \mid H)$ in Equation (1).

In KongohPlus, all possible genotype combinations are enumerated at each locus, and the probability of each combination under the specified proposition (i.e., the assumed number of unknown contributors) is calculated. These probabilities are derived from the genotype probabilities of each contributor. When appropriate, corrections for population substructure are applied using the Balding–Nichols model [4].

10.2.1. Genotype combination

In principle, KongohPlus performs the analysis by enumerating all possible genotype combinations at each locus. To reduce computational burden, however, users can define criteria for excluding implausible genotype combinations from the analysis. The available criteria are as follows:

- **Maximum BSR**: Excludes genotype combinations when the required back stutter ratio (BSR) exceeds the specified threshold.
- **Maximum FSR**: Excludes genotype combinations when the required forward stutter ratio (FSR) exceeds the specified threshold.
- **Maximum DSR**: Excludes genotype combinations when the required double-back stutter ratio (DSR) exceeds the specified threshold.

- **Maximum M2SR**: Excludes genotype combinations when the required minus 2-nt stutter ratio (M2SR) exceeds the specified threshold.
- **Maximum number of drop-in**: Specifies the maximum allowable number of drop-in peaks per locus. Genotype combinations requiring more drop-in peaks than the specified limit are excluded from the analysis.
- **Maximum height of drop-in**: Specifies the maximum peak height that may be explained as a drop-in event. Peaks exceeding the specified threshold are not considered as possible drop-in peaks.

10.2.2. Allele Frequencies

Genotype probability calculations require allele frequencies.

In KongohPlus, allele frequencies are treated as random variables following a Dirichlet distribution. A uniform prior is assumed, and the posterior mean of the Dirichlet distribution is used as the estimated allele frequency.

The frequency of allele a (p_a) is calculated as follows [5].

$$p_a = \frac{n_a + 1}{\sum_{a=1}^{k+1}(n_a + 1)} \quad (2)$$

where n_a is the number of observations of allele a in the population database, and k is the number of distinct alleles observed in the database.

The Dirichlet distribution is defined with $k+1$ categories, where the additional category represents unobserved alleles. In the conventional Dirichlet model, the dimensionality of the distribution is equal to the number of observed alleles (k). In KongohPlus, however, an additional category representing unobserved alleles is always included. This prevents the estimated frequencies of known alleles from changing depending on whether unobserved alleles are present in a particular analysis. Consequently, the dimensionality of the Dirichlet distribution is fixed at $k+1$ when estimating allele frequencies.

Because unobserved alleles are generally rare, considering a single unobserved-allele category is sufficient for most analyses. If two or more unobserved alleles are involved in a case, KongohPlus automatically increases the dimensionality of the Dirichlet distribution accordingly. As a result, the estimated frequencies of all alleles, including the observed alleles, differ slightly from those obtained in the standard analysis.

10.2.3. Treatment of Drop-out Alleles

When allelic drop-out occurs, the probabilities of unobserved alleles must be taken into account.

For example, suppose that only allele 11 is observed at a locus. In this case, the true genotype may be not only 11, 11 but also 11, Q, where Q represents any allele other than allele 11.

The allele frequency of Q is therefore given by the sum of the frequencies of all alleles other than allele 11: $p_Q = 1 - p_{11}$.

Assuming Hardy–Weinberg equilibrium, the probability of genotype 11, Q is given by $2p_{11}(1 - p_{11})$.

Similarly, when multiple alleles are observed at a locus, Q represents any allele other than the observed alleles. Its allele frequency is calculated by subtracting the sum of the observed allele frequencies from one.

10.3. Statistical Model for Peak Heights

This section describes the calculation of $f(CSP_l \mid G_{l,i}, M)$ in Equation (1).

The basic concept underlying the calculation of $f(CSP_l \mid G_{l,i}, M)$ is to compare the observed peak heights with the expected peak heights under a given set of conditions.

For example, consider the DNA profile shown in Figure 2, where the peaks corresponding to alleles 9 and 10 are relatively high, whereas those corresponding

to alleles 11 and 12 are relatively low.

Suppose that Scenario 1 assumes the major contributor has genotype 9,10 and the minor contributor has genotype 11,12. Under this assumption, the expected peak heights of alleles 9 and 10 are relatively high, whereas those of alleles 11 and 12 are relatively low. Because these expected peak heights are consistent with the observed peak heights, the genotype combination "9,10 + 11,12" receives a relatively large probabilistic weight.

Now consider Condition 2, in which the major contributor is assumed to have genotype 9,11 and the minor contributor is assumed to have genotype 10,12. In this case, the expected peak heights of alleles 9 and 11 are relatively high, whereas those of alleles 10 and 12 are relatively low. Because these expected peak heights differ substantially from the observed peak heights, the genotype combination "9,11 + 10,12" receives a relatively small probabilistic weight.

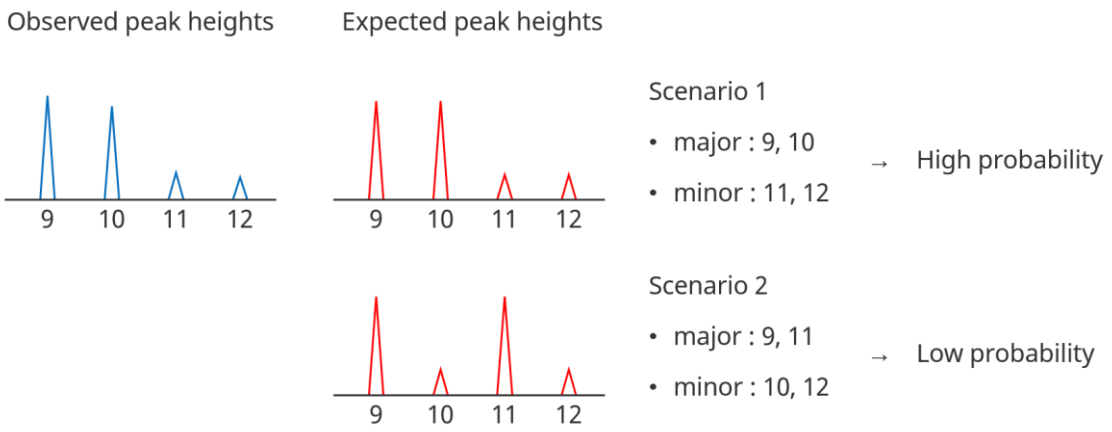


Figure 2. Conceptual illustration of peak-height evaluation

10.3.1. Expected Peak Heights

The expected peak height of each allelic and stutter product is calculated from the corresponding model parameters. The expected peak height of a heterozygous allele ($E_{l,r,a,c}$) is given by

$$E_{l,r,a,c} = \frac{TM_c D_c^{(f_a - \hat{f})/100} R_r \hat{A}_l}{1 + \hat{S}_{l,a}^{(\text{back})} + \hat{S}_{l,a}^{(\text{forward})} + \hat{S}_{l,a}^{(\text{double-back})} + \hat{S}_{l,a}^{(\text{minus 2-nt})}} \quad (3)$$

where $E_{l,r,a,c}$ represents the expected peak height of allele a originating from contributor c at locus l in the r-th PCR replicate. The parameters appearing in Equation (3) are classified into two categories: case-specific parameters, whose values vary from profile to profile, and laboratory-specific parameters, whose values are determined separately for each laboratory

case-specific parameter

T : Average peak height for a heterozygous allele (corresponding to the DNA quantity)

M_c : Mixture proportion of contributor c

D_c : Degree of DNA degradation of contributor c

R_r : Relative peak height of the r-th PCR replicate, where the first PCR replicate is defined as 1

laboratory-specific parameter

$\hat{A}_l$: PCR amplification efficiency at locus l (relative to the overall average, which is defined as 1)

$\hat{S}_{l,a}^{(\text{back})}$: Back stutter ratio for allele a at locus l

$\hat{S}_{l,a}^{(\text{forward})}$: Forward stutter ratio for allele a at locus l

$\hat{S}_{l,a}^{(\text{double-back})}$: Double-back stutter ratio for allele a at locus l

$\hat{S}_{l,a}^{(\text{minus 2-nt})}$: Minus 2-nt stutter ratio for allele a at locus l

The case-specific parameters are estimated by KongohPlus using maximum likelihood estimation for each DNA profile. In contrast, the laboratory-specific parameters are determined experimentally before analyzing casework samples.

Let f_a denote the fragment size of allele a, and let $\hat{f}$ denote the offset of the fragment size (100 bases by default). These variables are used to model the reduction in peak height caused by DNA degradation. The term $D_c^{(f_a-\hat{f})/100}$ indicates that the expected peak height at fragment size f_a is multiplied by $D_c^{(f_a-\hat{f})/100}$ relative to the expected peak height at the reference fragment size $\hat{f}$. Dividing the exponent by 100 allows the parameter D_c to be interpreted as the fold change in peak height for every 100-base increase in fragment size [6].

When multiple allelic products or multiple stutter products overlap at the same allele position, their expected peak heights are summed, and the total is regarded as the expected peak height at that position.

10.3.2. Probability Density of the Observed Peak Height

In Figure 2, the concept of calculating $f(CSP_l \mid G_{l,i}, M)$ in Equation (1) was introduced by comparing the expected peak height with the observed peak height. This section explains how this comparison is performed statistically.

Even if the expected peak height of a particular allele is calculated to be 1000 RFU, the observed peak height will not always be exactly 1000 RFU. Because PCR amplification is inherently stochastic, repeated amplification of the same DNA sample produces slightly different peak heights.

Therefore, rather than evaluating whether the expected and observed peak heights are identical, KongohPlus evaluates how likely the observed peak height is under the expected peak height.

Figure 3 conceptually illustrates this idea. If the observed peak height is close to the expected value, for example 1020 RFU, its probability density is relatively high, indicating that such an observation is likely under the assumed conditions. In contrast, if the observed peak height is far from the expected value, for example 300 RFU or 2500 RFU, its probability density is relatively low, indicating that such an

observation is unlikely under the assumed conditions. Thus, KongohPlus evaluates each observed peak height not by the difference between the observed and expected peak heights themselves, but by the probability density associated with that difference.

The probability density of an observed peak height is denoted by $f(O_{l,r,a} \mid G_{l,i}, M)$. Specifically, $f(O_{l,r,a} \mid G_{l,i}, M)$ represents the probability density of observing a peak height $O_{l,r,a}$ at the position of allele a in the r-th PCR replicate at locus l, given the genotype combination ($G_{l,i}$) and parameter set (M). It corresponds to the probability density in Figure 3.

The crime stain profile (CSP_l) at locus l, however, consists of multiple peak heights rather than a single observation. In practice, multiple allele positions exist within each locus, and when replicate PCR amplifications are performed, peak heights are observed for each allele position in each PCR replicate.

Therefore, the probability density of the complete set of observed peak heights at locus l is calculated as the product of the probability densities for all allele positions across all PCR replicates. Accordingly, $f(CSP_l \mid G_{l,i}, M)$ in Equation (1) is expressed as follows.

$$f(CSP_l \mid G_{l,i}, M) = \prod_r \prod_a f(O_{l,r,a} \mid G_{l,i}, M) \quad (4)$$

The following section describes how each $f(O_{l,r,a} \mid G_{l,i}, M)$ is calculated.

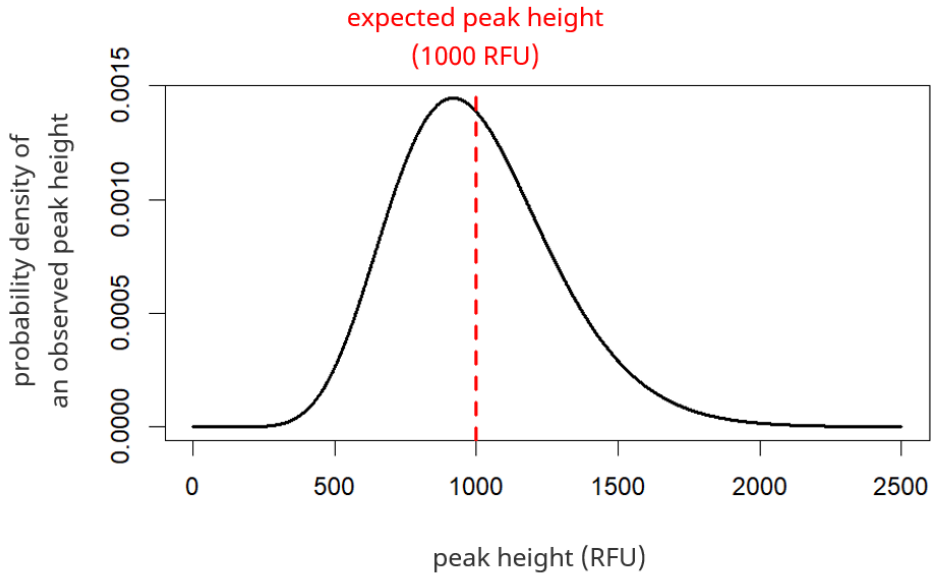


Figure 3 Conceptual illustration of the probability density of observed peak heights

10.3.3. Gamma Distribution

In KongohPlus, the probability density described in Figure 3 is modeled using a gamma distribution. This gamma distribution represents the variability of observed peak heights around a given expected peak height. By calculating the probability density of an observed peak height under the gamma distribution, KongohPlus quantitatively evaluates how likely the observed peak height is.

If the peak at allele a originates solely from the allelic product of contributor c, the probability density of the observed peak height is modeled by the following gamma distribution.

$$f\left(O_{l,r,a} \mid G_{l,i}, M\right) = \mathrm{Gamma}\left(\frac{E_{l,r,a,c}}{\hat{\eta}^{(\mathrm{allele})}}, \hat{\eta}^{(\mathrm{allele})}\right) \tag{5}$$

$E_{l,r,a,c}$: Expected peak height of the allelic product

$\hat{\eta}^{(\mathrm{allele})}$: Scale parameter of the gamma distribution for allelic products

Similarly, if the peak at the position of allele $a-1$ originates solely from the back stutter product generated from allele a of contributor c, the probability density of the observed peak height is modeled by the following gamma distribution.

$$f\left(O_{l,r,a-1} \mid G_{l,i}, M\right) = \mathrm{Gamma}\left(\frac{E_{l,r,a-1,c}}{\hat{\eta}^{(\mathrm{stutter})}}, \hat{\eta}^{(\mathrm{stutter})}\right) \quad (6)$$

$E_{l,r,a-1,c}$: Expected peak height of the back stutter product

$\hat{\eta}^{(\mathrm{stutter})}$: Scale parameter of the gamma distribution for all stutter products

The gamma-distribution scale parameters ($\hat{\eta}^{(\mathrm{allele})}$ and $\hat{\eta}^{(\mathrm{stutter})}$) are laboratory-specific parameters that are determined experimentally before casework analysis.

Equations (5) and (6) describe situations in which a peak consists of only a single allelic product or a single stutter product. In practice, however, multiple products may contribute to the same peak position. Such situations can be classified into the following three categories.

1. Multiple allelic products
   Example: allele 10 from contributor 1 and allele 10 from contributor 2.
2. Multiple stutter products
   Example: the forward stutter of allele 10 from contributor 1 and the back stutter of allele 12 from contributor 2.
3. Both allelic and stutter products
   Example: allele 10 from contributor 1 and the back stutter of allele 11 from contributor 2.

For the first two categories, all contributing products share the same scale parameter. Therefore, according to the reproductive property of the gamma distribution, the sum of these products also follows a gamma distribution. For

example, the sum of the allelic products originating from contributor 1 ($c = 1$) and contributor 2 ($c = 2$) is given by

$$f\left(O_{l,r,a} \mid G_{l,i}, M\right) = \mathrm{Gamma}\left(\frac{E_{l,r,a,1} + E_{l,r,a,2}}{\hat{\eta}^{(\mathrm{allele})}}, \hat{\eta}^{(\mathrm{allele})}\right) \tag{7}$$

In contrast, when an allelic product and a stutter product overlap, the two products have different scale parameters. Consequently, their sum does not strictly follow a gamma distribution.

To simplify the statistical model, KongohPlus approximates the distribution of the summed peak height by a single gamma distribution. Specifically, the mean and variance of the summed peak height are first calculated, and then a gamma distribution having the same mean and variance is determined. This approach is known as moment matching.

Suppose that one random variable follows $\mathrm{Gamma}(k_1, \theta_1)$ and another follows $\mathrm{Gamma}(k_2, \theta_2)$, and that their sum is approximated by $\mathrm{Gamma}(k_0, \theta_0)$.

The parameters k_0 and θ_0 are determined so that the mean and variance of the approximating gamma distribution are equal to those of the original sum. Because the mean and variance of independent random variables are additive,

$E(X_0) = E(X_1) + E(X_2)$

$Var(X_0) = Var(X_1) + Var(X_2)$

On the other hand, the mean and variance of a gamma distribution $\mathrm{Gamma}(k, \theta)$ are given by $E(X) = k\theta$ and $Var(X) = k\theta^2$, respectively. Therefore,

$k_0\theta_0 = k_1\theta_1 + k_2\theta_2$

$k_0\theta_0^2 = k_1\theta_1^2 + k_2\theta_2^2$

Solving these equations yields

$$k_0 = \frac{(k_1\theta_1 + k_2\theta_2)^2}{k_1\theta_1^2 + k_2\theta_2^2}$$

$$\theta_0 = \frac{k_1\theta_1^2 + k_2\theta_2^2}{k_1\theta_1 + k_2\theta_2}$$

Accordingly, the sum of the two gamma-distributed random variables can be approximated by the following gamma distribution.

$$\mathrm{Gamma}(k_0, \theta_0) = \mathrm{Gamma}\left(\frac{(k_1\theta_1 + k_2\theta_2)^2}{k_1\theta_1^2 + k_2\theta_2^2}, \frac{k_1\theta_1^2 + k_2\theta_2^2}{k_1\theta_1 + k_2\theta_2}\right) \quad (8)$$

This approximation is applied to the peak-height model in KongohPlus. For example, suppose that the allele a of contributor 1 ($c = 1$) and back stutter generated from allele a' of contributor 2 ($c = 2$) overlap at allele position a ($a = a' - 1$). The corresponding gamma distributions for the two products are as follows.

$$\mathrm{Gamma}(k_1, \theta_1) = \mathrm{Gamma}\left(\frac{E_{l,r,a,1}}{\hat{\eta}^{(\mathrm{allele})}}, \hat{\eta}^{(\mathrm{allele})}\right)$$

$$\mathrm{Gamma}(k_2, \theta_2) = \mathrm{Gamma}\left(\frac{E_{l,r,a'-1,2}}{\eta^{(\mathrm{stutter})}}, \eta^{(\mathrm{stutter})}\right)$$

Therefore,

$$k_1\theta_1 + k_2\theta_2 = E_{l,r,a,1} + E_{l,r,a'-1,2}$$

$$k_1\theta_1^2 + k_2\theta_2^2 = E_{l,r,a,1}\eta^{(allele)} + E_{l,r,a'-1,2}\,\eta^{(stutter)}$$

Substituting these values into Equation (8) yields a gamma distribution that approximates the distribution of the summed peak height.

10.3.4. Statistical Model for Drop-out

When allele a has dropped out, no observed peak height ($O_{l,r,a}$) is available. Therefore, the probability density of the observed peak height cannot be evaluated using the gamma distributions described in the previous sections.

Instead, KongohPlus assumes that the true peak height of allele a is below the analytical threshold (AT_l) at locus l. Accordingly, the probability of observing a drop-out allele is calculated as the probability that the peak height is less than the analytical threshold:

$$\Pr\left(O_{l,r,a} < AT_l \mid G_{l,i} M\right) = \int_0^{AT_l} \Pr\left(O_{l,r,a} = x \mid G_{l,i}, M\right) dx \tag{9}$$

For drop-out alleles, unobserved alleles are treated collectively as Q, as described in Section 10.2.3. When calculating the expected peak height of allele a using Equation (3), the fragment size (f_a) is assumed to be the median fragment size of all alleles registered for that locus.

10.3.5. Statistical Model for Drop-in

When allele a is considered to originate from a drop-in event, its expected peak height cannot be calculated using the method described in Section 10.3.1. Therefore, a separate statistical model is required to describe drop-in peaks.

In KongohPlus, the probability density that a drop-in event occurs at allele a and produces a peak with height $O_{l,r,a}$ is given by

$$f\left(O_{l,r,a}\right) = f\left(O_{l,r,a} \mid C\right) \Pr(C) = \text{Gamma}\left(\hat{\xi}^{(\text{dropin})}, \hat{\eta}^{(\text{dropin})}\right) \times \frac{\hat{F}^{(\text{dropin})}}{P^{(\text{dropin} \geq AT_l)}} \tag{10}$$

$\Pr(C)$: Probability of a drop-in event

$\hat{\xi}^{(\text{dropin})}$: Shape parameter of the gamma distribution for drop-in peak heights

$\hat{\eta}^{(\text{dropin})}$: Scale parameter of the gamma distribution for drop-in peak heights

$\hat{F}^{(\text{dropin})}$: Frequency of drop-in events observed in experimental data

$P^{(\text{dropin} \geq AT_l)}$: Probability that a drop-in peak height is greater than or equal to the analytical threshold at locus l

Here, $\text{Gamma}\left(\hat{\xi}^{(\text{dropin})}, \hat{\eta}^{(\text{dropin})}\right)$ represents the distribution of peak heights

generated by drop-in events. The probability of a drop-in event ($\Pr(C)$) is not simply defined as the observed frequency of drop-in events ($\hat{F}^{(\mathrm{dropin})}$); rather, it is adjusted to account for the probability that a drop-in peak exceeds the analytical threshold and is therefore detectable.

The parameters $\hat{\xi}^{(\mathrm{dropin})}, \hat{\eta}^{(\mathrm{dropin})}$, $\hat{F}^{(\mathrm{dropin})}$ are laboratory-specific parameters determined experimentally before casework analysis. Their estimation is described in Section 10.7.4.

10.4. Maximum Likelihood Estimation

10.4.1. Overview of Maximum Likelihood Estimation

In deconvolution, the objective of maximum likelihood estimation is to determine the case-specific parameter set that maximizes the likelihood function in Equation (1). KongohPlus uses the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) algorithm to perform this optimization [7]. CMA-ES is a stochastic optimization algorithm designed for continuous parameters. It searches for the optimal parameter set by generating multiple candidate solutions and iteratively adapting the search distribution toward regions with higher likelihood.

10.4.2. Multi-start Optimization

Because the likelihood function in Equation (1) is highly complex, it may contain multiple local optima. Consequently, maximum likelihood estimation starting from a single initial parameter set may fail to identify the global optimum.

To reduce this risk, KongohPlus employs a multi-start optimization strategy, in which maximum likelihood estimation is performed repeatedly using different initial parameter sets. Each optimization run is performed independently, and the maximum likelihood estimate is obtained for each run.

10.4.3. Evaluation of Consistency

The optimization results are regarded as consistent if a sufficiently large number of optimization runs produce likelihood values that are nearly identical to the

maximum likelihood.

Specifically, a result is considered consistent when its likelihood differs from the maximum likelihood by no more than a predefined threshold. The consistency of the optimization is then evaluated based on the number of such results.

If this criterion is not satisfied, KongohPlus displays a warning indicating that the optimization may have converged to local optima.

10.4.4. Settings

The settings related to maximum likelihood estimation are as follows.

- **Maximum number of iterations**
  Maximum number of iterations performed in a single optimization run.
- **Maximum number of MLE attempts**
  Number of independent optimization runs performed in the multi-start optimization.
- **Minimum number of consistent runs**
  Minimum number of optimization runs required for the result to be regarded as consistent.
- **Reproducibility threshold**
  Maximum allowable difference from the highest likelihood (log10 scale) for a run to be regarded as consistent.

These settings allow users to balance computation time and the stability of the optimization results. In addition, the upper and lower bounds of each parameter can be specified.

10.5. Probabilistic Genotyping

This section describes how probabilistic weights are assigned to genotype combinations at each locus.

Once the case-specific parameters have been estimated by maximum likelihood

estimation during deconvolution, the value of $f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H)$ can be calculated for each genotype combination. This value represents the extent to which a genotype combination explains the observed peaks and their heights.

However, the sum of $f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H)$ over all genotype combinations at a locus is not equal to one and therefore cannot be interpreted as a probability. KongohPlus therefore normalizes these values as follows.

$$\Pr(G_{l,i} \mid CSP_l, M, H) = \frac{f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H)}{\sum_i f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H)} \quad (11)$$

The normalized value, $\Pr(G_{l,i} \mid CSP_l, M, H)$ represents the probabilistic weight assigned to each genotype combination.

Table 1 provides an example. Because the values of $f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H)$ are on an arbitrary scale, their sum is not equal to one. After normalization, however, the values are converted to a scale whose total is one, allowing them to be interpreted as probabilistic weights.

Table 1 An example of probabilistic weights assigned to genotype combinations

major contributor	minor contributor	$f(CSP_l \mid G_{l,i}, M)$	$\Pr(G_{l,i} \mid H)$	$\Pr(G_{l,i} \mid CSP_l, M, H)$
9, 10	11, 12	4.50×10^{-10}	$2p_9p_{10} \times 2p_{11}p_{12}$	0.9
9, 11	10, 12	2.00×10^{-11}	$2p_9p_{11} \times 2p_{10}p_{12}$	0.04
9, 12	10, 11	1.50×10^{-11}	$2p_9p_{12} \times 2p_{10}p_{11}$	0.03
10, 11	9, 12	1.00×10^{-11}	$2p_{10}p_{11} \times 2p_9p_{12}$	0.02
10, 12	9, 11	4.95×10^{-12}	$2p_{10}p_{12} \times 2p_9p_{11}$	0.0099
11, 12	9, 10	5.00×10^{-14}	$2p_{11}p_{12} \times 2p_9p_{10}$	0.0001

10.6. Likelihood Ratio

Once the case-specific parameters have been estimated through deconvolution, the likelihood under each hypothesis can be calculated. This section describes how the likelihood ratio (LR) is calculated using these likelihoods.

The likelihood ratio (LR) is a measure of the extent to which the observed DNA profile supports one hypothesis over another. Based on the likelihood ratio framework described by Taylor et al. [3], KongohPlus calculates the LR using the parameter estimates obtained during deconvolution as follows:

$$\mathrm{LR} = \frac{f(CSP \mid H_1, M)}{f(CSP \mid H_2, M)} = \frac{\prod_l \sum_i f\left(CSP_l \mid G_{l,i}, M\right)\Pr\left(G_{l,i} \mid H_1\right)}{\prod_l \sum_i f\left(CSP_l \mid G_{l,i}, M\right)\Pr\left(G_{l,i} \mid H_2\right)} \tag{12}$$

The likelihoods in both the numerator and the denominator are calculated using the likelihood function given in Equation (1). However, KongohPlus uses the same case-specific parameter set for both hypotheses. In other words, the peak-height model estimated from the observed crime stain profile is assumed to be identical regardless of the hypothesis being evaluated. In contrast, the genotype probabilities $\Pr\left(G_{l,i} \mid H\right)$ are calculated separately for each hypothesis and therefore differ between the numerator and the denominator.

10.6.1. Example 1: No Known Contributors

Suppose that deconvolution has been performed on a two-person DNA mixture, yielding the results shown in Table 1. When the issue regards whether the DNA is from the suspect (with genotype 9, 10) or not, the competing propositions could be:

H_1 : The DNA originated from the suspect and someone else, unrelated to the suspect.

H_2 : The DNA originated from two individuals, unrelated to the suspect.

If the suspect is assigned as the major contributor or the minor contributor under H_1, the expected peak heights differ, resulting in different likelihoods. Therefore,

KongohPlus evaluates both possible contributor positions and the larger likelihood is used for H_1.

When the suspect is assigned as the major contributor, only the first genotype combination in Table 1 is consistent with the suspect's genotype (9,10). Under H_1, the suspect's genotype is fixed, and only the probability that the unknown individual has genotype 11,12 needs to be considered. Assuming Hardy–Weinberg equilibrium, this probability is $2p_{11}p_{12}$. The likelihood is therefore given by

$$f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H_1) = 4.50 \times 10^{-10} \times 2p_{11}p_{12} \tag{13}$$

When the suspect is assigned as the minor contributor, only the sixth genotype combination in Table 1 is consistent with the suspect's genotype. Again, only the probability that the unknown individual has genotype 11,12 needs to be considered, giving the following likelihood.

$$f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H_1) = 5.00 \times 10^{-14} \times 2p_{11}p_{12} \tag{14}$$

Comparing Equations (13) and (14), Equation (13) gives the larger likelihood and is therefore adopted as the likelihood under H_1.

In contrast, under H_2 the genotypes of both contributors are unknown. Therefore, the likelihood is calculated by summing over all possible genotype combinations.

$$\begin{aligned} & f(CSP_l \mid G_{l,i}, M)\Pr(G_{l,i} \mid H_2) \\ & = 4.50 \times 10^{-10} \times 2p_9p_{10} \times 2p_{11}p_{12} \\ & +2.00 \times 10^{-11} \times 2p_9p_{11} \times 2p_{10}p_{12} \\ & +1.50 \times 10^{-11} \times 2p_9p_{12} \times 2p_{10}p_{11} \\ & +1.00 \times 10^{-11} \times 2p_{10}p_{11} \times 2p_9p_{12} \\ & +4.95 \times 10^{-12} \times 2p_{10}p_{12} \times 2p_9p_{11} \\ & +5.00 \times 10^{-14} \times 2p_{11}p_{12} \times 2p_9p_{10} \end{aligned} \tag{15}$$

$= 2.00 \times 10^{-9} \times p_9 p_{10} p_{11} p_{12}$

Finally, the likelihood ratio is calculated as the ratio of Equation (13) to Equation (15).

$$\mathrm{LR} = \frac{4.50 \times 10^{-10} \times 2p_{11}p_{12}}{2.00 \times 10^{-9} \times p_9 p_{10} p_{11} p_{12}} = \frac{0.45}{p_9 p_{10}} \quad (16)$$

10.6.2. Example 2: With an Assumed Contributor

Next, consider a case in which part of the DNA profile is known to originate from the victim. For example, when the evidence consists of vaginal swabs collected from the victim, it is reasonable to assume that the victim is one of the contributors. Such a contributor is generally referred to as an assumed contributor.

Suppose that the victim has genotype 11,12 and the suspect has genotype 9,10, and that the issue regards whether the DNA is from the suspect or not, the competing propositions could be:

H_1 : The DNA originated from the victim and the suspect.

H_2 : The DNA originated from the victim and someone else, unrelated to the victim and the suspect.

When a contributor is common to both H_1 and H_2, KongohPlus first determines whether the observed DNA profile is better explained by assigning that contributor as the major contributor or the minor contributor. In this example, the likelihood is calculated under both assumptions, and the assignment giving the higher likelihood is adopted.

When the victim is assumed to be the major contributor, the likelihood is denoted by $5.00 \times 10^{-14} \times 2p_9 p_{10}$, whereas when the victim is assumed to be the minor contributor, the likelihood is denoted by $4.50 \times 10^{-10} \times 2p_9 p_{10}$. Because $4.50 \times 10^{-10} \times 2p_9 p_{10}$ is larger, the victim is treated as the minor contributor when calculating the likelihoods under both H_1 and H_2.

The likelihood ratio is therefore calculated as follows.

$$\mathrm{LR} = \frac{4.50 \times 10^{-10}}{4.50 \times 10^{-10} \times 2p_9p_{10}} = \frac{1}{2p_9p_{10}} \tag{17}$$

10.7. Estimation of Laboratory-Specific Parameters

This section describes how to estimate the laboratory-specific parameters that are determined experimentally before casework analysis.

To estimate the stutter ratios, locus-specific PCR amplification efficiencies, and the scale parameters of the gamma distributions, a sufficiently large number of single-source DNA profiles are required. Because the stutter ratios are estimated separately for each allele, it is desirable to use profiles obtained from many different individuals. In addition, because the scale parameters of the gamma distributions depend on the variability of peak heights, the experimental data should include profiles covering a wide range of PCR template DNA quantities.

To estimate the parameters related to drop-in, a sufficiently large number of non-template controls are required.

10.7.1. Stutter Ratios

This section defines the stutter ratio relative to the corresponding allele peak while accounting for differences in allele repeat structures, and describes how the average stutter ratio is estimated from experimental data.

KongohPlus considers four types of stutter ratios: the back stutter ratio ($\hat{S}_{l,a}^{(\mathrm{back})}$), forward stutter ratio ($\hat{S}_{l,a}^{(\mathrm{forward})}$), double-back stutter ratio ($\hat{S}_{l,a}^{(\mathrm{double-back})}$), and minus 2-nt stutter ratio ($\hat{S}_{l,a}^{(\mathrm{minus\ 2-nt})}$). Unless otherwise stated, these are collectively referred to as the stutter ratio ($S_{l,a}$). Although the stutter ratio varies from profile to profile, the objective is to estimate the average stutter ratio ($\hat{S}_{l,a}$) for each allele at each locus.

Statistical Model for Variation in Stutter Ratios

KongohPlus assumes that the stutter ratio follows the log-normal distribution given below.

$$S_{l,a} \sim \mathrm{LN}\left(\mu_{l,a}, \frac{\hat{\sigma}_l^2}{h_{l,a}}\right) \tag{18}$$

$\mu_{l,a}$: Mean of the logarithm of the stutter ratio for allele a at locus l

$h_{l,a}$: Total peak height of the allelic product and stutter products for allele a at locus l

σ_l^2 : Variance of the logarithm of the stutter ratio when $h_{l,a} = 1$

The variance of the logarithm of the stutter ratio is represented by $\hat{\sigma}_l^2/h_{l,a}$, indicating that the variability increases because of stochastic effects as $h_{l,a}$ decreases. This variance function is estimated from experimental data.

Statistical Model for the Relationship Between Alleles and Stutter Ratios

In general, the stutter ratio tends to increase as the allele number increases. However, when the repeat structure is complex, the stutter ratio is not always correlated with the allele number. In such cases, it is preferable to use a corrected allele value (e.g., the longest uninterrupted stretch: LUS [8]). In contrast, the expected minus 2-nt stutter ratio is nearly constant regardless of the allele number [9]. Furthermore, for some loci, the numbers of observed forward stutter and double-back stutter peaks are insufficient to estimate locus-specific models. In such cases, data from multiple loci are combined to estimate a common model.

Based on these considerations, KongohPlus implements the following five models describing the relationship between allele characteristics and the stutter ratio.

Model	Description
Linear (Allele)	The stutter ratio increases as the allele number increases.
Linear (Corrected Allele)	The stutter ratio increases as the corrected allele value increases.
Uniform	The stutter ratio is constant regardless of the allele.
Uniform (Multiple loci)	The stutter ratio is constant regardless of the allele, with data combined across multiple loci.
Not consider	The stutter ratio is not considered.

For the Linear (Allele) model, the relationship between the allele number and the mean of the logarithm of the stutter ratio is expressed as follows.

$$\mu_{l,a} = \log_e\left(\hat{\beta}_{0,l} + \hat{\beta}_{1,l} \times a\right) \tag{19}$$

where $\hat{\beta}_{0,l}$ and $\hat{\beta}_{1,l}$ denote the intercept and slope, respectively. Both parameters are estimated from experimental data.

When the estimated mean of the logarithm of the stutter ratio is less than or equal to zero ($\hat{\beta}_{0,l} + \hat{\beta}_{1,l} \times a \leq 0$), the logarithm is undefined. In such cases, KongohPlus sets the average stutter ratio to zero, corresponding to the assumption that no stutter peak is generated from that allele.

For the Uniform and Uniform (Multiple loci) models, the stutter ratio is assumed to be independent of the allele. Therefore, no slope or intercept is used, and the mean of the logarithm of the stutter ratio is estimated directly from the experimental data.

For the Linear (Corrected Allele) model, users must specify a corrected value for each allele. KongohPlus provides default corrected allele values for loci included in the GlobalFiler kit. The corrected values are based on the LUS model [8] for D8S1179,

D21S11, D2S441, TH01, FGA, D1S1656, D12S391, and D2S1338, and on the Multi-sequence model [10] for SE33. The following sections describe these models used to determine the default corrected allele values in KongohPlus.

LUS model

LUS is an abbreviation for longest uninterrupted stretch, which refers to the maximum number of consecutive repeat units within an STR sequence [8]. For many STR loci, the allele number corresponds to the repeat number. However, at some loci, the allele number does not necessarily correspond to the repeat number because the repeat region consists of multiple repeat motifs or other complex sequence structures. For such loci, the LUS provides a better predictor of the stutter ratio than the allele number.

When the allele number is corrected using the LUS, the mean of the logarithm of the stutter ratio is expressed as

$$\mu_{l,a} = \log_e\left(\hat{\beta}_{0,l} + \hat{\beta}_{1,l} \times wLUS_{l,a}\right) \tag{20}$$

where $wLUS_{l,a}$ denotes the weighted average LUS value for allele a at locus l, calculated as

$$wLUS_{l,a} = \frac{\sum_{i=1}^{N} n_i LUS_i}{\sum_{i=1}^{N} n_i} \tag{21}$$

where n_i は、the number of observations of the i-th sequence for the allele in a given population, and LUS_i is the LUS value of the i-th sequence.

For example, the NIST population database consisting of 1,036 individuals [11] contains the following sequence data for allele 15 at the D3S1358 locus.

Sequence	n_i	LUS_i
TCTA [TCTG]2 [TCTA]12	429	12
TCTA TCTG [TCTA]13	173	13
TCTA [TCTG]3 [TCTA]11	27	11
TCTA [TCTG]2 [TCTA]8 TCTG [TCTA]3	1	8
TCTA [TCTG]3 [TCTA]5 TCTG [TCTA]5	1	5

Accordingly, the weighted average LUS value for allele 15 at D3S1358 is calculated as

$$wLUS_{l,a} = \frac{429 \cdot 12 + 173 \cdot 13 + 27 \cdot 11 + 1 \cdot 8 + 1 \cdot 5}{429 + 173 + 27 + 1 + 1} \approx 12.2 \quad (22)$$

Multi-sequence model

For complex STR loci such as SE33, which contain multiple repeat motifs within an allele sequence, the LUS alone does not adequately explain the observed stutter ratios. To address this limitation, the Multi-sequence model was developed [10]. In this model, the corrected allele value is defined as the sum of the repeat numbers of all repeat motifs whose repeat numbers exceed a specified threshold.

When the allele number is corrected using the Multi-sequence model, the mean of the logarithm of the stutter ratio is expressed as follows.

$$\mu_{l,a} = \log_e\left(\hat{\beta}_{0,l} + \hat{\beta}_{1,l} \times wCA_{l,a}\right) \quad (23)$$

where $wCA_{l,a}$ denotes the weighted average corrected allele value for allele a at locus l, which is calculated as

$$wCA_{l,a} = \frac{\sum_{i=1}^{N} n_i CA_i}{\sum_{i=1}^{N} n_i} \tag{24}$$

Here, n_i is the number of observations of the i-th sequence for the allele in a given population, and CA_i is the corrected allele value for that sequence.

The corrected allele value (CA_i) is calculated as

$$CA_i = \sum_j \max(US_{i,j} - x_l, 0) \tag{25}$$

where $US_{i,j}$ is the repeat number of the j-th repeat motif in the i-th sequence, and x_l is the minimum repeat number required for a repeat motif to contribute to the corrected allele value.

For example, the following sequence data were obtained for allele 22 at SE33 from the NIST population database consisting of 1,036 individuals [12]. In this example, the corrected allele value (CA_i) was calculated using $x_l = 7$.

Sequence	n_i	CA_i $(x_l = 7)$
CT [CTTT]3 C [CTTT]22 CT [CTTT]3 CT [CTTT]2	24	15
CT [CTTT]3 CCTT C [CTTT]21 CT [CTTT]3 CT [CTTT]2	4	14
CT [CTTT]2 CCTT C [CTTT]12 TT [CTTT]10 CT [CTTT]5	1	8
CT [CTTT]2 C [CTTT]23 CT [CTTT]3 CT [CTTT]2	1	16

In this case, the weighted average corrected allele value for allele 22 at SE33 is calculated as follows.

$$wCA_{l,a} = \frac{24 \cdot 15 + 4 \cdot 14 + 1 \cdot 8 + 1 \cdot 16}{24 + 4 + 1 + 1} \approx 14.66667 \tag{26}$$

Parameter Estimation from Experimental Data

This section describes how the average stutter ratio ($\hat{S}_{l,a}$) for each allele at each locus is estimated from experimental data based on the statistical models described above.

For each experimentally obtained DNA profile, the observed stutter ratio at allele a of locus l (denoted by $S_{l,a}^{(\text{back,obs})}$, $S_{l,a}^{(\text{forward,obs})}$, $S_{l,a}^{(\text{double-back,obs})}$, and$S_{l,a}^{(\text{minus 2-nt,obs})}$) to indicate that it is an observed value) is calculated as follows.

$$S_{l,a}^{(\text{back,obs})} = \frac{O_{l,a-1}}{O_{l,a}}$$

$$S_{l,a}^{(\text{forward,obs})} = \frac{O_{l,a+1}}{O_{l,a}}$$

$$S_{l,a}^{(\text{double-back,obs})} = \frac{O_{l,a-2}}{O_{l,a}}$$

$$S_{l,a}^{(\text{minus 2-nt,obs})} = \frac{O_{l,a-0.2(\text{or } 0.8)}}{O_{l,a}}$$

$O_{l,a}$: Observed peak height of the allele a at locus l

$O_{l,a-1}$: Observed peak height of the back stutter derived from allele a at locus l

$O_{l,a+1}$: Observed peak height of the forward stutter derived from allele a at locus l

$O_{l,a-2}$: Observed peak height of the double-back stutter derived from allele a at locus l

$O_{l,a-0.2(\text{or } 0.8)}$: Observed peak height of the minus 2-nt stutter derived from allele a at locus l

The total peak height of the allelic product and the corresponding stutter product ($h_{l,a}^{(\text{obs})}$) is also calculated from the observed peak heights in the experimental data.

When an experimentally obtained DNA profile is heterozygous at a locus, the allelic product may overlap with a stutter product, or multiple stutter product s may overlap with one another. Such observations are excluded from the estimation of the average stutter ratio ($\hat{S}_{l,a}$). Specifically, only the stutter ratios marked with ✓ in the table below are used for parameter estimation.

$a' - a$ $a' - a$	allele	$S_{l,a}^{(\text{back,obs})}$	$S_{l,a}^{(\text{forward,obs})}$	$S_{l,a}^{(\text{double-back,obs})}$	$S_{l,a}^{(\text{minus 2-nt,obs})}$
1	a				
1	a'				
2	a				
2	a'		✓		✓
3	a	✓		✓	✓
3	a'	✓	✓		✓
0.2 or 0.8	a				
0.2 or 0.8	a'		✓	✓	
1.2 or 1.8	a	✓		✓	
1.2 or 1.8	a'	✓	✓		
2.2 or 2.8	a	✓	✓	✓	✓
2.2 or 2.8	a'	✓	✓	✓	✓
3 >	a	✓	✓	✓	✓
3 >	a'	✓	✓	✓	✓

Once the experimental data ($S_{l,a}^{(\text{back,obs})}$, $S_{l,a}^{(\text{forward,obs})}$, $S_{l,a}^{(\text{double-back,obs})}$, $S_{l,a}^{(\text{minus 2-nt,obs})}$, and $h_{l,a}^{(\text{obs})}$) have been obtained, the parameters of the statistical models are estimated by maximum likelihood estimation using the CMA-ES algorithm. Specifically, the following parameters are estimated:

- $\hat{\sigma}_l^2$ in Equation (18)
- $\hat{\beta}_{0,l}$ and $\hat{\beta}_{1,l}$ in Equations (19), (20), and (23)
- $\mu_{l,a}$ for the Uniform and Uniform (Multiple loci) models

The estimation procedure is the same as that used for deconvolution (see Section 10.4 for details).

After these parameters have been estimated, the average stutter ratio for each allele is calculated using Equation (19), (20), or (23) for the Linear (Allele) and Linear (Corrected Allele) models.

For the Uniform and Uniform (Multiple loci) models, the parameter $\mu_{l,a}$ estimated by the CMA-ES algorithm directly represents the average stutter ratio.

10.7.2. Locus-Specific Amplification Efficiency

This section describes how the locus-specific PCR amplification efficiencies are modeled using a Dirichlet distribution and how the average amplification efficiency for each locus is estimated from experimental data.

The PCR amplification efficiency at locus l (A_l) represents the amplification efficiency of that locus relative to the average amplification efficiency across all loci, which is defined as 1. Although the amplification efficiency varies from profile to profile, the objective of this section is to estimate the average amplification efficiency ($\hat{A}_l$) for each locus.

Statistical Model

In KongohPlus, the amplification efficiencies are first normalized so that their sum across all loci is equal to one. The normalized amplification efficiency (A_l^*) is

defined as

$$A_l^* = \frac{A_l}{\sum_{l=1}^{L} A_l} \tag{27}$$

This normalization is introduced because the amplification efficiencies are more conveniently modeled on a normalized scale when estimating the average amplification efficiency for each locus. KongohPlus assumes that the vector of normalized amplification efficiencies $(A_1^*, A_2^*, \ldots, A_L^*)$ follows a Dirichlet distribution.

$$(A_1^*, A_2^*, \ldots, A_L^*) \sim \mathrm{Dir}(\alpha_1, \alpha_2, \ldots, \alpha_L) \tag{28}$$

where $\alpha_1, \alpha_2, \ldots, \alpha_L$ are parameters estimated from experimental data.

Once α_l $(l = 1, 2, \ldots, L)$ has been estimated for each locus, the expected value of the normalized amplification efficiency is calculated as

$$\hat{A}_l^* = \frac{\alpha_l + 1}{\sum_{l=1}^{L}(\alpha_l + 1)} \tag{29}$$

This corresponds to the posterior mean under a uniform prior distribution $(\mathrm{Dir}(1, 1, \ldots, 1))$.

Finally, the average amplification efficiency for each locus $(\hat{A}_l)$ is estimated as

$$\hat{A}_l = \hat{A}_l^* \times L \tag{30}$$

Parameter Estimation from Experimental Data

For each experimentally obtained DNA profile, the normalized amplification efficiency $(\hat{A}_l^{*(\mathrm{obs})})$ is calculated as

$$\hat{A}_l^{*(\mathrm{obs})} = \frac{\sum_a O_{l,a}}{\sum_l \sum_a O_{l,a}} \tag{31}$$

where $O_{l,a}$ denotes the observed peak height at allele a of locus l.

After calculating $\hat{A}_l^{*(\mathrm{obs})}$ for all experimental DNA profiles, the parameter $\alpha_1, \alpha_2, \ldots, \alpha_L$ in Equation (28) is estimated by maximum likelihood estimation using the CMA-ES algorithm. The estimation procedure is the same as that used for deconvolution (see Section 10.4 for details).

Once α_l $(l = 1, 2, \ldots, L)$ has been estimated for each locus, the average amplification efficiency ($\hat{A}_l$) is obtained using Equations (29) and (30).

10.7.3. Scale Parameters of the Gamma Distributions

This section describes how the scale parameters of the gamma distributions are estimated from experimental data.

As described in Section 10.3.3, KongohPlus uses separate gamma distributions to model allelic products and stutter products. Accordingly, two scale parameters are estimated: one for allelic products ($\hat{\eta}^{(\mathrm{allele})}$) and one for stutter products ($\hat{\eta}^{(\mathrm{stutter})}$).

The estimation described below assumes that the experimental data satisfy the following conditions.

- Each DNA profile is obtained from a single-source sample.
- No DNA degradation is present.
- Each DNA profile is obtained from a single PCR amplification.

Under these assumptions, Equation (3) can be simplified as follows.

$$E_{l,a} = \frac{T\hat{A}_l}{1 + \hat{S}_{l,a}^{(\mathrm{back})} + \hat{S}_{l,a}^{(\mathrm{forward})} + \hat{S}_{l,a}^{(\mathrm{double-back})} + \hat{S}_{l,a}^{(\mathrm{minus\ 2-nt})}} \tag{32}$$

Except for T (the average peak height for a heterozygous allele), all parameters in Equation (32) are laboratory-specific parameters.

When the experimental DNA profiles satisfy the three conditions listed above, T can be calculated directly from the observed peak heights as follows.

$$T = \frac{1}{2L}\sum_{l=1}^{L}\sum_{a} O_{l,a} \tag{33}$$

That is, T is calculated by first obtaining the total peak height across all loci, averaging these totals over all loci, and then dividing the result by two to convert it to the average peak height for a heterozygous allele.

Therefore, the expected peak heights in Equation (32) can be calculated directly using experimental data.

The expected peak heights of stutter products are calculated using the same principle. When multiple allelic products or multiple stutter products overlap at the same allele position, their expected peak heights are summed, and the total is regarded as the expected peak height at that position.

The two scale parameters ($\hat{\eta}^{(\text{allele})}$ and $\hat{\eta}^{(\text{stutter})}$) are estimated by maximum likelihood estimation using the CMA-ES algorithm. The estimation procedure is the same as that used for deconvolution (see Section 10.4 for details).

10.7.4. Drop-in Parameters

This section describes how the laboratory-specific parameters used in the drop-in model presented in Section 10.3.5 are estimated from experimental data. To estimate the drop-in parameters, a sufficiently large number of non-template controls (NTCs) are analyzed.

The observed frequency of drop-in events ($\hat{F}^{(\text{dropin})}$) is calculated as follows.

$$\hat{F}^{(\text{dropin})} = \frac{n^{(\text{dropin})}}{n^{(\text{sample})} \times n^{(\text{marker})}} \tag{34}$$

$n^{(\text{dropin})}$: Number of observed drop-in peaks

$n^{(\text{sample})}$: Number of non-template controls

$n^{(\mathrm{marker})}$: Number of markers analyzed by KongohPlus

The peak heights of all observed drop-in peaks are then collected, and the shape parameter ($\hat{\xi}^{(\mathrm{dropin})}$) and scale parameter ($\hat{\eta}^{(\mathrm{dropin})}$) of the gamma distribution are estimated by maximum likelihood estimation using the CMA-ES algorithm. The estimation procedure is the same as that used for deconvolution (see Section 10.4 for details).

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